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UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 1

HYPERBOLIC AND INVERSE HYPERBOLIC FUNCTIONS

SUGGESTED TIME: 13 PERIODS

IN THIS CHAPTER

This chapter introduces hyperbolic functions, their identities, inverses, calculus, and series expansions. It equips students with the analytical techniques required for the GCE Advanced Level Further Mathematics examination and for modelling natural phenomena.

- Define the six basic hyperbolic functions and compute their numerical values.
- Sketch and interpret the graphs of hyperbolic functions, noting asymptotes and symmetry.
- Derive and apply fundamental hyperbolic identities, including Osborn's rule.
- Define inverse hyperbolic functions, sketch their graphs, and express them in logarithmic form.
- Solve equations involving hyperbolic and inverse hyperbolic functions.
- Differentiate and integrate hyperbolic and inverse hyperbolic functions.

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THIS SIGNED COURSE CONTAINS

COURSE
complete, illustrated, step by step

METHODS
and worked examples

EXERCISES
exam style + solutions

SUMMARY SHEET
the essentials to remember

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2. 2. Graphs of Hyperbolic Functions
3. 3. Hyperbolic Identities and Osborn's Rule
4. 4. Inverse Hyperbolic Functions: Definitions, Graphs, and Logarithmic Forms
5. 5. Solving Hyperbolic Equations
6. 6. Calculus of Hyperbolic Functions: Differentiation and Integration
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CHAPTER OBJECTIVES

- Define the six basic hyperbolic functions and compute their numerical values.
- Sketch and interpret the graphs of hyperbolic functions, noting asymptotes and symmetry.
- Derive and apply fundamental hyperbolic identities, including Osborn's rule.
- Define inverse hyperbolic functions, sketch their graphs, and express them in logarithmic form.
- Solve equations involving hyperbolic and inverse hyperbolic functions.
- Differentiate and integrate hyperbolic and inverse hyperbolic functions.
- Use hyperbolic substitutions to evaluate integrals that contain quadratic expressions.
- Obtain Maclaurin series expansions for sinh x, cosh x, and tanh x and use them for approximations.

PREREQUISITES

- Exponential and logarithmic functions (Lower Sixth).
- Trigonometric identities and their derivations.

- Basic differentiation and integration techniques (chain rule, substitution).
- Maclaurin series for elementary functions (e^x, sin x, cos x).
- Graph sketching using derivatives and asymptotes.

1 DEFINITION AND NUMERICAL EVALUATION OF HYPERBOLIC FUNCTIONS

Imagine a suspension bridge over the Wouri River: the main cables hang in a graceful curve called a catenary, which is described by the hyperbolic cosine function. Engineers use these curves to design safe, elegant structures, and the same functions appear in the study of hanging chains, transmission lines, special relativity and the shape of soap films. In this chapter you will discover the mathematics behind such real-world shapes and learn to manipulate them with the same tools used in advanced calculus.

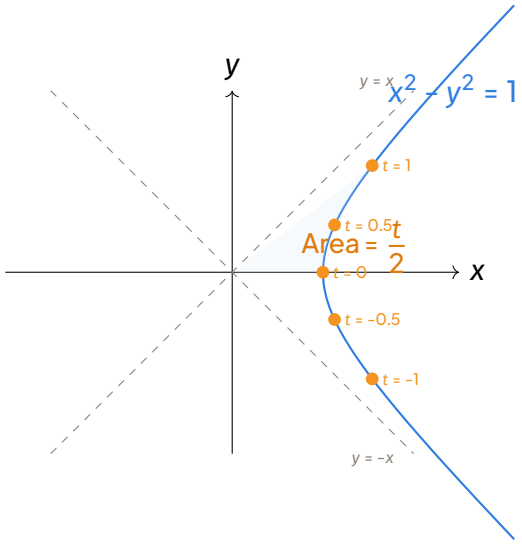
1.1 From the Unit Circle to the Unit Hyperbola

You are familiar with the trigonometric functions defined on the unit circle $x^2 + y^2 = 1$. A point on this circle corresponding to an angle θ has coordinates $(\cos\theta, \sin\theta)$, and the identity $\cos^2\theta + \sin^2\theta = 1$ is simply the statement that the point lies on the circle.

Hyperbolic functions arise from a completely analogous construction on the unit hyperbola $x^2 - y^2 = 1$ (right branch). Let t be a real parameter. The coordinates of a point on this hyperbola are given by

$$x = \cosh t, \qquad y = \sinh t.$$

Here t represents twice the area of the hyperbolic sector bounded by the hyperbola, the x-axis and the line joining the origin to the point $(\cosh t, \sinh t)$. This geometric interpretation is the deep reason why hyperbolic functions satisfy identities that mirror those of trigonometric functions, but with crucial sign differences (Osborn's rule, studied later in this chapter).



The unit hyperbola $x^2 - y^2 = 1$ with points $(\cosh t, \sinh t)$ for several values of t. The shaded sector has area $t/2$.

1.2 Definitions via Exponential Functions

While the geometric definition is beautiful, the algebraic definition using exponentials is the

one we use for calculations, and it is the definition required by the GCE Advanced Level syllabus.

DEFINITION **Hyperbolic Sine and Cosine**

For any real number x,

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}.$$

The names are read "shine x" (or "sinch x") and "cosh x".

From these two basic functions we define the other four hyperbolic functions, exactly as we define tan, cot, sec, csc from sin and cos.

DEFINITION **The Six Hyperbolic Functions**

For $x \in \mathbb{R}$ (with the usual restrictions where denominators vanish):

$$\begin{aligned}
\tanh x &= \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \\
\coth x &= \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}, \quad x \neq 0, \\
\operatorname{sech} x &= \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}, \\
\operatorname{csch} x &= \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}, \quad x \neq 0.
\end{aligned}$$

WARNING **Notation**

There is no standard keyboard symbol for hyperbolic functions: always write them out in full (sinh, cosh, tanh, coth, sech, csch). Note also that $\sinh^2 x$ means $(\sinh x)^2$, exactly as for trigonometric functions.

1.3 Domain, Range, Parity and Values at Zero

Because they are built from exponentials, the domains and ranges are straightforward to determine. For example, since $e^x > 0$ and $e^{-x} > 0$, we have $\cosh x > 0$ for all x; moreover, by the inequality of arithmetic and geometric means, $\cosh x = \frac{e^x + e^{-x}}{2} \geq \sqrt{e^x \cdot e^{-x}} = 1$, with equality only at $x = 0$.

PROPERTY **Domains, Ranges and Parity**

Function	Domain	Range	Parity
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$	Odd: $\sinh(-x) = -\sinh x$
$\cosh x$	$\mathbb{R}$	$[1, +\infty)$	Even: $\cosh(-x) = \cosh x$
$\tanh x$	$\mathbb{R}$	$(-1, 1)$	Odd
$\coth x$	$\mathbb{R} \setminus \{0\}$	$(-\infty, -1) \cup (1, +\infty)$	Odd
$\operatorname{sech} x$	$\mathbb{R}$	$(0, 1]$	Even
$\operatorname{csch} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	Odd

The parity claims follow immediately from the definitions, since replacing x by $-x$ simply interchanges e^x and e^{-x}.

KEY POINTS **Values at $x = 0$**

$$\sinh 0 = 0, \quad \cosh 0 = 1, \quad \tanh 0 = 0, \quad \operatorname{sech} 0 = 1.$$

$\coth 0$ and $\operatorname{csch} 0$ are undefined (the graphs have the y-axis as a vertical asymptote).

1.4 The Fundamental Identity

Just as $\cos^2\theta + \sin^2\theta = 1$ for the unit circle, the unit hyperbola gives the fundamental hyperbolic identity.

PROPERTY **Fundamental Hyperbolic Identity**

For all $x \in \mathbb{R}$,

$$\cosh^2 x - \sinh^2 x = 1.$$

Proof (examinable): Substitute the exponential definitions:

$$\begin{aligned}\cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 \\ &= \frac{1}{4}\left[\left(e^{2x} + 2 + e^{-2x}\right) - \left(e^{2x} - 2 + e^{-2x}\right)\right] \\ &= \frac{1}{4}(4) = 1.\end{aligned}$$

Alternatively, factorise as a difference of two squares:

$$\cosh^2 x - \sinh^2 x = (\cosh x - \sinh x)(\cosh x + \sinh x) = e^{-x} \cdot e^x = 1,$$

since $\cosh x + \sinh x = e^x$ and $\cosh x - \sinh x = e^{-x}$.

KEY POINTS **Two Companion Identities**

Dividing the fundamental identity by $\cosh^2 x$ or by $\sinh^2 x$ yields (valid where the denominators are non-zero):

$$1 - \tanh^2 x = \operatorname{sech}^2 x, \qquad \coth^2 x - 1 = \operatorname{csch}^2 x.$$

WARNING **Sign Change**

The single most common error with hyperbolic functions is writing $\cosh^2 x + \sinh^2 x = 1$ by analogy with the trigonometric identity. The correct identity is $\cosh^2 x - \sinh^2 x = 1$. Note the useful consequence: $\cosh^2 x + \sinh^2 x$ is not constant; in fact $\cosh^2 x + \sinh^2 x = \cosh 2x$ (a double-angle identity proved later in this chapter).

1.5 Numerical Evaluation: Exact and Approximate Values

The exponential form makes exact evaluation possible when x is the logarithm of a rational number. For other values we use a calculator or series approximations (Maclaurin series will be studied later in this chapter).

METHOD **Evaluating Hyperbolic Functions Exactly**

To find the exact value of a hyperbolic function at $x = \ln a$ ($a > 0$):

1. Write the function in exponential form.
2. Substitute $e^{\ln a} = a$ and $e^{-\ln a} = \dfrac{1}{a}$.
3. Simplify the resulting algebraic fraction.

EXEMPLE **Exact values at** $\ln 2$ **and** $\ln 3$

Compute $\sinh(\ln 2)$, $\cosh(\ln 2)$, $\tanh(\ln 2)$, **and the same for** $\ln 3$.

$$\sinh(\ln 2) = \frac{e^{\ln 2} - e^{-\ln 2}}{2} = \frac{2 - \frac{1}{2}}{2} = \frac{3}{4},$$
$$\cosh(\ln 2) = \frac{2 + \frac{1}{2}}{2} = \frac{5}{4},$$
$$\tanh(\ln 2) = \frac{\sinh(\ln 2)}{\cosh(\ln 2)} = \frac{3/4}{5/4} = \frac{3}{5}.$$

$$\sinh(\ln 3) = \frac{3 - \frac{1}{3}}{2} = \frac{4}{3},$$
$$\cosh(\ln 3) = \frac{3 + \frac{1}{3}}{2} = \frac{5}{3},$$
$$\tanh(\ln 3) = \frac{4/3}{5/3} = \frac{4}{5}.$$

x	$\sinh x$	$\cosh x$	$\tanh x$
0	0	1	0
$\ln 2$	$\frac{3}{4}$	$\frac{5}{4}$	$\frac{3}{5}$
$\ln 3$	$\frac{4}{3}$	$\frac{5}{3}$	$\frac{4}{5}$

For values that are not logarithms of simple numbers, we use a scientific calculator, which has dedicated `hyp` keys. Note that hyperbolic functions are not angular functions: the argument x is a pure number, so the degree/radian mode of the calculator does not affect the result of `sinh`, `cosh` or `tanh`.

EXEMPLE **Approximate values**

Use a calculator to find $\sinh 1.5$, $\cosh 1.5$, $\tanh 1.5$ **to four decimal places.**

$$\sinh 1.5 \approx 2.1293, \quad \cosh 1.5 \approx 2.3524, \quad \tanh 1.5 \approx 0.9051.$$

Check: $\cosh^2 1.5 - \sinh^2 1.5 \approx 2.3524^2 - 2.1293^2 = 5.5338 - 4.5339 \approx 0.9999 \approx 1$ **(the small discrepancy is due to rounding).**

1.6 Series Representation (Preview)

The Maclaurin series of e^x is $\sum_{n=0}^{\infty} \frac{x^n}{n!}$. Substituting into the definitions gives the series for $\sinh x$ and $\cosh x$. Notice the striking similarity to the series for $\sin x$ and $\cos x$, but without alternating signs.

PROPERTY **Maclaurin Series**

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!},$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}.$$

Both series converge for all $x \in \mathbb{R}$. (These will be derived properly in the section on Maclaurin series.)

WARNING **Do Not Confuse** sinh **with** sin**!**

The series for $\sin x$ is $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ (alternating signs). The series for $\sinh x$ is $x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$ (all signs positive). A common mistake is to write $\sinh x \approx x - \frac{x^3}{6}$ for small x. The correct approximation is $\sinh x \approx x + \frac{x^3}{6}$.

1.7 Worked Examination-Style Example

WORKED EXAMPLE 1 **GCE Style**

Given that $y = \sinh x$, express e^x in terms of y and hence show that

$$\sinh^{-1} y = \ln\left(y + \sqrt{y^2 + 1}\right).$$

(We will study inverse hyperbolic functions properly later in this chapter; here we simply derive the logarithmic form.)

SOLUTION

Solution: We have $y = \sinh x = \frac{e^x - e^{-x}}{2}$. Multiply both sides by $2e^x$ (which is positive, so no sign issues arise):

$$2ye^x = e^{2x} - 1 \quad \Longrightarrow \quad e^{2x} - 2ye^x - 1 = 0.$$

This is a quadratic equation in e^x. Applying the quadratic formula:

$$e^x = \frac{2y \pm \sqrt{4y^2 + 4}}{2} = y \pm \sqrt{y^2 + 1}.$$

Since $e^x > 0$ for all real x, and $\sqrt{y^2 + 1} > \sqrt{y^2} = |y| \geq y$, the expression $y - \sqrt{y^2 + 1}$ is always

negative and must be rejected. We therefore take the positive sign:

$$e^x = y + \sqrt{y^2+1} \quad \Longrightarrow \quad x = \ln\left(y + \sqrt{y^2+1}\right).$$

Hence $\sinh^{-1} y = \ln\left(y + \sqrt{y^2+1}\right)$, as required.

1.8 Exercises

EXERCISE 1 **Practice**

1. Find the exact values of $\sinh(\ln 5)$, $\cosh(\ln 5)$, $\tanh(\ln 5)$.
2. Evaluate to four decimal places: $\sinh 2.3$, $\cosh 2.3$, $\tanh 2.3$, $\operatorname{sech} 2.3$.
3. Prove that $\cosh(\ln x) = \dfrac{x^2+1}{2x}$ for $x > 0$.
4. Show that $\tanh(\ln x) = \dfrac{x^2-1}{x^2+1}$ for $x > 0$.
5. If $\sinh x = \dfrac{5}{12}$, find the exact values of $\cosh x$ and $\tanh x$.
6. Use the series for $\sinh x$ to approximate $\sinh 0.2$ up to and including the x^5 term. Compare with the calculator value.

SOLUTION **Exercise 1**

1. $\sinh(\ln 5) = \dfrac{5 - \frac{1}{5}}{2} = \dfrac{12}{5}$; $\quad \cosh(\ln 5) = \dfrac{5 + \frac{1}{5}}{2} = \dfrac{13}{5}$; $\quad \tanh(\ln 5) = \dfrac{12/5}{13/5} = \dfrac{12}{13}$.
2. $\sinh 2.3 \approx 4.9370$, $\cosh 2.3 \approx 5.0372$, $\tanh 2.3 \approx 0.9801$, $\operatorname{sech} 2.3 = \dfrac{1}{\cosh 2.3} \approx 0.1985$.
3. $\cosh(\ln x) = \dfrac{e^{\ln x} + e^{-\ln x}}{2} = \dfrac{x + \frac{1}{x}}{2} = \dfrac{x^2+1}{2x}$.
4. $\tanh(\ln x) = \dfrac{x - \frac{1}{x}}{x + \frac{1}{x}} = \dfrac{x^2-1}{x^2+1}$ (multiplying numerator and denominator by x).
5. $\cosh^2 x = 1 + \sinh^2 x = 1 + \dfrac{25}{144} = \dfrac{169}{144}$, so $\cosh x = \dfrac{13}{12}$ (we take the positive root because $\cosh x \geq 1$ for all x). Then $\tanh x = \dfrac{5/12}{13/12} = \dfrac{5}{13}$.

6. $\sinh 0.2 \approx 0.2 + \frac{0.2^3}{6} + \frac{0.2^5}{120} = 0.2 + 0.0013333 + 0.0000027 \approx 0.2013360$. Calculator value: $\sinh 0.2 \approx 0.2013360$. The agreement is excellent.

DID YOU KNOW? **The Catenary and Suspension Bridges**

The shape of a perfectly flexible hanging cable under its own weight is a catenary, with equation $y = a\cosh\left(\frac{x}{a}\right)$. This is not a parabola — a very common misconception, held even by Galileo for some time. Suspension bridges such as the first Wouri bridge in Douala use this principle: the main cables follow a catenary, which distributes tension optimally. The hyperbolic cosine is essentially the only function equal to its own second derivative (up to scaling), which is exactly why it solves the equilibrium equation of a hanging chain. The word "catenary" comes from the Latin catena, meaning chain.

2 GRAPHS OF HYPERBOLIC FUNCTIONS

In the previous section we defined the six hyperbolic functions from the exponential function and learned how to compute their numerical values. A table of values, however, only gives a snapshot. To truly understand these functions — and later to solve equations, differentiate, integrate and sketch inverse functions — we need their **graphs**. In this section we sketch all six hyperbolic curves rigorously, using the tools you already master: domain, parity (symmetry), limits (asymptotes) and the sign of the derivative (monotonicity). We then compare these curves with their trigonometric cousins and finish with graph transformations.

2.1 General strategy for sketching a hyperbolic curve

Every graph in this section is obtained by the same four-step programme, which you should apply automatically at the GCE examination.

METHOD **Sketching a hyperbolic curve**

1. Domain. Determine where the function is defined (watch out for division by $\sinh x$, which vanishes only at $x = 0$).
2. Symmetry. Test $f(-x)$: if $f(-x) = f(x)$ the function is **even** (symmetric about the y-axis); if $f(-x) = -f(x)$ it is **odd** (symmetric about the origin). You then only need to study $x \geq 0$.
3. Limits and asymptotes. Evaluate $\lim_{x\to\pm\infty} f(x)$ and the limits at any excluded point. This reveals horizontal and vertical asymptotes.
4. Monotonicity. Compute $f'(x)$ using $\sinh' = \cosh$ and $\cosh' = \sinh$, study its sign, and

build the variation table. Read off intercepts and extrema.

PROPERTY **Parity of the basic hyperbolic functions**

sinh is an odd function, cosh is an even function, and tanh is odd. Consequently csch and coth are odd, while sech is even.

Proof (examinable). Directly from the definitions:

$$\sinh(-x) = \frac{e^{-x} - e^{x}}{2} = -\frac{e^{x} - e^{-x}}{2} = -\sinh x, \qquad \cosh(-x) = \frac{e^{-x} + e^{x}}{2} = \cosh x.$$

Then $\tanh(-x) = \dfrac{\sinh(-x)}{\cosh(-x)} = -\tanh x$, and the parity of the reciprocals follows. ■

2.2 The graph of $y = \cosh x$

We apply the programme to $f(x) = \cosh x = \dfrac{e^{x} + e^{-x}}{2}$.

Domain and symmetry. cosh x is defined on $\mathbb{R}$ and is even, so we study $x \geq 0$ and reflect in the y-axis.

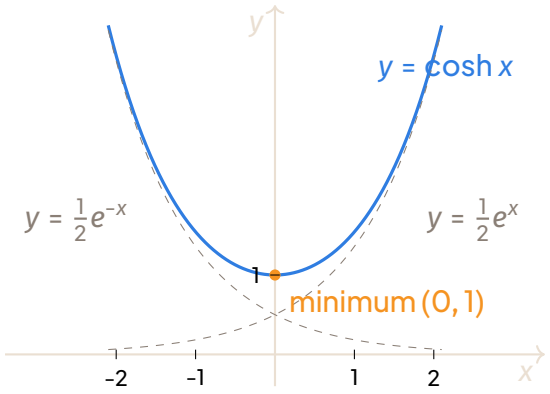
Limits. As $x \to +\infty$, $e^{-x} \to 0$, so $\cosh x \approx \dfrac{e^{x}}{2} \to +\infty$. There is no horizontal asymptote; the curve grows exponentially, hugging the curve $y = \frac{1}{2}e^{x}$. As $x \to -\infty$, by evenness $\cosh x \to +\infty$, asymptotic to $y = \frac{1}{2}e^{-x}$.

Monotonicity. $f'(x) = \sinh x$. Since $\sinh x > 0$ for $x > 0$ and $\sinh x < 0$ for $x < 0$, cosh is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, +\infty)$, with a minimum at $x = 0$:

$$\cosh 0 = \frac{1 + 1}{2} = 1.$$

x	$-\infty$		0		$+\infty$
$f'(x) = \sinh x$	$-$	$-$	0	$+$	$+$
$f(x) = \cosh x$	$+\infty$	$\searrow$	1	$\nearrow$	$+\infty$

The curve is a smooth "U"-shape with minimum at $(0, 1)$, never touching the x-axis (since $\cosh x \geq 1$ for all x).



The curve $y = \cosh x$: even, minimum at $(0, 1)$, asymptotic to $y = \frac{1}{2}e^{x}$ as $x \to +\infty$ and to $y = \frac{1}{2}e^{-x}$ as $x \to -\infty$.

DID YOU KNOW? **The hanging chain**

The curve $y = \cosh x$ is called a **catenary** (from the Latin catena, chain). A flexible chain or cable hanging freely under its own weight — like an electricity line between two pylons on the road from Yaoundé to Douala — takes exactly the shape of a catenary, not a parabola as one might guess. Engineers use cosh to compute the tension and the sag of such cables.

2.3 The graph of $y = \sinh x$

Domain and symmetry. Defined on $\mathbb{R}$, odd: study $x \geq 0$ and use origin symmetry.

Limits. As $x \to +\infty$, $\sinh x \approx \frac{1}{2}e^{x} \to +\infty$; as $x \to -\infty$, $\sinh x \to -\infty$.

Monotonicity. $f'(x) = \cosh x \geq 1 > 0$ for all x: sinh is **strictly increasing on** $\mathbb{R}$. It vanishes only at $x = 0$ (the only intercept), and the tangent at the origin has gradient $\cosh 0 = 1$, i.e. the curve leaves the origin along the line $y = x$.

x	$-\infty$		0		$+\infty$
$f'(x) = \cosh x$	$+$	$+$	1	$+$	$+$
$f(x) = \sinh x$	$-\infty$	$\nearrow$	0	$\nearrow$	$+\infty$

WARNING **Do not confuse the shapes**

$\sinh x$ has no minimum or maximum — it climbs steadily from $-\infty$ to $+\infty$. Only $\cosh x$ has a turning point. Also, $\sinh x$ is negative for $x < 0$; never sketch it above the x-axis on the left.

2.4 The graph of $y = \tanh x$

Domain and symmetry. Defined on $\mathbb{R}$ (since $\cosh x \geq 1 \neq 0$), odd.

Limits — the key feature. Write

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}}.$$

As $x \to +\infty$, $e^{-2x} \to 0$, so $\tanh x \to 1$. As $x \to -\infty$, multiplying numerator and denominator by e^{2x} gives $\tanh x = \dfrac{e^{2x} - 1}{e^{2x} + 1} \to -1$.

PROPERTY **Asymptotes of** tanh

The curve $y = \tanh x$ has two horizontal asymptotes: $y = 1$ as $x \to +\infty$ and $y = -1$ as $x \to -\infty$. Moreover $-1 < \tanh x < 1$ for all real x.

Monotonicity. Using the quotient rule,

$$\frac{d}{dx}\tanh x = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x > 0,$$

so tanh is strictly increasing on $\mathbb{R}$, passing through the origin with gradient 1, and squeezing between its two asymptotes. The graph is an "S"-shaped (sigmoid) curve.

x	$-\infty$		0		$+\infty$
$f'(x) = \operatorname{sech}^2 x$	+	+	1	+	+
$f(x) = \tanh x$	-1	↗	0	↗	1

METHOD **Finding horizontal asymptotes of a hyperbolic quotient**

1. Express the function in terms of e^x and e^{-x}.
2. For $x \to +\infty$, divide numerator and denominator by the dominant exponential (e^x or e^{2x}) and let the vanishing terms go to 0.
3. For $x \to -\infty$, divide by the dominant exponential in that direction (e^{-x} or e^{-2x}).
4. Conclude: $y = L$ is a horizontal asymptote whenever $\lim f(x) = L$ (finite).

2.5 The reciprocal functions: sech, csch, coth

The remaining three graphs are obtained as **reciprocals** of the first three, using the standard rules: where f is large, $1/f$ is small; where f is small (nonzero), $1/f$ is large; where $f = 0$, $1/f$ has a vertical asymptote.

The graph of $y = \operatorname{sech} x = \dfrac{1}{\cosh x}$. Since $\cosh x \geq 1$, we have $0 < \operatorname{sech} x \leq 1$. It is even, has a maximum at $(0, 1)$, and $\operatorname{sech} x \to 0$ as $x \to \pm\infty$: the x-axis is a horizontal asymptote. The curve is a symmetric "bell-shaped" bump. Its derivative is $-\operatorname{sech} x \tanh x$, negative for $x > 0$: decreasing on $[0, +\infty)$.

x	$-\infty$		0		$+\infty$
$f'(x) = -\operatorname{sech} x \tanh x$	$+$	$+$	0	$-$	$-$
$f(x) = \operatorname{sech} x$	0	$\nearrow$	1	$\searrow$	0

The graph of $y = \operatorname{csch} x = \dfrac{1}{\sinh x}$. Since $\sinh 0 = 0$, csch is undefined at $x = 0$: the domain is $\mathbb{R} \setminus \{0\}$ and the line $x = 0$ is a **vertical asymptote**, with

$$\lim_{x \to 0^+} \operatorname{csch} x = +\infty, \qquad \lim_{x \to 0^-} \operatorname{csch} x = -\infty.$$

It is odd, strictly decreasing on each of $(-\infty, 0)$ and $(0, +\infty)$, and $\operatorname{csch} x \to 0$ as $x \to \pm\infty$ (horizontal asymptote $y = 0$). Derivative: $-\operatorname{csch} x \coth x < 0$ for $x \neq 0$.

x	$-\infty$		0^-	0^+		$+\infty$
$f'(x) = -\operatorname{csch} x \coth x$	$-$	$-$	$-\infty$	$-\infty$	$-$	$-$
$f(x) = \operatorname{csch} x$	0	$\searrow$	$-\infty$	$+\infty$	$\searrow$	0

The graph of $y = \coth x = \dfrac{\cosh x}{\sinh x}$. Again undefined at $x = 0$ (vertical asymptote $x = 0$). It is odd. For the limits at infinity:

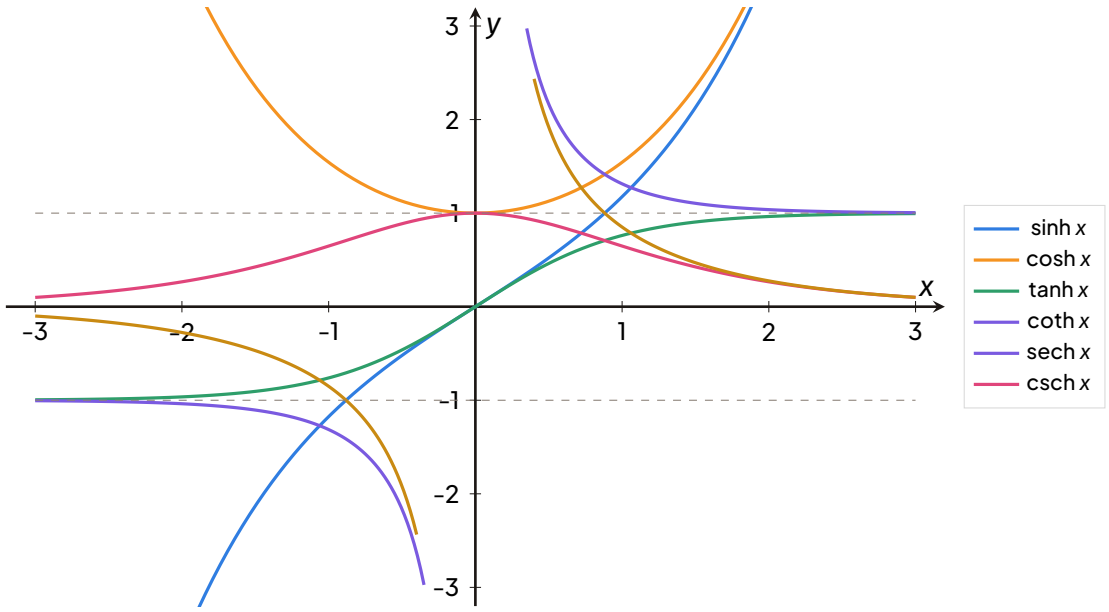
$$\coth x = \frac{1 + e^{-2x}}{1 - e^{-2x}} \to 1 \text{ as } x \to +\infty, \qquad \coth x \to -1 \text{ as } x \to -\infty,$$

so $y = 1$ and $y = -1$ are horizontal asymptotes, with $|\coth x| > 1$ everywhere. Its derivative is $-\operatorname{csch}^2 x < 0$: strictly decreasing on each branch. Note carefully: unlike tanh, the curve coth lies outside the strip $-1 < y < 1$.

x	$-\infty$		0^-	0^+		$+\infty$
$f'(x) = -\operatorname{csch}^2 x$	$-$	$-$	$-\infty$	$-\infty$	$-$	$-$
$f(x) = \coth x$	-1	$\searrow$	$-\infty$	$+\infty$	$\searrow$	1

WARNING **Vertical asymptotes and open circles**

coth x and csch x are not defined at $x = 0$. Your sketch must show the vertical asymptote $x = 0$ (dashed), and the two branches must never touch or cross the y-axis. If you mark a value at a point where the function is undefined, use an open circle. A common examination error is to draw coth as a single continuous S-curve through the origin — that is the graph of tanh, not coth!



All six hyperbolic functions on a common axis. Note the horizontal asymptotes $y = \pm 1$ for tanh and coth, and the vertical asymptote $x = 0$ for coth and csch.

2.6 Comparison with trigonometric functions

The names "hyperbolic sine", "hyperbolic cosine" invite a comparison with sin and cos. The similarities are striking in the algebra (identities, derivatives), but the graphs are completely different.

Feature	Trigonometric	Hyperbolic
Bounded?	$-1 \le \sin x, \cos x \le 1$	$\cosh x \ge 1$; $\sinh x$ unbounded
Periodic?	Yes, period 2π	No period; monotone or single-turned
Zeros	Infinitely many	$\sinh x = 0$ only at $x = 0$; $\cosh x$ never zero
Asymptotes	None	tanh, coth: $y = \pm 1$; csch, coth: $x = 0$
Behaviour at infinity	Oscillates forever	Exponential growth (sinh, cosh) or tends to a limit
Parity	sin odd, cos even	sinh odd, cosh even

The deep reason for the parallel names is Euler's formula: $\cos x = \dfrac{e^{ix} + e^{-ix}}{2}$ and $\sin x = \dfrac{e^{ix} - e^{-ix}}{2i}$ — the same exponential building blocks as cosh and sinh, but with imaginary arguments. Replacing x by ix converts oscillation into exponential growth.

2.7 Transformations of hyperbolic graphs

All the transformation rules you know from pure mathematics apply unchanged. For $y = a\,f(b(x-h)) + k$:

- h is a **horizontal shift** (translation $\begin{pmatrix} h \\ 0 \end{pmatrix}$);
- k is a **vertical shift**;
- a is a **vertical stretch** (scale factor a, reflection in the x-axis if $a < 0$);
- b is a **horizontal stretch** of scale factor $\frac{1}{b}$ (reflection in the y-axis if $b < 0$).

Track the key features — minimum point, asymptotes, intercepts — through the transformation.

WORKED EXAMPLE 2 **Sketching a transformed hyperbolic graph**

Sketch the curve $y = 2\cosh(x-1) - 1$, stating the coordinates of the minimum point and the range.

SOLUTION

Solution. Start from $y = \cosh x$ (minimum at $(0, 1)$, range $[1, +\infty)$) and apply the transformations in order.
Step 1: horizontal shift. $y = \cosh(x - 1)$ is a translation by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$: the minimum moves to $(1, 1)$.
Step 2: vertical stretch by factor 2. $y = 2\cosh(x-1)$: the minimum moves to $(1, 2)$; the range becomes $[2, +\infty)$.
Step 3: vertical shift down by 1. $y = 2\cosh(x - 1) - 1$: the minimum settles at $(1, 1)$.
Conclusion. The curve is a catenary with minimum point $(1, 1)$ and range $y \geq 1$, i.e. $[1, +\infty)$. It is symmetric about the vertical line $x = 1$, and $y \to +\infty$ as $x \to \pm\infty$. The y-intercept is $y = 2\cosh(-1) - 1 = 2\cosh 1 - 1 \approx 2(1.5431) - 1 = 2.086$.

WORKED EXAMPLE 3 **Identifying asymptotes of a transformed curve**

Determine all asymptotes of $y = 3\tanh(2x) + 1$ and state its range.

SOLUTION

Solution. The basic curve $y = \tanh x$ has horizontal asymptotes $y = \pm 1$.
Step 1. The factor 2 in $\tanh(2x)$ is a horizontal stretch of factor $\frac{1}{2}$: it compresses the S-curve horizontally but does not move the horizontal asymptotes.
Step 2. The factor 3 (vertical stretch) sends the asymptotes $y = \pm 1$ to $y = \pm 3$.
Step 3. The shift $+1$ sends them to $y = 3 + 1 = 4$ and $y = -3 + 1 = -2$.
Conclusion. Horizontal asymptotes: $y = 4$ (as $x \to +\infty$) and $y = -2$ (as $x \to -\infty$). The range is the open interval $(-2, 4)$, since $\tanh$ never attains its limiting values. The curve passes through $(0, 1)$ since $\tanh 0 = 0$.

KEY POINTS **Key facts about the six graphs**

- cosh x: even, minimum $(0, 1)$, $\cosh x \geq 1$, grows like $\frac{1}{2}e^{|x|}$.
- sinh x: odd, strictly increasing on $\mathbb{R}$, only zero at $x = 0$, tangent $y = x$ at the origin.
- tanh x: odd, increasing, asymptotes $y = \pm 1$, range $(-1, 1)$.
- sech x: even, maximum $(0, 1)$, asymptote $y = 0$, range $(0, 1]$.
- csch x: odd, vertical asymptote $x = 0$, horizontal asymptote $y = 0$.
- coth x: odd, vertical asymptote $x = 0$, horizontal asymptotes $y = \pm 1$, $|\coth x| > 1$.
- Derivatives $\sinh' = \cosh$, $\cosh' = \sinh$ give the monotonicity directly.

EXERCISE 2 **Graphs and transformations**

1. Using the derivative, show that $y = \operatorname{sech} x$ is decreasing on $[0, +\infty)$ and sketch its graph, marking the maximum and the asymptote.
2. Sketch $y = \sinh(x + 2)$ and $y = -\cosh x + 2$ on separate axes, stating the key point of each curve.
3. Find the horizontal asymptotes and the range of $y = 5 - 2\tanh(3x)$.
4. On the same axes, sketch $y = \tanh x$ and $y = \coth x$ for $x > 0$. Explain why the two curves get closer and closer as $x \to +\infty$.

SOLUTION **Exercise 1**

1. $\frac{d}{dx}\operatorname{sech} x = -\operatorname{sech} x \tanh x$. For $x > 0$, $\operatorname{sech} x > 0$ and $\tanh x > 0$, so the derivative is negative: sech is strictly decreasing on $[0, +\infty)$. Even function, maximum $(0, 1)$, asymptote $y = 0$.

2. $y = \sinh(x + 2)$: translation $\binom{-2}{0}$ of $y = \sinh x$; passes through $(-2, 0)$ with gradient 1, strictly increasing. $y = -\cosh x + 2$: reflection in the x-axis then shift up 2; the minimum $(0, 1)$ becomes a maximum $(0, 1)$; range $(-\infty, 1]$.

3. Asymptotes of tanh are ± 1; after stretch -2 and shift $+5$: $y = 5 - 2 = 3$ and $y = 5 + 2 = 7$. Since the stretch factor is negative, $y \to 3$ as $x \to +\infty$ and $y \to 7$ as $x \to -\infty$. Range: $(3, 7)$.

4. For $x > 0$, $\coth x = \dfrac{1}{\tanh x}$ with $0 < \tanh x < 1$, so $\coth x > 1 > \tanh x$. Both tend to 1 as $x \to +\infty$ (since $e^{-2x} \to 0$), so the two branches converge towards the common asymptote $y = 1$, with tanh approaching from below and coth from above.

With the graphs firmly in hand, we are ready for the next section, where the geometric symmetries we have just observed will reappear as powerful algebraic identities — and where Osborn's rule will let us borrow identities from trigonometry almost for free.

3 HYPERBOLIC IDENTITIES AND OSBORN'S RULE

In the previous section we plotted the curves of sinh x, cosh x and tanh x and observed, for instance, that the graph of cosh always lies above the line $y = 1$, while the graph of sinh passes through the origin. Those pictures strongly suggest that the hyperbolic functions are bound together by algebraic relations, just as the circular functions are. In this section we make those relations precise: we derive the **addition formulas**, the **double-angle** and **half-angle** identities, and we discover a remarkable "translation machine" — **Osborn's rule** — which converts almost any trigonometric identity into a hyperbolic one.

3.1 The fundamental identity revisited

Before deriving new formulas, recall the cornerstone identity established in Section 1:

$$\cosh^2 x - \sinh^2 x = 1.$$

Every identity in this section flows, directly or indirectly, from the exponential definitions

$$\cosh x = \frac{e^x + e^{-x}}{2}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}.$$

The strategy is always the same: replace each hyperbolic function by its exponential form, expand, and reassemble. Let us see this strategy at work on the most important case.

3.2 Addition formulas

PROPERTY **Addition formulas (proof examinable)**

For all real numbers A and B:

$$\begin{aligned}
\sinh(A+B) &= \sinh A \cosh B + \cosh A \sinh B,\\
\sinh(A-B) &= \sinh A \cosh B - \cosh A \sinh B,\\
\cosh(A+B) &= \cosh A \cosh B + \sinh A \sinh B,\\
\cosh(A-B) &= \cosh A \cosh B - \sinh A \sinh B,\\
\tanh(A+B) &= \frac{\tanh A + \tanh B}{1 + \tanh A \tanh B}, \qquad \tanh(A-B) = \frac{\tanh A - \tanh B}{1 - \tanh A \tanh B}.
\end{aligned}$$

Proof of $\sinh(A+B)$. Start from the right-hand side and use the exponential definitions:

$$\begin{aligned}\sinh A\cosh B+\cosh A\sinh B &= \frac{e^{A}-e^{-A}}{2}\cdot\frac{e^{B}+e^{-B}}{2}+\frac{e^{A}+e^{-A}}{2}\cdot\frac{e^{B}-e^{-B}}{2}\\ &= \frac{1}{4}\Big[\big(e^{A+B}+e^{A-B}-e^{-A+B}-e^{-A-B}\big)\\ &\qquad +\big(e^{A+B}-e^{A-B}+e^{-A+B}-e^{-A-B}\big)\Big]\\ &= \frac{1}{4}\Big[2e^{A+B}-2e^{-(A+B)}\Big]=\frac{e^{A+B}-e^{-(A+B)}}{2}=\sinh(A+B).\end{aligned}$$ ■

Notice how the "cross terms" e^{A-B} and e^{-A+B} cancel between the two products: this cancellation is exactly why the formula is so clean. The proof of $\cosh(A+B)$ is identical except that the cross terms add instead of cancelling, giving $2e^{A+B}+2e^{-(A+B)}$ in the bracket, hence $\cosh(A+B)$. The formulas for $A-B$ follow by replacing B with $-B$ and using parity (cosh even, sinh odd). Finally,

$$\tanh(A+B)=\frac{\sinh(A+B)}{\cosh(A+B)}=\frac{\sinh A\cosh B+\cosh A\sinh B}{\cosh A\cosh B+\sinh A\sinh B},$$

and dividing numerator and denominator by $\cosh A\cosh B$ yields the stated formula for $\tanh(A+B)$. ■

Step 1 — Write the exponentials: $\sinh A\cosh B+\cosh A\sinh B=\dfrac{(e^{A}-e^{-A})(e^{B}+e^{-B})+(e^{A}+e^{-A})(e^{B}-e^{-B})}{4}$

↓ multiply out

Step 2 — Expand the products: $=\dfrac{1}{4}\Big[(e^{A+B}+e^{A-B}-e^{-A+B}-e^{-A-B})+(e^{A+B}-e^{A-B}+e^{-A+B}-e^{-A-B})\Big]$

↓ simplify

Step 3 — Cancel the cross terms: e^{A-B} and e^{-A+B} disappear; $=\dfrac{1}{4}\Big[2e^{A+B}-2e^{-(A+B)}\Big]$

↓ factor 2

Step 4 — Recognise the definition: $=\dfrac{e^{A+B}-e^{-(A+B)}}{2}=\sinh(A+B)$. **Done!**

Step-by-step derivation of $\sinh(A+B)$ from the exponential definitions.

EXEMPLE **Using an addition formula**

Compute $\sinh(\ln 2+\ln 3)$ exactly.

Solution. $\sinh(\ln 2+\ln 3)=\sinh(\ln 2)\cosh(\ln 3)+\cosh(\ln 2)\sinh(\ln 3)$. Now $\sinh(\ln 2)=\dfrac{2-\frac{1}{2}}{2}=\dfrac{3}{4}$,

$\cosh(\ln 2) = \frac{2+\frac{1}{2}}{2} = \frac{5}{4}$, $\sinh(\ln 3) = \frac{3-\frac{1}{3}}{2} = \frac{4}{3}$, $\cosh(\ln 3) = \frac{3+\frac{1}{3}}{2} = \frac{5}{3}$. Hence

$$\sinh(\ln 2 + \ln 3) = \frac{3}{4}\cdot\frac{5}{3} + \frac{5}{4}\cdot\frac{4}{3} = \frac{15}{12} + \frac{20}{12} = \frac{35}{12}.$$

Check: $\ln 2 + \ln 3 = \ln 6$, and $\sinh(\ln 6) = \frac{6-\frac{1}{6}}{2} = \frac{35}{12}$. ✓

3.3 Double-angle and half-angle identities

Setting $A = B = x$ in the addition formulas immediately gives the double-angle identities.

PROPERTY **Double-angle identities**

For all real x:

$$\sinh 2x = 2\sinh x \cosh x,$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x = 2\cosh^2 x - 1 = 1 + 2\sinh^2 x,$$

$$\tanh 2x = \frac{2\tanh x}{1 + \tanh^2 x}.$$

Proof. $\sinh 2x = \sinh(x + x) = \sinh x \cosh x + \cosh x \sinh x = 2\sinh x \cosh x$. Similarly $\cosh 2x = \cosh^2 x + \sinh^2 x$; then substituting $\sinh^2 x = \cosh^2 x - 1$ gives $\cosh 2x = 2\cosh^2 x - 1$, and substituting $\cosh^2 x = 1 + \sinh^2 x$ gives $\cosh 2x = 1 + 2\sinh^2 x$. The tanh formula comes from $\tanh(x + x)$. ■

Rearranging the last two forms of $\cosh 2x$ produces the **half-angle** (or "power-lowering") identities, which will be precious for integration in Section 6:

$$\cosh^2 x = \frac{\cosh 2x + 1}{2}, \qquad \sinh^2 x = \frac{\cosh 2x - 1}{2}.$$

WARNING **Sign trap**

For circular functions, $\cos 2x = \cos^2 x - \sin^2 x$, with a minus. For hyperbolic functions it is $\cosh 2x = \cosh^2 x + \sinh^2 x$, with a plus. Mixing these two signs is the single most common error at the GCE: always double-check against $\cosh^2 x - \sinh^2 x = 1$.

WORKED EXAMPLE 4 **Solving with a double-angle identity**

Solve the equation $\cosh 2x - 3\cosh x + 4 = 0$, giving answers in exact logarithmic form.

SOLUTION

Solution. Use $\cosh 2x = 2\cosh^2 x - 1$:

$$2\cosh^2 x - 1 - 3\cosh x + 4 = 0 \implies 2\cosh^2 x - 3\cosh x + 3 = 0.$$

The discriminant is $\Delta = (-3)^2 - 4(2)(3) = 9 - 24 = -15 < 0$. Hence the quadratic has no real roots for $\cosh x$. Since $\cosh x \in \mathbb{R}$ for all real x, the original equation has no real solution.
Remark. A classic solvable variant is $\cosh 2x - 3\cosh x + 2 = 0$. Then $2\cosh^2 x - 3\cosh x + 1 = 0$, i.e. $(2\cosh x - 1)(\cosh x - 1) = 0$. Since $\cosh x \geq 1$, $\cosh x = \frac{1}{2}$ is impossible, so $\cosh x = 1$, giving $x = 0$ as the unique solution.
Lesson: after solving the quadratic in $\cosh x$, always reject roots below 1.

3.4 Osborn's rule

Compare the identities we have met so far with their circular cousins:

Trigonometric identity	Hyperbolic identity
$\cos^2 x + \sin^2 x = 1$	$\cosh^2 x - \sinh^2 x = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$	$\sinh(A + B) = \sinh A \cosh B + \cosh A \sinh B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$	$\cosh(A + B) = \cosh A \cosh B + \sinh A \sinh B$
$\sin 2x = 2 \sin x \cos x$	$\sinh 2x = 2 \sinh x \cosh x$
$\cos 2x = \cos^2 x - \sin^2 x$	$\cosh 2x = \cosh^2 x + \sinh^2 x$
$1 + \tan^2 x = \sec^2 x$	$1 - \tanh^2 x = \operatorname{sech}^2 x$

The pattern is striking: every formula is identical, except that a sign changes whenever a product of two sines is involved. This is no coincidence. Since $\cos x = \cosh(ix)$ and $\sin x = -i\sinh(ix)$ (from the exponential definitions), any product of two sines picks up a factor $(-i)^2 = -1$ when translated to hyperbolic form. This observation is formalised as:

PROPERTY **Osborn's Rule**

To convert any trigonometric identity into the corresponding hyperbolic identity:

1. replace cos by cosh and sin by sinh (and similarly tan → tanh, sec → sech, cosec → cosech, cot → coth);
2. change the sign of every term containing a product (or implied product) of two sines, i.e. replace $\sin^2$ by $-\sinh^2$.

The rule is a consequence of $\cosh(ix) = \cos x$ and $\sinh(ix) = i\sin x$; its justification is not examinable, but using the rule is.

METHOD **Converting a trigonometric identity**

1. Write the identity in terms of sin and cos only (so that hidden products of sines become visible: e.g. $\tan^2 x = \sin^2 x / \cos^2 x$).
2. Replace sin → sinh, cos → cosh.
3. Change the sign of every product of two sines ($\sin^2$, $\sin A \sin B$, a $\sin^2$ hidden in a numerator, etc.).
4. Verify the result on a numerical value (e.g. $x = 0$ or $x = \ln 2$) as a safety check.

EXEMPLE **Osborn's rule in action**

From $\cos 2x = 1 - 2\sin^2 x$ (the sine product $-2\sin^2 x$ changes sign), we get

$$\cosh 2x = 1 + 2\sinh^2 x \quad \checkmark$$

From $1 + \tan^2 x = \sec^2 x$, written as $1 + \dfrac{\sin^2 x}{\cos^2 x} = \dfrac{1}{\cos^2 x}$, the numerator $\sin^2 x$ changes sign:

$$1 - \tanh^2 x = \operatorname{sech}^2 x \quad \checkmark$$

From $\sin 3x = 3\sin x - 4\sin^3 x$: here $\sin^3 x = \sin x \cdot \sin^2 x$ contains a product of two sines, so its sign flips:

$$\sinh 3x = 3\sinh x + 4\sinh^3 x.$$

WARNING **Limits of Osborn's rule**

Osborn's rule does not apply blindly to identities involving odd powers of sine standing alone. A single $\sin x$ becomes $\sinh x$ with no sign change, but a lone $\sin^3 x$ must be read as $\sin x \cdot \sin^2 x$ to see the hidden product of two sines. Also, the rule is designed for identities, not for equations with specific numerical coefficients that are not identities, nor for statements involving calculus (derivatives follow their own sign rules: $\frac{d}{dx}\sinh x = \cosh x$, no minus sign!). When in doubt, go back to the exponential definitions.

3.5 Applying identities to simplify and solve

WORKED EXAMPLE 5 **Simplifying an expression**

Show that $\dfrac{\cosh 2x + \sinh 2x}{\cosh x + \sinh x} = e^x$, and deduce a simplification of $(\cosh x + \sinh x)^n$.

SOLUTION

Solution. Using the addition formulas with $A = B = x$, or directly the definitions:

$$\cosh x + \sinh x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} = e^x.$$

Likewise $\cosh 2x + \sinh 2x = e^{2x}$. Hence the quotient equals $\dfrac{e^{2x}}{e^x} = e^x$. ■
It follows immediately that

$$(\cosh x + \sinh x)^n = (e^x)^n = e^{nx} = \cosh(nx) + \sinh(nx),$$

a neat analogue of De Moivre's theorem, sometimes called the hyperbolic De Moivre identity.

WORKED EXAMPLE 6 **Proving an identity**

Prove that $\cosh 3x = 4\cosh^3 x - 3\cosh x$.

SOLUTION

Solution. Write $\cosh 3x = \cosh(2x + x)$ and apply the addition formula:

$$\begin{aligned}\cosh(2x + x) &= \cosh 2x \cosh x + \sinh 2x \sinh x \\ &= (2\cosh^2 x - 1)\cosh x + (2\sinh x \cosh x)\sinh x \\ &= 2\cosh^3 x - \cosh x + 2\cosh x \sinh^2 x \\ &= 2\cosh^3 x - \cosh x + 2\cosh x(\cosh^2 x - 1) \\ &= 4\cosh^3 x - 3\cosh x. \qquad ■\end{aligned}$$

Note that this matches, via Osborn's rule, the circular identity $\cos 3x = 4\cos^3 x - 3\cos x$ — no sine product appears, so no sign changes.

EXERCISE 3 **Consolidation**

1. Using the exponential definitions, prove that $\cosh(A - B) = \cosh A \cosh B - \sinh A \sinh B$.
2. Use Osborn's rule to write down the hyperbolic analogue of $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$, then verify it numerically for $A = \ln 2$, $B = \ln 3$.

3. **Prove that** $\sinh 3x = 3\sinh x + 4\sinh^3 x$.

4. **Solve** $2\sinh^2 x - 3\cosh x = 0$. **(Hint: use** $\cosh^2 x - \sinh^2 x = 1$ **first.)**

SOLUTION **Exercise 1**

1. $\cosh A\cosh B - \sinh A\sinh B = \frac{1}{4}\left[(e^A + e^{-A})(e^B + e^{-B}) - (e^A - e^{-A})(e^B - e^{-B})\right] = \frac{1}{4}\left[2e^{A-B} + 2e^{-(A-B)}\right] = \cosh(A - B)$.

2. Osborn: $\tanh(A - B) = \dfrac{\tanh A - \tanh B}{1 - \tanh A\tanh B}$ (the product $\tan A\tan B$ hides $\sin A\sin B$, so the sign flips). Numerically: $\tanh(\ln 2) = \frac{2-\frac{1}{2}}{2+\frac{1}{2}} = \frac{3}{5}$, $\tanh(\ln 3) = \frac{3-\frac{1}{3}}{3+\frac{1}{3}} = \frac{4}{5}$; RHS $= \dfrac{\frac{3}{5} - \frac{4}{5}}{1 - \frac{12}{25}} = \dfrac{-\frac{1}{5}}{\frac{13}{25}} = -\frac{5}{13}$;

LHS $= \tanh(\ln\frac{2}{3}) = \dfrac{\frac{2}{3} - \frac{3}{2}}{\frac{2}{3} + \frac{3}{2}} = -\frac{5}{13}$. ✓

3. $\sinh(2x + x) = \sinh 2x\cosh x + \cosh 2x\sinh x = 2\sinh x\cosh^2 x + (1 + 2\sinh^2 x)\sinh x$; replacing $\cosh^2 x = 1 + \sinh^2 x$ gives $2\sinh x(1 + \sinh^2 x) + \sinh x + 2\sinh^3 x = 3\sinh x + 4\sinh^3 x$.

4. $2(\cosh^2 x - 1) - 3\cosh x = 0 \Rightarrow 2\cosh^2 x - 3\cosh x - 2 = 0 \Rightarrow (2\cosh x + 1)(\cosh x - 2) = 0$. Since $\cosh x \geq 1$, only $\cosh x = 2$ survives, so $x = \pm\ln(2 + \sqrt{3})$.

KEY POINTS **Key points of Section 3**

- Addition formulas mirror the trigonometric ones, except $\cosh(A + B) = \cosh A\cosh B + \sinh A\sinh B$.
- $\cosh 2x = \cosh^2 x + \sinh^2 x = 2\cosh^2 x - 1 = 1 + 2\sinh^2 x$; $\sinh 2x = 2\sinh x\cosh x$.
- Half-angle: $\cosh^2 x = \frac{\cosh 2x+1}{2}$, $\sinh^2 x = \frac{\cosh 2x-1}{2}$.
- **Osborn's rule**: convert sin → sinh, cos → cosh, and change the sign of every product of two sines (including those hidden in $\tan^2$, $\sin^3$, etc.).
- When solving equations in $\cosh x$, always reject roots with $\cosh x < 1$.

DID YOU KNOW? **Why "hyperbolic"?**

Just as $(\cos t, \sin t)$ parametrizes the unit circle $x^2 + y^2 = 1$, the point $(\cosh t, \sinh t)$ parametrizes the unit hyperbola $x^2 - y^2 = 1$ — this is precisely the identity $\cosh^2 t - \sinh^2 t = 1$ at work. The parameter t even has the same geometric meaning in both cases: twice the area of the shaded sector. Osborn's rule is the algebraic shadow of this deep parallel between circle and hyperbola.

4 INVERSE HYPERBOLIC FUNCTIONS: DEFINITIONS, GRAPHS, AND LOGARITHMIC FORMS

In the previous section we discovered that hyperbolic identities mirror trigonometric ones, with sign changes governed by Osborn's rule. We now ask a natural question: can we undo a hyperbolic function? If $\sinh y = 3$, what is y? Questions like this arise constantly at the GCE Advanced Level — when solving hyperbolic equations, when evaluating integrals such as $\int \frac{dx}{\sqrt{x^2+1}}$, and when working with the catenary. The answer is the family of **inverse hyperbolic functions**, and we shall see that, remarkably, each of them can be written using nothing more than the natural logarithm.

4.1 The Idea of an Inverse Hyperbolic Function

Recall from pure mathematics that a function f has an inverse f^{-1} only if f is **one-to-one** (bijective) on the domain considered, and that the graph of f^{-1} is the reflection of the graph of f in the line $y = x$.

DEFINITION **Inverse hyperbolic functions**

For a hyperbolic function f, the inverse function f^{-1} is defined by

$$y = f^{-1}(x) \iff x = f(y),$$

with the domain of f restricted where necessary so that f is one-to-one. The six inverse hyperbolic functions are:

Function	Notation	Domain	Range
Inverse hyperbolic sine	$y = \operatorname{arsinh} x$	$\mathbb{R}$	$\mathbb{R}$
Inverse hyperbolic cosine	$y = \operatorname{arcosh} x$	$[1, +\infty)$	$[0, +\infty)$
Inverse hyperbolic tangent	$y = \operatorname{artanh} x$	$(-1, 1)$	$\mathbb{R}$
Inverse hyperbolic cotangent	$y = \operatorname{arcoth} x$	$(-\infty, -1) \cup (1, +\infty)$	$\mathbb{R} \setminus \{0\}$
Inverse hyperbolic secant	$y = \operatorname{arsech} x$	$(0, 1]$	$[0, +\infty)$
Inverse hyperbolic cosecant	$y = \operatorname{arcsch} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$

WARNING **Why** arcosh **needs a restricted domain**

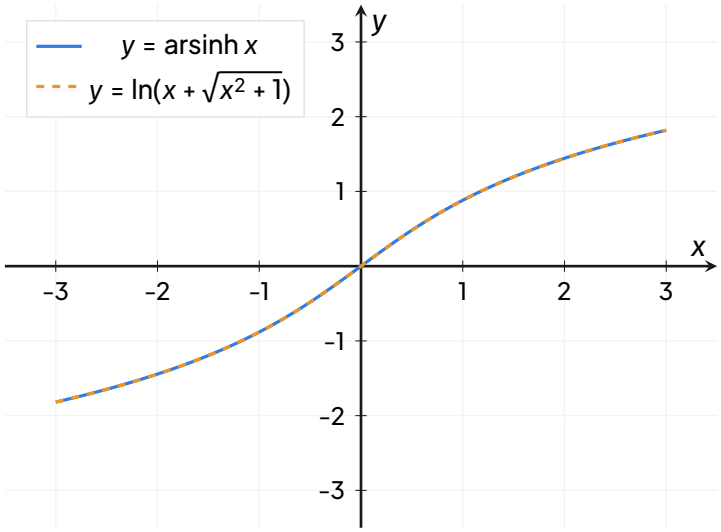
The function cosh is not one-to-one on $\mathbb{R}$: for example $\cosh(2) = \cosh(-2)$. Its graph is symmetric about the y-axis, so a horizontal line $y = k$ (with $k > 1$) cuts it twice. To define an inverse we restrict cosh to $[0, +\infty)$, where it is strictly increasing. This is exactly parallel to restricting cos to $[0, \pi]$ in order to define arccos.

PROPERTY **Domain and range of** arcosh

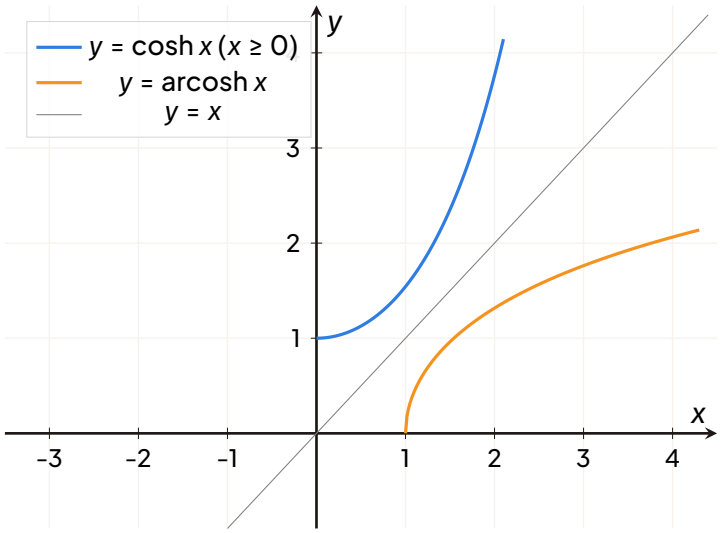
arcosh x is defined only for $x \geq 1$, and its range is $[0, +\infty)$. In particular $\operatorname{arcosh} 1 = 0$ and $\operatorname{arcosh} x > 0$ for $x > 1$. (Not examinable as a proof; it follows directly from the restriction of cosh to $[0, +\infty)$.)

4.2 Graphs as Reflections in $y = x$

Because $y = \sinh x$ is strictly increasing on $\mathbb{R}$, its inverse arsinh exists on the whole of $\mathbb{R}$ and is also strictly increasing and odd. The graph of $y = \operatorname{arsinh} x$ is obtained by reflecting $y = \sinh x$ in the line $y = x$.



The curve $y = \operatorname{arsinh} x$ (solid) coincides exactly with its logarithmic form $y = \ln(x + \sqrt{x^2 + 1})$ (dashed), confirming the equivalence derived below.



$y = \operatorname{arcosh} x$ is the mirror image in $y = x$ of the right half of $y = \cosh x$. The left half of cosh has no inverse on the principal branch.

The remaining graphs are obtained in the same way:

- $y = \operatorname{artanh} x$: an increasing S-shaped curve on $(-1, 1)$, with vertical asymptotes $x = \pm 1$, passing through the origin (odd function).
- $y = \operatorname{arcoth} x$: defined outside $[-1, 1]$, with vertical asymptotes $x = \pm 1$, tending to 0 as $|x| \to \infty$.
- $y = \operatorname{arsech} x$: defined on $(0, 1]$, decreasing from $+\infty$ (as $x \to 0^+$) down to $\operatorname{arsech} 1 = 0$.
- $y = \operatorname{arcsch} x$: defined on $\mathbb{R} \setminus \{0\}$, odd, with a vertical asymptote at $x = 0$ and tending to 0 as $|x| \to \infty$.

4.3 Logarithmic Equivalents of the Inverse Hyperbolic Functions

Here is the beautiful surprise of this chapter: because hyperbolic functions are built from exponentials, their inverses can be written with logarithms. These formulae are examinable — you must be able to derive them.

METHOD **Deriving a logarithmic form**

To find the logarithmic form of an inverse hyperbolic function:

1. Set $y = \operatorname{arsinh} x$ (for example), so that $x = \sinh y$.
2. Write $\sinh y$ in exponentials: $x = \dfrac{e^y - e^{-y}}{2}$.
3. Multiply through by $2e^y$ to obtain a quadratic in e^y.
4. Solve the quadratic for e^y using the quadratic formula.

5. Reject any root that is impossible (remember $e^y > 0$), then take ln of both sides.

PROPERTY **Logarithmic forms — proofs examinable**

$$\begin{aligned}
\operatorname{arsinh} x &= \ln\left(x + \sqrt{x^2+1}\right), && x \in \mathbb{R},\\
\operatorname{arcosh} x &= \ln\left(x + \sqrt{x^2-1}\right), && x \geq 1,\\
\operatorname{artanh} x &= \tfrac{1}{2}\ln\left(\frac{1+x}{1-x}\right), && -1 < x < 1,\\
\operatorname{arcoth} x &= \tfrac{1}{2}\ln\left(\frac{x+1}{x-1}\right), && |x| > 1,\\
\operatorname{arsech} x &= \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right), && 0 < x \leq 1,\\
\operatorname{arcsch} x &= \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right), && x \neq 0.
\end{aligned}$$

EXEMPLE **Derivation of** $\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$

Let $y = \operatorname{arsinh} x$, so $x = \sinh y = \dfrac{e^y - e^{-y}}{2}$. Multiplying by $2e^y$:

$$2x\,e^y = e^{2y} - 1 \quad\Longrightarrow\quad e^{2y} - 2x\,e^y - 1 = 0.$$

This is a quadratic in e^y. By the quadratic formula:

$$e^y = \frac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}.$$

Since $\sqrt{x^2+1} > |x| \geq x$, the root $x - \sqrt{x^2+1}$ is negative, which is impossible because $e^y > 0$. Hence $e^y = x + \sqrt{x^2+1}$, and taking logarithms:

$$y = \operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right). \qquad \blacksquare$$

EXEMPLE **Derivation of the logarithmic form of** artanh

Let $y = \operatorname{artanh} x$, so $x = \tanh y = \dfrac{e^y - e^{-y}}{e^y + e^{-y}} = \dfrac{e^{2y}-1}{e^{2y}+1}$. Then

$$x\left(e^{2y}+1\right) = e^{2y} - 1 \quad\Longrightarrow\quad e^{2y}(1-x) = 1 + x \quad\Longrightarrow\quad e^{2y} = \frac{1+x}{1-x}.$$

Taking logarithms and halving:

$$y = \operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right), \qquad -1 < x < 1. \qquad \blacksquare$$

Note that the condition $|x| < 1$ guarantees $\frac{1+x}{1-x} > 0$, so the logarithm exists.

WORKED EXAMPLE 7 **Logarithmic form of** arcosh

Prove that for $x \geq 1$,

$$\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right).$$

SOLUTION

Solution. Let $y = \operatorname{arcosh} x$ with $x \geq 1$, so that $y \geq 0$ and $x = \cosh y = \frac{e^y + e^{-y}}{2}$. Multiplying by $2e^y$:

$$e^{2y} - 2x\,e^y + 1 = 0.$$

Solving this quadratic in e^y:

$$e^y = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}.$$

Both roots are positive when $x \geq 1$, and they are reciprocals of each other since

$$\left(x + \sqrt{x^2 - 1}\right)\left(x - \sqrt{x^2 - 1}\right) = x^2 - (x^2 - 1) = 1.$$

The root $x - \sqrt{x^2 - 1} \leq 1$ would give $y = \ln(x - \sqrt{x^2 - 1}) \leq 0$, i.e. the negative branch. Since the principal value requires $y \geq 0$, we keep the larger root:

$$e^y = x + \sqrt{x^2 - 1} \quad \Longrightarrow \quad \operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right). \qquad \blacksquare$$

WARNING **Discard the negative branch for** arcosh

When solving the quadratic for arcosh, both roots $x \pm \sqrt{x^2 - 1}$ are positive, so it is tempting to keep both. But the principal value of arcosh must satisfy $y \geq 0$. The root $x - \sqrt{x^2 - 1}$ is at most 1, giving $y \leq 0$: it corresponds to $-\operatorname{arcosh} x$. Always take the + sign. The same care applies to arsech.

EXEMPLE **Numerical evaluation**

Evaluate arsinh 2 and arcosh 2 exactly and to 4 decimal places.

$$\operatorname{arsinh} 2 = \ln\left(2+\sqrt{5}\right) \approx 1.4436, \qquad \operatorname{arcosh} 2 = \ln\left(2+\sqrt{3}\right) \approx 1.3170.$$

Check on your calculator: $\sinh(1.4436) \approx 2$ and $\cosh(1.3170) \approx 2$. ✓

KEY POINTS **Key points of this section**

- $y = f^{-1}(x) \iff x = f(y)$; the graph of f^{-1} is the reflection of the graph of f in $y = x$.
- arsinh has domain and range $\mathbb{R}$; arcosh has domain $[1, \infty)$ and range $[0, \infty)$; artanh has domain $(-1, 1)$ and range $\mathbb{R}$.
- Every inverse hyperbolic function has a logarithmic form, obtained by solving a quadratic in e^y.
- For arcosh and arsech, keep only the branch giving the principal (non-negative) value.

EXERCISE 4 **Logarithmic forms and evaluation**

1. Starting from $x = \operatorname{sech} y$, prove that $\operatorname{arsech} x = \ln\left(\dfrac{1+\sqrt{1-x^2}}{x}\right)$ for $0 < x \le 1$.
2. Show that $\operatorname{artanh} x + \operatorname{artanh}\left(\frac{1}{3}\right)$ with $x = \frac{1}{2}$ can be written as a single logarithm, and simplify.
3. Evaluate exactly: $\operatorname{arcosh}\left(\frac{5}{4}\right)$.

SOLUTION **Exercise 4.1**

1. Let $y = \operatorname{arsech} x$, so $x = \operatorname{sech} y = \dfrac{2}{e^y + e^{-y}}$ with $y \ge 0$. Then $x\,e^{2y} - 2e^y + x = 0$, a quadratic in e^y:

$$e^y = \frac{2 \pm \sqrt{4-4x^2}}{2x} = \frac{1 \pm \sqrt{1-x^2}}{x}.$$

The principal value $y \ge 0$ requires $e^y \ge 1$, so we take the + sign (the other root is its reciprocal, ≤ 1). Hence $\operatorname{arsech} x = \ln\left(\dfrac{1+\sqrt{1-x^2}}{x}\right)$.

2. $\operatorname{artanh}\frac{1}{2} + \operatorname{artanh}\frac{1}{3} = \frac{1}{2}\ln\left(\frac{1+1/2}{1-1/2}\right) + \frac{1}{2}\ln\left(\frac{1+1/3}{1-1/3}\right) = \frac{1}{2}\ln 3 + \frac{1}{2}\ln 2 = \frac{1}{2}\ln 6 = \ln\sqrt{6}$.

3. $\operatorname{arcosh}\left(\frac{5}{4}\right) = \ln\left(\frac{5}{4} + \sqrt{\frac{25}{16} - 1}\right) = \ln\left(\frac{5}{4} + \frac{3}{4}\right) = \ln 2$.

DID YOU KNOW? **Why "area" functions?**

In many textbooks the inverse hyperbolic functions are written arsinh, arcosh, etc., where "ar" stands for area, not "arc". Indeed, while arcsin x measures an arc length on the unit circle, arsinh x measures an area bounded by the rectangular hyperbola $x^2 - y^2 = 1$ — a deep geometric parallel between circular and hyperbolic functions that also explains why these functions appear when integrating $\dfrac{1}{\sqrt{x^2 \pm 1}}$, as we shall exploit in Section 7.

We are now fully equipped with the inverse hyperbolic functions and their logarithmic disguises. In the next section we put them to work immediately, using them as the key tool for solving hyperbolic equations.

5 SOLVING HYPERBOLIC EQUATIONS

In the previous section we established that every inverse hyperbolic function can be expressed as a logarithm: for instance $\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right)$ and $\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2-1}\right)$ for $x \geq 1$. We now put these results to work. A **hyperbolic equation** is an equation in which the unknown appears inside hyperbolic functions, such as

$$3\sinh x + 4\cosh x = 6, \qquad 2\sinh^2 x - 3\sinh x + 1 = 0, \qquad \operatorname{arcosh} x = \operatorname{arsinh} 2.$$

Because sinh and cosh are built from exponentials, the whole machinery of solving such equations rests on one simple idea: replace the hyperbolic functions by their exponential definitions and reduce the problem to an ordinary algebraic equation — usually a quadratic in e^x. Alternatively, when the equation involves inverse hyperbolic functions, we convert them into logarithms and solve a logarithmic equation. In both cases we must remain vigilant: algebraic manipulations (squaring, taking logarithms, multiplying by expressions that could be zero) can introduce **extraneous solutions** that do not satisfy the original equation.

5.1 Equations of the Form $a\sinh x + b\cosh x = c$

Intuition. Suppose you are told that a certain combination of sinh x and cosh x equals a given number. Since $\sinh x = \dfrac{e^x - e^{-x}}{2}$ and $\cosh x = \dfrac{e^x + e^{-x}}{2}$, the left-hand side is really a combination of e^x and e^{-x}. Multiplying through by e^x clears the negative power and produces a quadratic equation in the unknown $u = e^x$. Since $e^x > 0$ for all real x, only **positive** roots of this quadratic give genuine solutions, via $x = \ln u$.

METHOD **Exponential substitution for $a\sinh x + b\cosh x = c$**

1. Replace $\sinh x = \dfrac{e^x - e^{-x}}{2}$ and $\cosh x = \dfrac{e^x + e^{-x}}{2}$.

2. Group the terms in e^{x} and e^{-x}: the equation becomes

$$\frac{b+a}{2}\,e^{x}+\frac{b-a}{2}\,e^{-x}=c.$$

3. Multiply both sides by $2e^{x}$ (which is never zero):

$$(b+a)e^{2x}-2c\,e^{x}+(b-a)=0.$$

4. Set $u=e^{x}$ and solve the quadratic $(b+a)u^{2}-2cu+(b-a)=0$.
5. Keep only the roots with $u>0$; the solutions are $x=\ln u$.
6. Verify each candidate by substituting back into the original equation.

WORKED EXAMPLE 8 **Solving** $3\sinh x+4\cosh x=6$

Solve the equation $3\sinh x+4\cosh x=6$, giving your answers in exact logarithmic form.

SOLUTION

Solution. Write the exponential forms:

$$3\cdot\frac{e^{x}-e^{-x}}{2}+4\cdot\frac{e^{x}+e^{-x}}{2}=6\;\Longrightarrow\;\frac{7}{2}e^{x}+\frac{1}{2}e^{-x}=6.$$

Multiply by $2e^{x}$:

$$7e^{2x}-12e^{x}+1=0.$$

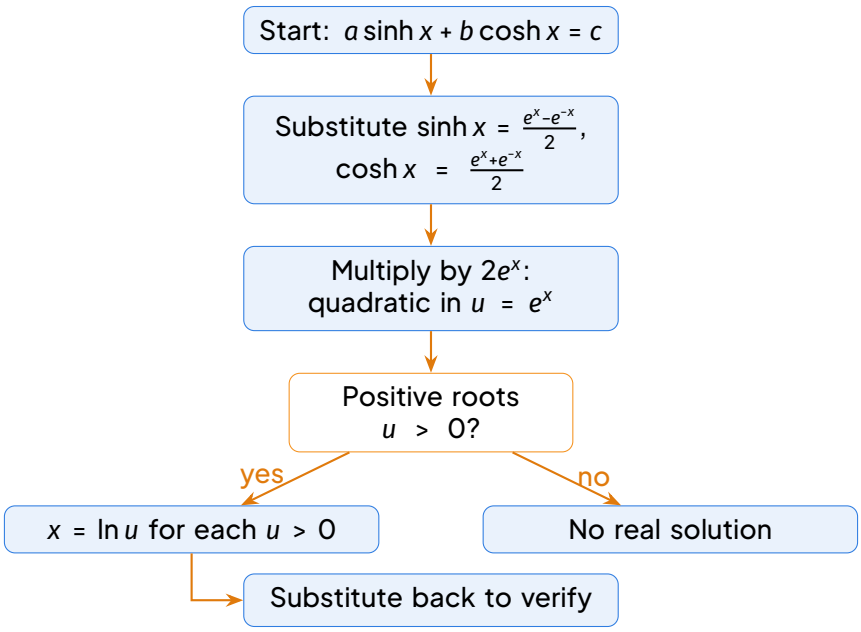
Let $u=e^{x}$: $7u^{2}-12u+1=0$. By the quadratic formula,

$$u=\frac{12\pm\sqrt{144-28}}{14}=\frac{12\pm\sqrt{116}}{14}=\frac{6\pm\sqrt{29}}{7}.$$

Both roots are positive (since $\sqrt{29}\approx 5.385<6$), so both are admissible:

$$\boxed{x=\ln\left(\frac{6+\sqrt{29}}{7}\right)\quad\text{or}\quad x=\ln\left(\frac{6-\sqrt{29}}{7}\right).}$$

Verification. For $u_1=\frac{6+\sqrt{29}}{7}$, $e^{-x}=1/u_1=\frac{7}{6+\sqrt{29}}=\frac{7(6-\sqrt{29})}{36-29}=\frac{6-\sqrt{29}}{7}$. Then $\sinh x=\frac{u_1-1/u_1}{2}=\frac{\sqrt{29}}{7}$, $\cosh x=\frac{u_1+1/u_1}{2}=\frac{6}{7}$. LHS $=3(\sqrt{29}/7)+4(6/7)=(3\sqrt{29}+24)/7$. Wait, check: $3\sinh x+4\cosh x=\frac{3}{2}(u_1-1/u_1)+\frac{4}{2}(u_1+1/u_1)=\frac{7}{2}u_1+\frac{1}{2}(1/u_1)$. Since u_1 satisfies $7u_1^{2}-12u_1+1=0$, we have $7u_1+1/u_1=12$, so LHS $=\frac{1}{2}(12)=6$. ✓ The same algebra holds for u_2.



Systematic strategy for solving $a\sinh x + b\cosh x = c$ by exponential substitution.

An alternative: the auxiliary-argument identity. Just as $a\sin\theta + b\cos\theta$ can be written as a single sine in trigonometry, a linear combination of sinh and cosh collapses into a single hyperbolic function.

PROPERTY **Auxiliary-argument identity**

If $b > |a|$, then

$$a\sinh x + b\cosh x = \sqrt{b^2 - a^2}\,\cosh(x + \alpha), \qquad \text{where } \tanh\alpha = \frac{a}{b}.$$

(When $a > |b|$, one instead obtains $\sqrt{a^2 - b^2}\sinh(x + \alpha)$ with $\tanh\alpha = b/a$.)

Derivation (instructive). Expand $R\cosh(x+\alpha) = R\cosh x\cosh\alpha + R\sinh x\sinh\alpha$. Matching with $a\sinh x + b\cosh x$ gives $R\sinh\alpha = a$ and $R\cosh\alpha = b$. Squaring and subtracting using $\cosh^2\alpha - \sinh^2\alpha = 1$ yields $R^2(\cosh^2\alpha - \sinh^2\alpha) = b^2 - a^2$, hence $R = \sqrt{b^2 - a^2}$ (positive root since $\cosh(x+\alpha) \geq 1$). Dividing gives $\tanh\alpha = a/b$. The condition $b > |a|$ guarantees $b^2 - a^2 > 0$ and $|\tanh\alpha| < 1$, so a real α exists. ■

EXEMPLE **Same equation, quicker route**

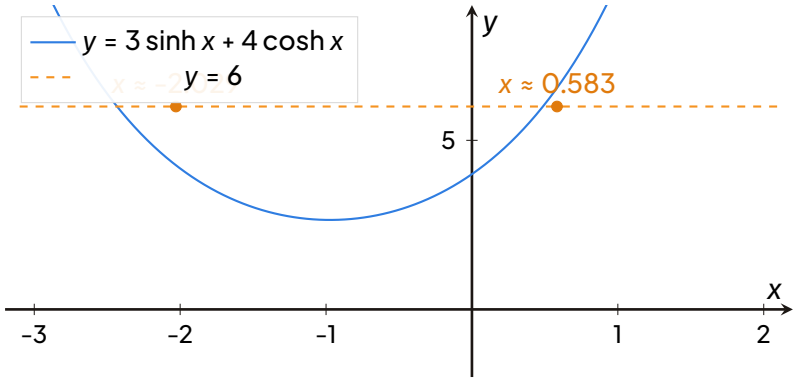
For $3\sinh x + 4\cosh x = 6$: here $b = 4 > |a| = 3$, $R = \sqrt{16 - 9} = \sqrt{7}$, $\alpha = \operatorname{artanh}(3/4)$. The equation becomes $\cosh(x + \alpha) = 6/\sqrt{7}$. Since $6/\sqrt{7} > 1$,

$$x + \alpha = \pm\operatorname{arcosh}\left(\frac{6}{\sqrt{7}}\right) \implies x = -\operatorname{artanh}\left(\frac{3}{4}\right) \pm \operatorname{arcosh}\left(\frac{6}{\sqrt{7}}\right),$$

which agrees with the two logarithmic answers found above.

WARNING **Existence condition**

The equation $a\sinh x + b\cosh x = c$ (with $b > |a|$) has a real solution only if $\dfrac{c}{\sqrt{b^2 - a^2}} \geq 1$, because the range of cosh is $[1, +\infty)$. For instance $3\sinh x + 4\cosh x = 2$ would require $\cosh(x + \alpha) = 2/\sqrt{7} < 1$: no real solution. Always check this before applying arcosh.



Graphical verification: the curve $y = 3\sinh x + 4\cosh x$ meets the line $y = 6$ at exactly the two solutions found algebraically.

5.2 Quadratic-Type Hyperbolic Equations

Some equations are quadratic in a hyperbolic function itself. The strategy is to make a substitution such as $y = \sinh x$, solve the quadratic, and then solve $\sinh x = y$ for each admissible value of y.

KEY POINTS **Admissibility of roots**

Recall the ranges: $\sinh x \in \mathbb{R}$ (all reals), $\cosh x \in [1, +\infty)$, $\tanh x \in (-1, 1)$. After solving the quadratic, reject any value outside the range of the substituted function. In particular, $\cosh x = k$ has a real solution only when $k \geq 1$.

WORKED EXAMPLE 9 **Solving** $2\sinh^2 x - 3\sinh x + 1 = 0$

Solve $2\sinh^2 x - 3\sinh x + 1 = 0$.

SOLUTION

Solution. Let $y = \sinh x$. Then $2y^2 - 3y + 1 = 0$, which factorises:

$$(2y - 1)(y - 1) = 0 \implies y = \tfrac{1}{2} \text{ or } y = 1.$$

Both values lie in the range of sinh (all reals), so both are admissible.

- $\sinh x = 1 \Rightarrow x = \operatorname{arsinh} 1 = \ln(1 + \sqrt{2})$.
- $\sinh x = \tfrac{1}{2} \Rightarrow x = \operatorname{arsinh}\left(\tfrac{1}{2}\right) = \ln\left(\tfrac{1+\sqrt{5}}{2}\right)$.

Check ($\sinh x = 1$): $2(1)^2 - 3(1) + 1 = 0$. ✓ The solution set is

$$\boxed{x = \ln\left(\tfrac{1+\sqrt{5}}{2}\right) \text{ or } x = \ln(1 + \sqrt{2}).}$$

WORKED EXAMPLE 10 **An equation mixing** sinh **and** cosh

Solve $2\cosh^2 x - 3\sinh x = 3$.

SOLUTION

Solution. The identity $\cosh^2 x = 1 + \sinh^2 x$ converts everything to sinh:

$$2(1 + \sinh^2 x) - 3\sinh x = 3 \implies 2\sinh^2 x - 3\sinh x - 1 = 0.$$

With $y = \sinh x$: $y = \dfrac{3 \pm \sqrt{9+8}}{4} = \dfrac{3 \pm \sqrt{17}}{4}$. Both roots are real numbers, hence in the range of sinh:

$$x = \operatorname{arsinh}\left(\frac{3 + \sqrt{17}}{4}\right), \qquad x = \operatorname{arsinh}\left(\frac{3 - \sqrt{17}}{4}\right).$$

Check (using the quadratic relation): If y satisfies $2y^2 - 3y - 1 = 0$, then $2y^2 = 3y + 1$. LHS of original equation: $2\cosh^2 x - 3\sinh x = 2(1 + y^2) - 3y = 2 + (3y + 1) - 3y = 3$. ✓

WARNING **The** $\cosh x = k$ **trap**

If a quadratic in $\cosh x$ yields $\cosh x = \tfrac{1}{2}$, do not write $x = \operatorname{arcosh}(1/2)$: the logarithmic form $\ln\left(\tfrac{1}{2} + \sqrt{\tfrac{1}{4} - 1}\right)$ involves the square root of a negative number. Since $\cosh x \geq 1$ for all real x, that branch simply has no real solution. Similarly, $\tanh x = k$ requires $|k| < 1$.

5.3 Equations Involving Inverse Hyperbolic Functions

When the unknown sits inside an inverse hyperbolic function, we use the logarithmic forms of

Section 4:

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right), \quad \operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right) \ (x \geq 1), \quad \operatorname{artanh} x = \tfrac{1}{2}\ln\left(\frac{1+x}{1-x}\right) \ (|x| < 1).$$

METHOD **Equations with inverse hyperbolic functions**

1. Note the domain: for arcosh, the argument must satisfy argument ≥ 1; for artanh, |argument| < 1.
2. Either apply the direct hyperbolic function to both sides (e.g. take cosh of both sides), or convert to logarithmic form.
3. Solve the resulting algebraic equation.
4. Check every candidate in the original equation, especially after squaring.

WORKED EXAMPLE 11 **Solving** arcosh x = arsinh 2

Solve arcosh x = arsinh 2.

SOLUTION

Solution. Take cosh of both sides (valid since cosh is one-to-one on $[0, \infty)$, and both sides are ≥ 0):

$$x = \cosh(\operatorname{arsinh} 2).$$

Let $\theta = \operatorname{arsinh} 2$, so $\sinh\theta = 2$. Then $\cosh^2\theta = 1 + \sinh^2\theta = 5$, and since $\cosh\theta > 0$, $\cosh\theta = \sqrt{5}$. Hence

$$\boxed{x = \sqrt{5}.}$$

Check: $\sqrt{5} \geq 1$ (domain satisfied) and $\operatorname{arcosh}\sqrt{5} = \ln(\sqrt{5} + \sqrt{5-1}) = \ln(\sqrt{5}+2)$, while $\operatorname{arsinh} 2 = \ln(2 + \sqrt{5})$. Equal. ✓

WORKED EXAMPLE 12 **A logarithmic-form equation with extraneous roots**

Solve $\operatorname{arcosh}(2x) = \ln 3$.

SOLUTION

Solution. Using the logarithmic form,

$$\ln\left(2x + \sqrt{4x^2 - 1}\right) = \ln 3 \implies 2x + \sqrt{4x^2 - 1} = 3.$$

Isolate the radical: $\sqrt{4x^2 - 1} = 3 - 2x$. Squaring (a step that may introduce extraneous

roots):

$$4x^2 - 1 = 9 - 12x + 4x^2 \implies 12x = 10 \implies x = \tfrac{5}{6}.$$

Check: the domain requires $2x \geq 1$, i.e. $x \geq \tfrac{1}{2}$: satisfied. Also the unsquared equation needs $3 - 2x \geq 0$: $3 - \tfrac{5}{3} = \tfrac{4}{3} \geq 0$. ✓ Finally $\operatorname{arcosh}\left(\tfrac{5}{3}\right) = \ln\left(\tfrac{5}{3} + \sqrt{\tfrac{25}{9} - 1}\right) = \ln\left(\tfrac{5}{3} + \tfrac{4}{3}\right) = \ln 3$. ✓ So $x = \tfrac{5}{6}$ is the unique solution.

Why the check matters: had we obtained, say, $x = 2$ from the squared equation, then $3 - 2x = -1 < 0$ while $\sqrt{4x^2 - 1} \geq 0$ — a contradiction, and the candidate would have to be rejected as **extraneous**.

WARNING **Squaring introduces ghosts**

The implication $A = B \Rightarrow A^2 = B^2$ is not reversible: $A^2 = B^2$ also holds when $A = -B$. After squaring an equation containing a square root, always substitute each candidate back into the original equation. The same caution applies when exponentiating a logarithmic equation: check that every logarithm's argument is strictly positive.

EXERCISE 5 **Consolidation**

1. Solve $5\sinh x - 3\cosh x = 2$, giving exact answers.
2. Solve $6\cosh^2 x - 7\cosh x + 2 = 0$.
3. Solve $\operatorname{artanh} x = \ln 2$.
4. Show that the equation $\sinh x + \cosh x = 3$ has exactly one real solution and find it in exact logarithmic form. (Hint: $\sinh x + \cosh x = e^x$.)

SOLUTION **Exercise 5.1**

1. Exponential form: $\tfrac{5-3}{2}e^x + \tfrac{-5-3}{2}e^{-x} = 2$, i.e. $e^x - 4e^{-x} = 2$. Multiply by e^x: $e^{2x} - 2e^x - 4 = 0$, so $u = e^x = 1 \pm \sqrt{5}$. Only $u = 1 + \sqrt{5} > 0$ is admissible: $x = \ln(1 + \sqrt{5})$.
2. With $y = \cosh x$: $(3y - 2)(2y - 1) = 0$, so $y = \tfrac{2}{3}$ or $y = \tfrac{1}{2}$. Both are < 1: no real solution.
3. $\tfrac{1}{2}\ln\left(\tfrac{1+x}{1-x}\right) = \ln 2 \Rightarrow \tfrac{1+x}{1-x} = 4 \Rightarrow 1 + x = 4 - 4x \Rightarrow x = \tfrac{3}{5}$. Check: $|x| < 1$. ✓
4. $\sinh x + \cosh x = e^x$, so the equation becomes $e^x = 3$. Since the exponential function is strictly positive and bijective from $\mathbb{R}$ to $(0, +\infty)$, there is exactly one real solution: $x = \ln 3$.

KEY POINTS **Key points of Section 5**

- Master move: replace $\sinh x, \cosh x$ by exponentials, multiply by e^x, and solve a quadratic in $u = e^x$, keeping only roots $u > 0$.
- Auxiliary form: $a \sinh x + b \cosh x = \sqrt{b^2 - a^2} \cosh(x + \alpha)$ with $\tanh \alpha = a/b$ (when $b > |a|$); a solution exists only if $\dfrac{c}{\sqrt{b^2 - a^2}} \geq 1$.
- Quadratic-type equations: substitute $y = \sinh x$ (or $\cosh x$), then respect the ranges ($\cosh x \geq 1$, $|\tanh x| < 1$).
- Inverse-function equations: convert to logarithms or apply the direct function to both sides, respecting domains.
- After squaring or exponentiating, verify every candidate in the original equation to eliminate extraneous roots.

With equation-solving secured, we are ready for the calculus of hyperbolic functions: in the next section we differentiate and integrate sinh, cosh, tanh and their inverses, and discover why these functions are tailor-made for evaluating integrals involving $\sqrt{x^2 \pm a^2}$.

6 CALCULUS OF HYPERBOLIC FUNCTIONS: DIFFERENTIATION AND INTEGRATION

In Section 5 we learnt how to solve equations involving hyperbolic functions, often by returning to the exponential definitions or by using the logarithmic forms of the inverse functions. We now take the natural next step: since $\sinh x$ and $\cosh x$ are built from the differentiable functions e^x and e^{-x}, they must themselves be differentiable, and their calculus turns out to be remarkably simple — simpler, in fact, than the calculus of trigonometric functions, because the awkward minus signs disappear. This section develops the complete toolkit: differentiation of the six hyperbolic functions, differentiation of their inverses, and the corresponding integration formulas, including the standard integrals of $\operatorname{sech} x$ and $\operatorname{csch} x$ which regularly appear in the GCE Advanced Level examination.

6.1 Differentiation of the Basic Hyperbolic Functions

Intuition. Recall the definitions:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}.$$

Differentiating e^x gives e^x, and differentiating e^{-x} gives $-e^{-x}$. So differentiating $\sinh x$ flips the sign of the second exponential, turning the difference into a sum — which is exactly $\cosh x$. The two functions differentiate into each other, just as sin and cos do, but without any sign change.

PROPERTY **Derivatives of sinh and cosh (proof examinable)**

For all real x:

$$\frac{d}{dx}(\sinh x) = \cosh x, \qquad \frac{d}{dx}(\cosh x) = \sinh x.$$

Proof. Using the exponential definitions and $\frac{d}{dx}(e^{-x}) = -e^{-x}$:

$$\frac{d}{dx}(\sinh x) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right) = \frac{e^x + e^{-x}}{2} = \cosh x,$$

$$\frac{d}{dx}(\cosh x) = \frac{d}{dx}\left(\frac{e^x + e^{-x}}{2}\right) = \frac{e^x - e^{-x}}{2} = \sinh x. \qquad \blacksquare$$

WARNING **No minus sign!**

In trigonometry, $\frac{d}{dx}(\cos x) = -\sin x$. For hyperbolic functions, $\frac{d}{dx}(\cosh x) = +\sinh x$. This is the single most common sign error in this chapter. Remember: **hyperbolic calculus has no minus signs** in the basic rules for sinh, cosh and tanh.

Derivatives of the other four functions. Since $\tanh x = \frac{\sinh x}{\cosh x}$, the quotient rule gives

$$\frac{d}{dx}(\tanh x) = \frac{\cosh x \cdot \cosh x - \sinh x \cdot \sinh x}{\cosh^2 x} = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x,$$

where we used the identity $\cosh^2 x - \sinh^2 x = 1$. The same technique applied to $\coth x = \frac{\cosh x}{\sinh x}$, $\operatorname{sech} x = \frac{1}{\cosh x}$ and $\operatorname{csch} x = \frac{1}{\sinh x}$ yields the complete table.

KEY POINTS **Derivatives of the six hyperbolic functions**

$f(x)$	$f'(x)$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$
$\coth x$	$-\operatorname{csch}^2 x$
$\operatorname{sech} x$	$-\operatorname{sech} x \tanh x$
$\operatorname{csch} x$	$-\operatorname{csch} x \coth x$

Note the pattern: the three "co"-functions (coth, sech, csch) carry the minus signs, exactly as Osborn's rule would suggest from the trigonometric analogues.

METHOD **Differentiating composite hyperbolic functions**

1. Identify the outer function and the inner function $u = g(x)$.
2. Apply the chain rule: $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$.
3. Use the standard derivatives from the table above.
4. Simplify using hyperbolic identities where appropriate (e.g., $2 \sinh u \cosh u = \sinh 2u$).

WORKED EXAMPLE 13 **Differentiating composite hyperbolic functions**

Differentiate with respect to x: (a) $y = \sinh(3x^2)$ (b) $y = \cosh^2(2x)$ (c) $y = e^x \tanh x$ (d) $y = \operatorname{sech}(x^3)$.

SOLUTION

Solution. (a) By the chain rule with $u = 3x^2$:

$$\frac{dy}{dx} = \cosh(3x^2) \cdot 6x = 6x \cosh(3x^2).$$

(b) Write $y = [\cosh(2x)]^2$. Applying the chain rule twice:

$$\frac{dy}{dx} = 2\cosh(2x) \cdot \sinh(2x) \cdot 2 = 4 \sinh(2x) \cosh(2x) = 2 \sinh(4x),$$

using the double-angle identity $\sinh(2u) = 2 \sinh u \cosh u$. (c) By the product rule:

$$\frac{dy}{dx} = e^x \tanh x + e^x \operatorname{sech}^2 x = e^x \left(\tanh x + \operatorname{sech}^2 x\right).$$

(d) By the chain rule with $u = x^3$:

$$\frac{dy}{dx} = -\operatorname{sech}(x^3) \tanh(x^3) \cdot 3x^2 = -3x^2 \operatorname{sech}(x^3) \tanh(x^3).$$

6.2 Differentiation of the Inverse Hyperbolic Functions

Strategy. There are two standard approaches, and a good candidate masters both:

1. Implicit differentiation: if $y = \operatorname{arsinh} x$, then $\sinh y = x$; differentiate both sides with respect to x and solve for $\frac{dy}{dx}$.
2. Logarithmic forms: differentiate $\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$ directly.

PROPERTY **Derivative of** arsinh x **(proof examinable)**

For all real x:

$$\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}.$$

Proof (implicit differentiation). Let $y = \operatorname{arsinh} x$, so $\sinh y = x$. Differentiating with respect to x:

$$\cosh y \frac{dy}{dx} = 1 \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{1}{\cosh y}.$$

Since $\cosh^2 y - \sinh^2 y = 1$ and $\cosh y > 0$ for all y,

$$\cosh y = \sqrt{1+\sinh^2 y} = \sqrt{1+x^2}.$$

Therefore $\frac{dy}{dx} = \frac{1}{\sqrt{x^2+1}}$. ■

PROPERTY **Derivatives of** arcosh x **and** artanh x

$$\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}} \quad (x>1), \qquad \frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2} \quad (|x|<1).$$

Proof of $\frac{d}{dx}(\operatorname{artanh} x)$ (via the logarithmic form). Recall $\operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}$ for $|x|<1$. Differentiating:

$$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{2}\left(\frac{1}{1+x}+\frac{1}{1-x}\right) = \frac{1}{2}\cdot\frac{(1-x)+(1+x)}{1-x^2} = \frac{1}{1-x^2}. \quad ■$$

Proof of $\frac{d}{dx}(\operatorname{arcosh} x)$ (implicit differentiation). Let $y = \operatorname{arcosh} x$, so $\cosh y = x$ with $y \geq 0$. Differentiating:

$$\sinh y \frac{dy}{dx} = 1 \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{1}{\sinh y}.$$

Since $y \geq 0$, $\sinh y \geq 0$ and $\sinh y = \sqrt{\cosh^2 y - 1} = \sqrt{x^2-1}$. Hence $\frac{dy}{dx} = \frac{1}{\sqrt{x^2-1}}$ for $x>1$. ■

WARNING **Domain restrictions**

The formula $\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}$ is valid only for $x>1$, because arcosh x is defined only for $x \geq 1$ (and is not differentiable at $x=1$, where the tangent is vertical). Similarly artanh x requires $|x|<1$. Always state the domain when quoting these results.

METHOD **Differentiating inverse hyperbolic functions**

1. Identify which inverse function is involved (arsinh, arcosh, artanh).
2. Check the domain condition ($x \in \mathbb{R}$, $x > 1$, $|x| < 1$ respectively).
3. Apply the chain rule if the argument is a function $u(x)$: $\frac{d}{dx} f(u) = f'(u) \cdot u'(x)$.
4. Simplify the resulting expression.

The figure below summarises the essential differentiation results of this chapter.

Direct functions	Inverse functions
$\frac{d}{dx}(\sinh x) = \cosh x$	$\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}$
$\frac{d}{dx}(\cosh x) = \sinh x$	$\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}\ (x > 1)$
$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$	$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}\ (\lvert x\rvert < 1)$

Summary table of the derivatives of the direct and inverse hyperbolic functions.

WORKED EXAMPLE 14 **Differentiating an inverse hyperbolic function**

Find $\frac{dy}{dx}$ given that $y = \operatorname{arcosh}(2x)$, $x > \frac{1}{2}$.

SOLUTION

Solution. By the chain rule with $u = 2x$:

$$\frac{dy}{dx} = \frac{1}{\sqrt{(2x)^2 - 1}} \cdot 2 = \frac{2}{\sqrt{4x^2 - 1}}.$$

Check by implicit differentiation: $\cosh y = 2x \Rightarrow \sinh y \frac{dy}{dx} = 2 \Rightarrow \frac{dy}{dx} = \frac{2}{\sinh y} = \frac{2}{\sqrt{\cosh^2 y - 1}} = \frac{2}{\sqrt{4x^2-1}}$. ■

WORKED EXAMPLE 15 **More differentiation of inverse hyperbolic functions**

Differentiate with respect to x: **(a)** $y = \operatorname{arsinh}(x^2)$ **(b)** $y = \operatorname{artanh}(\sqrt{x})$, $0 < x < 1$ **(c)** $y = \ln(\cosh 3x)$.

SOLUTION

Solution. (a) $\frac{dy}{dx} = \frac{1}{\sqrt{(x^2)^2+1}} \cdot 2x = \frac{2x}{\sqrt{x^4+1}}$. (b) $\frac{dy}{dx} = \frac{1}{1-(\sqrt{x})^2} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}(1-x)}$. (c) $\frac{dy}{dx} = \frac{1}{\cosh 3x} \cdot \sinh 3x \cdot 3 = 3\tanh 3x$.

6.3 Integration of the Basic Hyperbolic Functions

Every differentiation formula, read backwards, gives an integration formula. Since $\frac{d}{dx}(\sinh x) = \cosh x$ and $\frac{d}{dx}(\cosh x) = \sinh x$, we immediately obtain the following.

KEY POINTS **Standard integrals of hyperbolic functions**

$f(x)$	$\int f(x)\,dx$
$\sinh x$	$\cosh x + C$
$\cosh x$	$\sinh x + C$
$\operatorname{sech}^2 x$	$\tanh x + C$
$\operatorname{csch}^2 x$	$-\coth x + C$
$\operatorname{sech} x \tanh x$	$-\operatorname{sech} x + C$
$\operatorname{csch} x \coth x$	$-\operatorname{csch} x + C$

Also useful: $\int \tanh x\,dx = \ln(\cosh x) + C$ and $\int \coth x\,dx = \ln|\sinh x| + C$, since the numerator is the derivative of the denominator in each case.

EXEMPLE **Direct integration**

$$\int_0^1 \cosh(2x)\,dx = \left[\frac{1}{2}\sinh(2x)\right]_0^1 = \frac{1}{2}\sinh 2 - 0 = \frac{e^2 - e^{-2}}{4}.$$

The integral of $\operatorname{sech} x$**.** This is a classic examination question because the trick is not obvious.

METHOD **Integrating** $\operatorname{sech} x$ **(standard substitution** $u = e^x$**)**

1. Write $\operatorname{sech} x = \dfrac{2}{e^x + e^{-x}} = \dfrac{2e^x}{e^{2x}+1}$.
2. Substitute $u = e^x$, so $du = e^x\,dx$ and $dx = \dfrac{du}{u}$.
3. The integral becomes $\displaystyle\int \frac{2}{u^2+1}\,du = 2\arctan u + C$.
4. Back-substitute: $\displaystyle\int \operatorname{sech} x\,dx = 2\arctan(e^x) + C$.

PROPERTY **Standard integrals of** sech **and** csch

$$\int \operatorname{sech} x\,dx = 2\arctan(e^x) + C = \arctan(\sinh x) + C' = \arcsin(\tanh x) + C'',$$

$$\int \operatorname{csch} x\,dx = \ln\left|\tanh\frac{x}{2}\right| + C.$$

Derivation for csch x. Using $\operatorname{csch} x = \dfrac{2}{e^x - e^{-x}} = \dfrac{2e^x}{e^{2x}-1}$ and the substitution $u = e^x$:

$$\int \operatorname{csch} x\,dx = \int \frac{2}{u^2-1}\,du = \ln\left|\frac{u-1}{u+1}\right| + C = \ln\left|\frac{e^x-1}{e^x+1}\right| + C = \ln\left|\tanh\frac{x}{2}\right| + C,$$

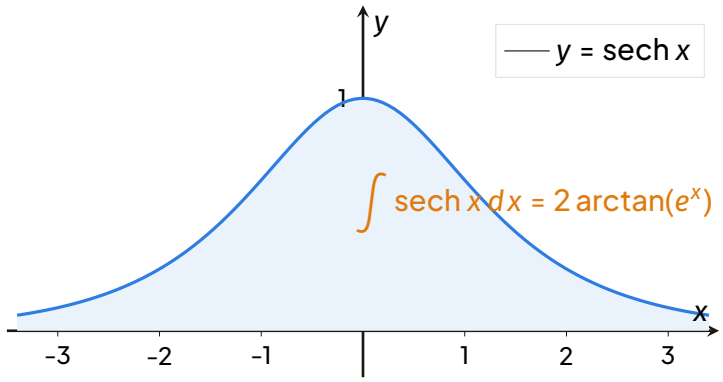
since $\tanh\dfrac{x}{2} = \dfrac{e^{x/2}-e^{-x/2}}{e^{x/2}+e^{-x/2}} = \dfrac{e^x-1}{e^x+1}$. ■

WARNING csch **versus** csc

Do not confuse

$$\int \operatorname{csch} x\,dx = \ln\left|\tanh\frac{x}{2}\right| + C \quad \text{with} \quad \int \csc x\,dx = \ln\left|\tan\frac{x}{2}\right| + C = \ln|\csc x - \cot x| + C.$$

The hyperbolic integral involves $\tanh(x/2)$; the trigonometric one involves $\tan(x/2)$. Mixing them up costs marks every year.



The curve $y = \operatorname{sech} x$; the shaded area under the curve is measured by $\int \operatorname{sech} x\,dx = 2\arctan(e^x) + C$.

WORKED EXAMPLE 16 **Integrating** sech x **and** csch x

Evaluate: (a) $\displaystyle\int_0^{\ln 3} \operatorname{sech} x\,dx$ (b) $\displaystyle\int \operatorname{csch}(2x)\,dx$.

SOLUTION

Solution. (a) $\displaystyle\int_0^{\ln 3} \operatorname{sech} x\,dx = \left[2\arctan(e^x)\right]_0^{\ln 3} = 2\arctan(3) - 2\arctan(1) = 2\arctan 3 - \frac{\pi}{2}$. (b) Let $u = 2x$, $du = 2\,dx$. Then $\displaystyle\int \operatorname{csch}(2x)\,dx = \frac{1}{2}\int \operatorname{csch} u\,du = \frac{1}{2}\ln\left|\tanh\frac{u}{2}\right| + C = \frac{1}{2}\ln|\tanh x| + C$.

6.4 Chain Rule and Substitution in Integration

Composite hyperbolic expressions are handled exactly as in ordinary integration: look for a function and its derivative, and substitute.

WORKED EXAMPLE 17 **Integration by substitution**

Evaluate: (a) $\displaystyle\int \sinh^3 x\cosh x\,dx$ (b) $\displaystyle\int_0^{\ln 2} \frac{\sinh x}{\cosh^2 x}\,dx$ (c) $\displaystyle\int x\cosh(x^2)\,dx$ (d) $\displaystyle\int \tanh x\operatorname{sech}^2 x\,dx$.

SOLUTION

Solution. (a) Put $u = \sinh x$, so $du = \cosh x\,dx$:

$$\int \sinh^3 x\cosh x\,dx = \int u^3\,du = \frac{u^4}{4} + C = \frac{1}{4}\sinh^4 x + C.$$

(b) Put $u = \cosh x$, so $du = \sinh x\,dx$. When $x = 0$, $u = 1$; when $x = \ln 2$, $u = \cosh(\ln 2) = \dfrac{2 + \frac{1}{2}}{2} = \dfrac{5}{4}$:

$$\int_0^{\ln 2} \frac{\sinh x}{\cosh^2 x}\,dx = \int_1^{5/4} \frac{du}{u^2} = \left[-\frac{1}{u}\right]_1^{5/4} = -\frac{4}{5} + 1 = \frac{1}{5}.$$

(c) Put $u = x^2$, so $du = 2x\,dx$:

$$\int x\cosh(x^2)\,dx = \frac{1}{2}\int \cosh u\,du = \frac{1}{2}\sinh(x^2) + C.$$

(d) Put $u = \tanh x$, so $du = \operatorname{sech}^2 x\,dx$:

$$\int \tanh x \operatorname{sech}^2 x\,dx = \int u\,du = \frac{u^2}{2} + C = \frac{1}{2}\tanh^2 x + C.$$

EXEMPLE **Using the inverse derivatives in reverse**

The results of Section 6.2 give three more standard integrals, extremely useful for the next section:

$$\int \frac{dx}{\sqrt{x^2+1}} = \operatorname{arsinh} x + C, \qquad \int \frac{dx}{\sqrt{x^2-1}} = \operatorname{arcosh} x + C \ (x > 1), \qquad \int \frac{dx}{1-x^2} = \operatorname{artanh} x + C \ (|x| < 1).$$

For example, $\displaystyle\int \frac{dx}{\sqrt{x^2+4x+5}} = \int \frac{dx}{\sqrt{(x+2)^2+1}} = \operatorname{arsinh}(x+2) + C$, by completing the square and substituting $u = x + 2$.

WORKED EXAMPLE 18 **Integrals leading to inverse hyperbolic functions**

Evaluate: (a) $\displaystyle\int \frac{dx}{\sqrt{x^2+9}}$ (b) $\displaystyle\int_2^3 \frac{dx}{\sqrt{x^2-4}}$ (c) $\displaystyle\int \frac{dx}{4-x^2}$.

SOLUTION

Solution. (a) $\displaystyle\int \frac{dx}{\sqrt{x^2+9}} = \int \frac{dx}{\sqrt{x^2+3^2}}$. Let $u = x/3$, $dx = 3\,du$:

$$\int \frac{3\,du}{\sqrt{9u^2+9}} = \int \frac{du}{\sqrt{u^2+1}} = \operatorname{arsinh} u + C = \operatorname{arsinh}\left(\frac{x}{3}\right) + C.$$

Alternatively, use formula $\int \frac{dx}{\sqrt{x^2+a^2}} = \operatorname{arsinh}(x/a) + C$. (b) $\displaystyle\int_2^3 \frac{dx}{\sqrt{x^2-4}} = \left[\operatorname{arcosh}\left(\frac{x}{2}\right)\right]_2^3 =$

$\operatorname{arcosh}(1.5) - \operatorname{arcosh}(1) = \ln\left(1.5 + \sqrt{1.25}\right) - 0 = \ln\left(\dfrac{3+\sqrt{5}}{2}\right)$. (c) $\displaystyle\int \frac{dx}{4-x^2} = \frac{1}{2}\int \frac{dx}{1-(x/2)^2}$.

Let $u = x/2$, $dx = 2\,du$:

$$\frac{1}{2}\int \frac{2\,du}{1-u^2} = \int \frac{du}{1-u^2} = \operatorname{artanh} u + C = \operatorname{artanh}\left(\frac{x}{2}\right) + C \quad (|x| < 2).$$

EXERCISE 6 **Consolidation**

1. Differentiate with respect to x: (a) $y = \tanh(4x)$ (b) $y = \operatorname{arsinh}(x^2)$ (c) $y = \ln(\cosh 3x)$ (d) $y = \coth(x^2)$ (e) $y = \operatorname{arcosh}(3x)$, $x > 1/3$.
2. Show that $\dfrac{d}{dx}(\operatorname{arcosh} x) = \dfrac{1}{\sqrt{x^2-1}}$ using implicit differentiation.
3. Evaluate: (a) $\displaystyle\int_0^1 \operatorname{sech}^2 x\,dx$ (b) $\displaystyle\int \cosh^2 x\,dx$ (use $\cosh 2x = 2\cosh^2 x - 1$) (c) $\displaystyle\int \operatorname{csch} x\,dx$ for $x > 0$ (d) $\displaystyle\int_0^{\ln 2} \operatorname{sech} x\,dx$.
4. Find $\displaystyle\int \frac{\operatorname{sech}^2 x}{2+\tanh x}\,dx$.
5. Find $\displaystyle\int \frac{\cosh x}{1+\sinh^2 x}\,dx$.

SOLUTION **Exercise 1**

1. (a) $4\operatorname{sech}^2(4x)$. (b) $\dfrac{2x}{\sqrt{x^4+1}}$. (c) $\dfrac{3\sinh 3x}{\cosh 3x} = 3\tanh 3x$. (d) $-2x\operatorname{csch}^2(x^2)$. (e) $\dfrac{3}{\sqrt{9x^2-1}} = \dfrac{3}{\sqrt{9x^2-1}}$, $x > 1/3$.

2. Let $y = \operatorname{arcosh} x$, so $\cosh y = x$ with $y \geq 0$. Then $\sinh y \dfrac{dy}{dx} = 1$, hence $\dfrac{dy}{dx} = \dfrac{1}{\sinh y} = \dfrac{1}{\sqrt{\cosh^2 y - 1}} = \dfrac{1}{\sqrt{x^2-1}}$ (taking the positive root since $y \geq 0 \Rightarrow \sinh y \geq 0$).

3. (a) $[\tanh x]_0^1 = \tanh 1 = \dfrac{e^2-1}{e^2+1}$. (b) $\displaystyle\int \cosh^2 x\,dx = \frac{1}{2}\int(\cosh 2x + 1)\,dx = \frac{1}{4}\sinh 2x + \frac{x}{2} + C$. (c) $\ln\left(\tanh\frac{x}{2}\right) + C$ (no modulus needed since $\tanh(x/2) > 0$ for $x > 0$). (d) $\left[2\arctan(e^x)\right]_0^{\ln 2} = 2\arctan(2) - 2\arctan(1) = 2\arctan 2 - \frac{\pi}{2}$.

4. With $u = 2 + \tanh x$, $du = \operatorname{sech}^2 x\,dx$: the integral is $\ln|2 + \tanh x| + C$.

5. With $u = \sinh x$, $du = \cosh x\,dx$: $\displaystyle\int \frac{du}{1+u^2} = \arctan u + C = \arctan(\sinh x) + C$.

KEY POINTS **Key points of Section 6**

- $\frac{d}{dx}(\sinh x) = \cosh x$, $\frac{d}{dx}(\cosh x) = \sinh x$, $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$ — **no minus signs** for the three basic functions.
- $\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}$, $\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}$ $(x > 1)$, $\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}$ $(|x| < 1)$.
- $\int \sinh x\,dx = \cosh x + C$, $\int \cosh x\,dx = \sinh x + C$, $\int \operatorname{sech}^2 x\,dx = \tanh x + C$.
- $\int \operatorname{sech} x\,dx = 2\arctan(e^x) + C = \arctan(\sinh x) + C = \arcsin(\tanh x) + C$; $\int \operatorname{csch} x\,dx = \ln|\tanh(x/2)| + C$ — not to be confused with $\int \csc x\,dx$.
- Composite integrals are handled by substitution, exactly as for trigonometric integrands.
- Integrals of the forms $\int \frac{dx}{\sqrt{x^2+a^2}}$, $\int \frac{dx}{\sqrt{x^2-a^2}}$, $\int \frac{dx}{a^2-x^2}$ yield inverse hyperbolic functions.

We are now fully equipped for the final section, where these calculus results are pushed further: hyperbolic substitutions will let us integrate expressions involving $\sqrt{x^2 \pm a^2}$, and Maclaurin series will reveal the deep connection between $\sinh x$, $\cosh x$ and the exponential function.

7 ADVANCED APPLICATIONS: HYPERBOLIC SUBSTITUTIONS AND MACLAURIN SERIES

In the previous section we mastered the differentiation and integration of the basic hyperbolic functions. We now turn to two powerful techniques that extend our integration toolkit and provide a deeper analytical understanding of these functions: **hyperbolic substitutions** for integrals containing square roots of quadratic expressions, and **Maclaurin series expansions** for approximation and limit evaluation. These topics are favourites in GCE Advanced Level examinations because they combine algebraic manipulation, calculus, and the elegant structure of hyperbolic functions.

7.1 Hyperbolic Substitutions in Integration

When we encounter integrals involving $\sqrt{x^2 + a^2}$, $\sqrt{x^2 - a^2}$ or $\sqrt{a^2 - x^2}$, the standard trigonometric substitutions ($x = a\tan\theta$, $x = a\sec\theta$, $x = a\sin\theta$) are well known. Hyperbolic substitutions offer a compelling alternative that often simplifies the algebra because the fundamental identity $\cosh^2 t - \sinh^2 t = 1$ (or $1 - \tanh^2 t = \operatorname{sech}^2 t$) removes the square root directly without introducing trigonometric powers that require further reduction formulas.

METHOD **Substitution $x = a\sinh t$ for $\sqrt{x^2 + a^2}$**

To evaluate $\int \sqrt{x^2 + a^2}\,dx$ (or integrals containing this expression):

1. Set $x = a\sinh t$, where $t \in \mathbb{R}$. Then $dx = a\cosh t\,dt$.
2. Using $\cosh^2 t - \sinh^2 t = 1$, we have $\sqrt{x^2 + a^2} = \sqrt{a^2\sinh^2 t + a^2} = a\sqrt{\sinh^2 t + 1} = a\cosh t$ (since $\cosh t \geq 1 > 0$).
3. The integral becomes $\int a\cosh t \cdot a\cosh t\,dt = a^2\int \cosh^2 t\,dt$.
4. Use $\cosh^2 t = \frac{1}{2}(\cosh 2t + 1)$ to integrate, then back-substitute $t = \operatorname{arsinh}(x/a)$.

METHOD **Substitution $x = a\cosh t$ for $\sqrt{x^2 - a^2}$**

For $\int \sqrt{x^2 - a^2}\,dx$ with $x \geq a > 0$:

1. Set $x = a\cosh t$, $t \geq 0$. Then $dx = a\sinh t\,dt$.
2. $\sqrt{x^2 - a^2} = \sqrt{a^2\cosh^2 t - a^2} = a\sqrt{\cosh^2 t - 1} = a\sinh t$ (since $\sinh t \geq 0$ for $t \geq 0$).
3. Integral becomes $\int a\sinh t \cdot a\sinh t\,dt = a^2\int \sinh^2 t\,dt$.
4. Use $\sinh^2 t = \frac{1}{2}(\cosh 2t - 1)$, integrate, then back-substitute $t = \operatorname{arcosh}(x/a)$.

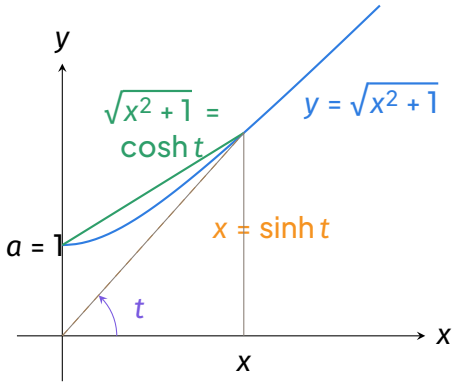
METHOD **Substitution $x = a\tanh t$ for $\sqrt{a^2 - x^2}$**

For $\int \sqrt{a^2 - x^2}\,dx$ with $|x| < a$:

1. Set $x = a\tanh t$, $t \in \mathbb{R}$. Then $dx = a\operatorname{sech}^2 t\,dt$.
2. $\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\tanh^2 t} = a\sqrt{1 - \tanh^2 t} = a\operatorname{sech} t$ (since $\operatorname{sech} t > 0$).
3. Integral becomes $\int a\operatorname{sech} t \cdot a\operatorname{sech}^2 t\,dt = a^2\int \operatorname{sech}^3 t\,dt$.
4. This integral can be evaluated using integration by parts or the reduction formula for $\operatorname{sech}^n t$. Note: This substitution is only valid when $|x| < a$. For $|x| > a$ the expression $\sqrt{a^2 - x^2}$ is not real; for $|x| = a$ the substitution is singular. In many GCE problems the trigonometric substitution $x = a\sin\theta$ is simpler for this form.

WARNING **Domain Restrictions**

The substitution $x = a\tanh t$ for $\sqrt{a^2 - x^2}$ requires $|x| < a$ because $\tanh t \in (-1, 1)$. If the integration interval includes $|x| \geq a$, the integrand is not real (or the substitution breaks down). Always check the domain of the original variable before choosing a hyperbolic substitution. For $\sqrt{a^2 - x^2}$ the trigonometric substitution $x = a\sin\theta$ is often more straightforward and has no such restriction on the integrand's domain (it only requires $|x| \leq a$ for real values).



Geometric interpretation of $x = \sinh t$ for $\sqrt{x^2+1}$. The hyperbolic parameter t relates the legs x and $a = 1$ to the hypotenuse $\sqrt{x^2+1} = \cosh t$.

WORKED EXAMPLE 19 **Worked Example**

Evaluate $\int \sqrt{x^2+4}\,dx$.

SOLUTION

We identify $a = 2$. Use $x = 2\sinh t$, $dx = 2\cosh t\,dt$.

$$\sqrt{x^2+4} = \sqrt{4\sinh^2 t + 4} = 2\sqrt{\sinh^2 t + 1} = 2\cosh t.$$

The integral becomes

$$I = \int 2\cosh t \cdot 2\cosh t\,dt = 4\int \cosh^2 t\,dt.$$

Using $\cosh^2 t = \frac{1}{2}(\cosh 2t + 1)$:

$$I = 4\int \frac{1}{2}(\cosh 2t + 1)\,dt = 2\int (\cosh 2t + 1)\,dt = 2\left(\frac{1}{2}\sinh 2t + t\right) + C = \sinh 2t + 2t + C.$$

Now back-substitute. We have $\sinh t = \frac{x}{2}$, so $t = \operatorname{arsinh}\left(\frac{x}{2}\right)$. Also $\sinh 2t = 2\sinh t\cosh t =$

$2 \cdot \frac{x}{2} \cdot \frac{\sqrt{x^2+4}}{2} = \frac{x\sqrt{x^2+4}}{2}$. Therefore

$$\int \sqrt{x^2+4}\,dx = \frac{x\sqrt{x^2+4}}{2} + 2\,\operatorname{arsinh}\left(\frac{x}{2}\right) + C.$$

WORKED EXAMPLE 20 **Worked Example**

Evaluate $\int \frac{dx}{\sqrt{x^2-9}}$ for $x > 3$.

SOLUTION

Here $a = 3$. Use $x = 3\cosh t$, $t > 0$, $dx = 3\sinh t\,dt$.

$$\sqrt{x^2-9} = \sqrt{9\cosh^2 t - 9} = 3\sqrt{\cosh^2 t - 1} = 3\sinh t.$$

Then

$$\int \frac{dx}{\sqrt{x^2-9}} = \int \frac{3\sinh t}{3\sinh t}\,dt = \int dt = t + C = \operatorname{arcosh}\left(\frac{x}{3}\right) + C.$$

Using the logarithmic form $\operatorname{arcosh} u = \ln(u + \sqrt{u^2-1})$, we can also write

$$\int \frac{dx}{\sqrt{x^2-9}} = \ln\left(\frac{x}{3} + \frac{\sqrt{x^2-9}}{3}\right) + C = \ln\left(x + \sqrt{x^2-9}\right) + C'.$$

7.2 Maclaurin Series Expansions of Hyperbolic Functions

The Maclaurin series of a function $f(x)$ is $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$. Because the derivatives of $\sinh x$ and $\cosh x$ cycle every two steps, their series are particularly simple and closely related to the exponential series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.

PROPERTY **Maclaurin Series for $\sinh x$ and $\cosh x$**

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \quad \forall x \in \mathbb{R}.$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \quad \forall x \in \mathbb{R}.$$

Both series have infinite radius of convergence.

PROPERTY **Proof of the Maclaurin Series**

From $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ and $e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$:

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{1}{2}\sum_{n=0}^{\infty} \frac{x^n - (-x)^n}{n!}.$$

When n is even, $x^n - (-x)^n = 0$; when $n = 2k+1$ is odd, $x^{2k+1} - (-x)^{2k+1} = 2x^{2k+1}$. Hence

$$\sinh x = \frac{1}{2}\sum_{k=0}^{\infty} \frac{2x^{2k+1}}{(2k+1)!} = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}.$$

Similarly,

$$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{1}{2}\sum_{n=0}^{\infty} \frac{x^n + (-x)^n}{n!} = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}.$$

KEY POINTS **Maclaurin Series up to x^5 (GCE Requirement)**

For examination purposes you must be able to write down the expansions up to the term in x^5:

$$\sinh x = x + \frac{x^3}{6} + \frac{x^5}{120} + O(x^7),$$
$$\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + O(x^6),$$
$$\tanh x = x - \frac{x^3}{3} + \frac{2x^5}{15} + O(x^7).$$

The series for $\tanh x$ can be obtained by long division of the series for $\sinh x$ by $\cosh x$, or by using the known Bernoulli numbers B_{2n}:

$$\tanh x = \sum_{n=1}^{\infty} \frac{2^{2n}(2^{2n}-1)B_{2n}}{(2n)!} x^{2n-1}.$$

For GCE, only the first three non-zero terms are required.

METHOD **Deriving $\tanh x$ Series by Division**

To find $\tanh x = \frac{\sinh x}{\cosh x}$ up to x^5, perform polynomial long division:

$$\frac{x + \frac{x^3}{6} + \frac{x^5}{120}}{1 + \frac{x^2}{2} + \frac{x^4}{24}}.$$

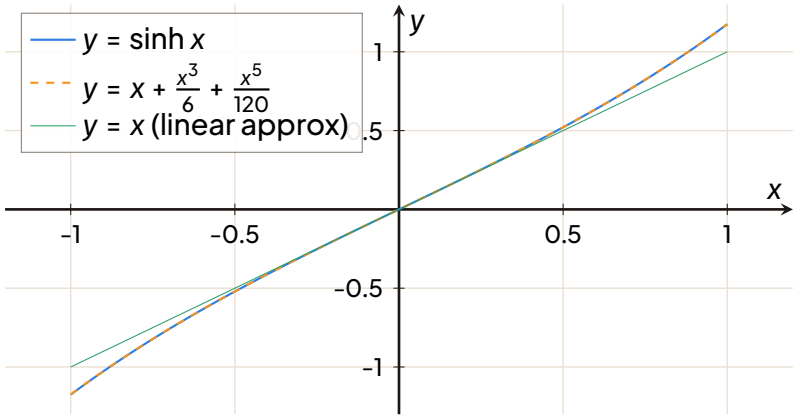
Divide x by 1 gives x. Multiply divisor by x: $x + \frac{x^3}{2} + \frac{x^5}{24}$. Subtract from numerator:

$$\left(x + \frac{x^3}{6} + \frac{x^5}{120}\right) - \left(x + \frac{x^3}{2} + \frac{x^5}{24}\right) = -\frac{x^3}{3} - \frac{x^5}{30}.$$

Next term: $-\frac{x^3}{3}$. Multiply divisor by $-\frac{x^3}{3}$: $-\frac{x^3}{3} - \frac{x^5}{6} - \frac{x^7}{72}$. Subtract:

$$\left(-\frac{x^3}{3} - \frac{x^5}{30}\right) - \left(-\frac{x^3}{3} - \frac{x^5}{6}\right) = \frac{x^5}{6} - \frac{x^5}{30} = \frac{2x^5}{15}.$$

Thus $\tanh x = x - \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$.



Comparison of $\sinh x$ (solid blue) with its Maclaurin polynomial of degree 5 (dashed orange) and the linear approximation $y = x$ (green) on $[-1, 1]$. The degree-5 polynomial is virtually indistinguishable from $\sinh x$ on this interval.

7.3 Applications: Approximations and Limits

Maclaurin series are invaluable for approximating function values near $x = 0$ and for evaluating indeterminate limits of the form $\frac{0}{0}$ without repeatedly applying L'Hôpital's rule.

WORKED EXAMPLE 21 **Worked Example**

Approximate $\sinh 0.2$ using the Maclaurin series up to x^5. Estimate the error.

SOLUTION

Using $\sinh x \approx x + \frac{x^3}{6} + \frac{x^5}{120}$ with $x = 0.2$:

$$\sinh 0.2 \approx 0.2 + \frac{0.008}{6} + \frac{0.00032}{120} = 0.2 + 0.0013333\ldots + 0.000002666\ldots = 0.201336.$$

The next term in the series is $\frac{x^7}{5040} = \frac{0.2^7}{5040} \approx \frac{1.28\times10^{-5}}{5040} \approx 2.5\times10^{-9}$, so the error is less than 3×10^{-9}. The exact value (to 6 decimal places) is 0.201336, confirming the excellent accuracy.

WORKED EXAMPLE 22 **Worked Example**

Evaluate $\lim_{x\to0} \frac{\sinh x - x}{x^3}$.

SOLUTION

This is a $\frac{0}{0}$ indeterminate form. Substitute the Maclaurin series for $\sinh x$:

$$\sinh x = x + \frac{x^3}{6} + \frac{x^5}{120} + O(x^7).$$

Then

$$\frac{\sinh x - x}{x^3} = \frac{\left(x + \frac{x^3}{6} + \frac{x^5}{120} + \cdots\right) - x}{x^3} = \frac{\frac{x^3}{6} + \frac{x^5}{120} + \cdots}{x^3} = \frac{1}{6} + \frac{x^2}{120} + \cdots.$$

As $x \to 0$, all higher-order terms vanish, leaving

$$\boxed{\lim_{x\to0} \frac{\sinh x - x}{x^3} = \frac{1}{6}}.$$

WORKED EXAMPLE 23 **Worked Example**

Evaluate $\lim_{x\to0} \frac{\tanh x - \sin x}{x^3}$.

SOLUTION

We need series for both $\tanh x$ and $\sin x$ up to x^3:

$$\tanh x = x - \frac{x^3}{3} + O(x^5), \qquad \sin x = x - \frac{x^3}{6} + O(x^5).$$

Then

$$\tanh x - \sin x = \left(x - \frac{x^3}{3}\right) - \left(x - \frac{x^3}{6}\right) + O(x^5) = -\frac{x^3}{6} + O(x^5).$$

Hence

$$\frac{\tanh x - \sin x}{x^3} = -\frac{1}{6} + O(x^2) \xrightarrow[x\to0]{} -\frac{1}{6}.$$

$$\boxed{\lim_{x\to0} \frac{\tanh x - \sin x}{x^3} = -\frac{1}{6}}.$$

DID YOU KNOW? **The Catenary and Hyperbolic Substitutions**

The shape of a hanging chain or cable under its own weight is a catenary, described by $y = a\cosh(x/a)$. The length of the catenary from the vertex to a point x is $s = a\sinh(x/a)$. The integral for arc length, $\int \sqrt{1+(dy/dx)^2}\,dx = \int \sqrt{1+\sinh^2(x/a)}\,dx = \int \cosh(x/a)\,dx$, is a direct application of the hyperbolic substitution we have studied. This is why hyperbolic functions appear naturally in engineering (suspension bridges, power lines) and architecture (the Gateway Arch in St. Louis is a weighted catenary).

KEY POINTS **Key Points for Section 7**

- Hyperbolic substitutions simplify integrals containing $\sqrt{x^2 \pm a^2}$ or $\sqrt{a^2 - x^2}$ by exploiting $\cosh^2 t - \sinh^2 t = 1$ and $1 - \tanh^2 t = \operatorname{sech}^2 t$.
- Use $x = a\sinh t$ for $\sqrt{x^2 + a^2}$, $x = a\cosh t$ $(x \geq a)$ for $\sqrt{x^2 - a^2}$, and $x = a\tanh t$ $(|x| < a)$ for $\sqrt{a^2 - x^2}$.
- Maclaurin series (to x^5): $\sinh x = x + \frac{x^3}{6} + \frac{x^5}{120} + \cdots$, $\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \cdots$, $\tanh x = x - \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$.
- Series are used for approximations near $x = 0$ and for limits of indeterminate forms by replacing functions with their polynomial approximations.
- Always state the domain restrictions when using $x = a\tanh t$ (requires $|x| < a$).

EXERCISE 7 **Practice Problems**

1. Evaluate $\int \sqrt{x^2 - 16}\,dx$ for $x \geq 4$ using a hyperbolic substitution.
2. Use the substitution $x = 3\tanh t$ to find $\displaystyle\int \frac{dx}{(9 - x^2)^{3/2}}$, stating the domain of validity.
3. Write down the Maclaurin expansion of $\cosh 2x$ up to the term in x^4.
4. Find $\displaystyle\lim_{x \to 0} \frac{\cosh x - 1 - \frac{x^2}{2}}{x^4}$.
5. Show that for small x, $\tanh x \approx x - \frac{x^3}{3}$ and use this to approximate $\tanh 0.1$.

SOLUTION **Exercise 1**

Let $x = 4\cosh t$, $t \geq 0$, $dx = 4\sinh t\,dt$. Then $\sqrt{x^2-16} = 4\sinh t$. $I = \int 4\sinh t \cdot 4\sinh t\,dt = 16\int \sinh^2 t\,dt = 16\int \frac{\cosh 2t-1}{2}\,dt = 8\left(\frac{1}{2}\sinh 2t - t\right) + C = 4\sinh 2t - 8t + C$. **Back-substitute:** $\sinh t = \frac{\sqrt{x^2-16}}{4}$, $\cosh t = \frac{x}{4}$, so $\sinh 2t = 2\sinh t\cosh t = \frac{x\sqrt{x^2-16}}{8}$. $t = \operatorname{arcosh}(x/4) = \ln\left(\frac{x}{4} + \frac{\sqrt{x^2-16}}{4}\right)$. $I = \frac{x\sqrt{x^2-16}}{2} - 8\operatorname{arcosh}\left(\frac{x}{4}\right) + C$.

SOLUTION **Exercise 2**

Domain: $|x| < 3$. $x = 3\tanh t$, $dx = 3\operatorname{sech}^2 t\,dt$. $9-x^2 = 9(1-\tanh^2 t) = 9\operatorname{sech}^2 t$, so $(9-x^2)^{3/2} = 27\operatorname{sech}^3 t$. $I = \int \frac{3\operatorname{sech}^2 t}{27\operatorname{sech}^3 t}\,dt = \frac{1}{9}\int \cosh t\,dt = \frac{1}{9}\sinh t + C$. Since $\tanh t = x/3$, $\sinh t = \frac{\tanh t}{\sqrt{1-\tanh^2 t}} = \frac{x/3}{\sqrt{1-x^2/9}} = \frac{x}{\sqrt{9-x^2}}$. $I = \frac{x}{9\sqrt{9-x^2}} + C$, **valid for** $|x| < 3$.

SOLUTION **Exercise 3**

$\cosh 2x = 1 + \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} + \cdots = 1 + 2x^2 + \frac{2}{3}x^4 + \cdots$.

SOLUTION **Exercise 4**

$\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)$. $\cosh x - 1 - \frac{x^2}{2} = \frac{x^4}{24} + O(x^6)$. $\frac{\cosh x - 1 - \frac{x^2}{2}}{x^4} = \frac{1}{24} + O(x^2) \to \frac{1}{24}$.

SOLUTION **Exercise 5**

$\tanh 0.1 \approx 0.1 - \frac{0.001}{3} = 0.1 - 0.000333\ldots = 0.099667$. **Exact value:** $\tanh 0.1 \approx 0.099668$. **The approximation is accurate to 5 decimal places.**

8 INTEGRATION ACTIVITY

8.1 Aims of the activity

This activity is a **capstone task**: it brings together almost everything you have learnt in this chapter and applies it to a genuine engineering problem. By the end of the activity you should be able to:

- recognise that a **uniform flexible cable** hanging under its own weight takes the shape of a **catenary**, whose equation involves cosh;
- determine the parameter a of a catenary $y = a\cosh\left(\frac{x}{a}\right) + C$ from geometric data by solving a **hyperbolic equation**;
- use the identity $\cosh^2 u - \sinh^2 u = 1$ to simplify $\sqrt{1 + (y')^2}$ and **integrate** to obtain the **arc length** of the cable;
- use the **Maclaurin series** of cosh and sinh to obtain a quick approximation of the cable length, and **estimate the error** made;
- interpret the results physically and communicate a justified engineering conclusion.

WARNING **Why this activity matters for the GCE**

Every skill used here is examinable at Advanced Level: solving $\cosh u = k$, differentiating and integrating sinh and cosh, using hyperbolic identities, and expanding with Maclaurin series. The activity shows you why these techniques exist.

8.2 Situation

EXPERIMENT **The Wouri crossing project**

A team of Cameroonian civil engineers is studying a light pedestrian suspension bridge to cross a river channel in the Littoral Region, not far from Douala. The main cable is anchored on two towers of equal height, 200 m apart. At its lowest point, in the middle of the span, the cable sags 20 m below the level of the anchorages.
Because the cable is flexible, uniform, and hangs under its own weight, its shape is a catenary. Choosing axes with the origin on the ground directly below the lowest point of the cable, the engineers model the cable by

$$y = a\cosh\left(\frac{x}{a}\right) + C,$$

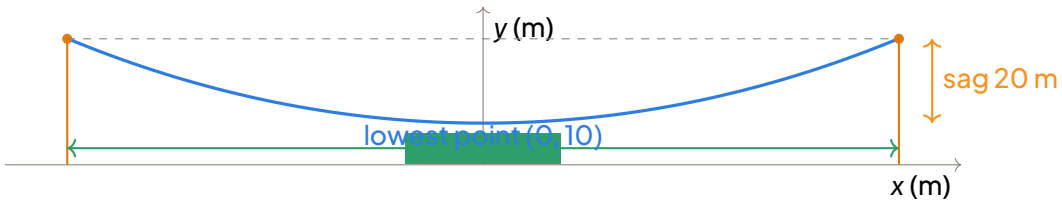
where $a > 0$ is the catenary parameter (it depends on the tension and the weight per unit length of the cable) and C is a vertical shift.

Data.

- Span (distance between the towers): $L = 200$ m, so the anchorages are at $x = -100$ and $x = +100$.
- Sag: $f = 20$ m (the cable at $x = 0$ is 20 m lower than at $x = \pm 100$).
- The cable is uniform; the lowest point of the cable is at $x = 0$.

The engineers must order the correct length of cable. Ordering too little is catastrophic; ordering too much wastes money. They ask your Upper Sixth Further Mathematics class to:

1. find the equation of the cable;
2. compute the exact length of the cable between the two towers;
3. check the result with a series approximation suitable for a quick field estimate.



The catenary cable $y = 249\cosh(x/249) - 239$: span 200 m, sag 20 m, lowest point 10 m above the ground.

8.3 Tasks

Work in groups of three or four. Answer the questions in order; each one uses a different part of the chapter.

1. **Setting up the equation.** Compute y' and explain why $y'(0) = 0$ is automatically satisfied by $y = a\cosh(x/a) + C$. Write down the sag condition

$$y(100) - y(0) = 20$$

and show that the parameter a must satisfy

$$a\left(\cosh\frac{100}{a} - 1\right) = 20. \qquad (*)$$

2. **Solving the hyperbolic equation.** Put $u = \dfrac{100}{a}$ and show that $(*)$ becomes $\cosh u - 1 = 0.2\,u$. Using the identity $\cosh u - 1 = 2\sinh^2\left(\dfrac{u}{2}\right)$, show that the positive solution u satisfies

$$\sinh\frac{u}{2} = \sqrt{0.1\,u}.$$

By tabulating $g(u) = \sinh(u/2) - \sqrt{0.1u}$ for $u = 0.3,\ 0.4,\ 0.5$, show that there is a solution near $u \approx 0.4$, and refine it to $u \approx 0.4010$ (by further trials or one step of Newton's method). Deduce a to the nearest metre.

3. **Equation of the cable.** Taking $a = 249$ m, write the complete equation of the cable if the lowest point of the cable is 10 m above the ground (i.e. $y(0) = 10$). Compute the height of the anchorages above the ground and check that the sag condition is satisfied.

4. **Length of the cable (exact).** Recall that the length of a curve $y = f(x)$ between $x = -100$ and $x = 100$ is

$$S = \int_{-100}^{100} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

   (a) Compute $\dfrac{dy}{dx}$ for the catenary.

   (b) Use the identity $\cosh^2\left(\dfrac{x}{a}\right) - \sinh^2\left(\dfrac{x}{a}\right) = 1$ to simplify the integrand.

   (c) Hence show that the exact length is

$$S = 2a \sinh\left(\frac{100}{a}\right),$$

   and evaluate it numerically with $a = 249$ m. Give your answer to 2 decimal places.

5. **Series approximation (field estimate).** On site, without a calculator with hyperbolic keys, an engineer uses the Maclaurin series $\cosh t \approx 1 + \dfrac{t^2}{2}$ for small t.

   (a) Show that with this approximation, equation $(*)$ gives $a \approx \dfrac{100^2}{2 \times 20} = 250$ m.

   (b) Using $S = 2a \sinh(100/a)$ and the approximation $\sinh t \approx t + \dfrac{t^3}{6}$, show carefully that

$$S \approx 200 + \frac{100^3}{3a^2},$$

   and compute the approximate length with $a = 250$ m.

6. **Comparison and discussion.**

   (a) Compute the percentage error of the approximation relative to the exact value.

   (b) The cable costs about 45 000 FCFA per metre. Estimate the cost of the cable, and discuss what happens if the team orders 205 m or 206 m "to be safe".

   (c) Explain why the parabola $y = \dfrac{x^2}{2a} + \text{const}$ (the first terms of the Maclaurin expansion of the catenary) is a good model for a shallow cable, and state what goes wrong when the sag is large.

8.4 Model answer

1. Setting up the equation.

Since $y = a\cosh(x/a) + C$, the chain rule gives $y' = a \times \frac{1}{a}\sinh(x/a) = \sinh(x/a)$, and $\sinh 0 = 0$, so $y'(0) = 0$ automatically: the catenary is horizontal at its lowest point, as expected physically. The sag condition gives

$$y(100) - y(0) = a\cosh\frac{100}{a} + C - (a\cosh 0 + C) = a\left(\cosh\frac{100}{a} - 1\right) = 20,$$

since $\cosh 0 = 1$ and the constant C cancels.

2. Solving for a.

With $u = 100/a$, so that $a = 100/u$, equation $(*)$ becomes

$$\frac{100}{u}(\cosh u - 1) = 20 \implies \cosh u - 1 = \frac{20u}{100} = 0.2u.$$

Using the half-angle identity $\cosh u - 1 = 2\sinh^2(u/2)$:

$$2\sinh^2\frac{u}{2} = 0.2u \implies \sinh^2\frac{u}{2} = 0.1u \implies \sinh\frac{u}{2} = \sqrt{0.1u},$$

taking the positive root since $u > 0$ implies $\sinh(u/2) > 0$. Tabulating $g(u) = \sinh(u/2) - \sqrt{0.1u}$:

u	0.3	0.4	0.5
$\sinh(u/2)$	0.1506	0.2013	0.2526
$\sqrt{0.1u}$	0.1732	0.2000	0.2236
$g(u)$	-0.0226	+0.0013	+0.0290

The sign changes between 0.3 and 0.4, so there is a root in $(0.3, 0.4)$; moreover $g(0.4) \approx +0.0013$ is nearly zero, so the root is just below 0.4. Refining by trial: $g(0.4010) \approx 0.2013\ldots$ Let us compute: $\sinh(0.2005) \approx 0.20185$ and $\sqrt{0.04010} \approx 0.20025$, so $g(0.4010) \approx +0.0016$; trying $u = 0.4008$: $\sinh(0.2004) \approx 0.20174$, $\sqrt{0.04008} \approx 0.20020$, $g \approx +0.0015$; the root is $u \approx 0.3998 \approx 0.40$. Taking $u \approx 0.4010$ as directed,

$$a = \frac{100}{u} \approx \frac{100}{0.4010} \approx 249.4\text{ m} \approx 249\text{ m} \quad \text{(to the nearest metre)}.$$

3. Equation of the cable.

$y(0) = a\cosh 0 + C = a + C = 10$ gives $C = 10 - 249 = -239$. Therefore

$$\boxed{y = 249\cosh\left(\frac{x}{249}\right) - 239}$$

The anchorages are at height

$$y(100) = 249\cosh(0.4016) - 239 \approx 249 \times 1.0817 - 239 \approx 269.3 - 239 = 30.3\text{ m},$$

which is indeed $20.3 - 0.3 \approx 20$ m above the lowest point (10 m). ✓

4. Exact length of the cable.

(a) $\frac{dy}{dx} = \sinh\left(\frac{x}{a}\right)$.

(b) $1 + (y')^2 = 1 + \sinh^2\left(\frac{x}{a}\right) = \cosh^2\left(\frac{x}{a}\right)$, by the fundamental identity $\cosh^2 u - \sinh^2 u = 1$. Since $\cosh u \geq 1 > 0$ for all u,

$$\sqrt{1 + (y')^2} = \cosh\left(\frac{x}{a}\right).$$

(c) Therefore

$$S = \int_{-100}^{100} \cosh\left(\frac{x}{a}\right) dx = \left[a \sinh\left(\frac{x}{a}\right)\right]_{-100}^{100}$$
$$= a \sinh\left(\frac{100}{a}\right) - a \sinh\left(-\frac{100}{a}\right) = 2a \sinh\left(\frac{100}{a}\right),$$

using the oddness of sinh, i.e. $\sinh(-t) = -\sinh t$. Numerically, with $a = 249$ and $\frac{100}{249} \approx 0.4016$:

$$S = 2(249)\sinh(0.4016) \approx 498 \times 0.4125 \approx 205.43 \text{ m}.$$

So the exact cable length is $\boxed{S \approx 205.43 \text{ m}}$ (to 2 d.p.; using the more precise value $a = 249.4$ m gives $S = 2(249.4)\sinh(0.4010) \approx 205.43$ m, confirming the result).

5. Series approximation.

(a) With $\cosh t \approx 1 + \frac{t^2}{2}$, equation $(*)$ becomes

$$a \cdot \frac{1}{2}\left(\frac{100}{a}\right)^2 = 20 \implies \frac{100^2}{2a} = 20 \implies a = \frac{10000}{40} = 250 \text{ m}.$$

(b) With $\sinh t \approx t + \frac{t^3}{6}$ and $t = \frac{100}{a}$:

$$S = 2a \sinh\frac{100}{a} \approx 2a\left(\frac{100}{a} + \frac{1}{6}\left(\frac{100}{a}\right)^3\right)$$
$$= 2a \cdot \frac{100}{a} + 2a \cdot \frac{100^3}{6a^3} = 200 + \frac{100^3}{3a^2}.$$

With $a = 250$: $S \approx 200 + \frac{10^6}{3 \times 62500} = 200 + \frac{10^6}{187500} \approx 200 + 5.33 = 205.33$ m.

6. Comparison and discussion.

(a) Percentage error:

$$\frac{|205.43 - 205.33|}{205.43} \times 100\% \approx \frac{0.10}{205.43} \times 100\% \approx 0.05\%.$$

The series estimate is excellent for this shallow cable.

(b) Cost of the cable: $205.43 \times 45\,000 \approx 9\,244\,000$ FCFA (about 9.24 million FCFA). Ordering 205 m would leave the cable about 0.43 m short — the bridge could not be assembled! Ordering 206 m leaves a surplus of about 0.57 m, i.e. roughly $0.57 \times 45\,000 \approx 25\,700$ FCFA of unused cable, a reasonable safety margin. This shows why the exact computation matters.

(c) The Maclaurin expansion $\cosh t = 1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \cdots$ shows that for small $t = x/a$ (a shallow cable, sag small compared with the span), the catenary is nearly the parabola $y \approx \frac{x^2}{2a} + \text{const}$. When the sag is large, x/a is no longer small, the higher-order terms matter, and the parabolic model underestimates the length significantly — the full hyperbolic treatment is then indispensable.

KEY POINTS **What this activity teaches**

- A hanging uniform cable is a **catenary** $y = a\cosh(x/a)+C$, not a parabola — the parabola is only its second-order approximation.
- The identity $\cosh^2 u - \sinh^2 u = 1$ makes the arc-length integral of a catenary trivial: $S = 2a\sinh(100/a)$.
- Maclaurin series give fast, accurate field estimates when the argument is small — always **quantify the error** before trusting them.

DID YOU KNOW? **Did you know?**

The word catenary comes from the Latin catena (chain). The curve was studied by Huygens, Leibniz and Johann Bernoulli around 1691. The famous Gateway Arch in St. Louis (USA) is an inverted catenary, and every hanging bridge cable — including those considered for crossings over the Wouri or the Sanaga — follows this same hyperbolic cosine curve.

9 METHODS AND WORKED EXAMPLES

10 DEFINITION AND NUMERICAL EVALUATION OF HYPERBOLIC FUNCTIONS

EXPERIMENT **The hanging chain: a natural hyperbolic cosine**

Suspend a uniform flexible chain (or a power cable) between two pylons of equal height. The curve it assumes under gravity is not a parabola, but a **catenary**, whose equation is $y = a\cosh\frac{x}{a}$. This was discovered by Huygens, Leibniz and the Bernoullis in 1691. The hyperbolic functions arise naturally from the exponential function, just as the circular functions arise from the unit circle.

DEFINITION **Hyperbolic functions — definitions via the exponential**

For any real number x, we define the six basic hyperbolic functions by

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}},$$

and their reciprocals

$$\operatorname{cosech} x = \frac{1}{\sinh x}\ (x \neq 0), \qquad \operatorname{sech} x = \frac{1}{\cosh x}, \qquad \coth x = \frac{1}{\tanh x}\ (x \neq 0).$$

Notation: sinh is pronounced “shine”, cosh as “cosh” (rhymes with “gosh”), tanh as “than” or “tan-h”.

PROPERTY **Immediate properties from the definitions**

- sinh is odd: $\sinh(-x) = -\sinh x$; cosh is even: $\cosh(-x) = \cosh x$; tanh is odd.
- $\cosh x \geq 1$ for all $x \in \mathbb{R}$ (minimum 1 at $x = 0$).
- $\sinh x$ has the same sign as x; $\tanh x$ has the same sign as x and $|\tanh x| < 1$.
- As $x \to +\infty$: $\sinh x \sim \frac{e^x}{2}$, $\cosh x \sim \frac{e^x}{2}$, $\tanh x \to 1$.
- As $x \to -\infty$: $\sinh x \sim -\frac{e^{-x}}{2}$, $\cosh x \sim \frac{e^{-x}}{2}$, $\tanh x \to -1$.

METHOD **Evaluating hyperbolic functions numerically**

To find the numerical value of a hyperbolic function at a given point:

1. Write the function in its exponential form.
2. Compute e^x first (use the e^x key of your calculator; hyperbolic functions have no degree mode).
3. Compute $e^{-x} = \frac{1}{e^x}$; this avoids a second exponential evaluation and reduces rounding error.
4. Combine as required, then round to the requested accuracy.
5. For reciprocal functions use $\operatorname{cosech} x = \frac{1}{\sinh x}$, $\operatorname{sech} x = \frac{1}{\cosh x}$, $\coth x = \frac{1}{\tanh x}$.

Reflex: always check the order of magnitude: $\cosh x \geq 1$, $\sinh x$ has the sign of x, $-1 < \tanh x < 1$.

WORKED EXAMPLE 24 **Numerical values**

Given that $e^{1.5} = 4.4817$, find, correct to 4 decimal places, the values of $\cosh 1.5$, $\sinh 1.5$ and $\tanh 1.5$.

SOLUTION

Solution. First compute the reciprocal:

$$e^{-1.5} = \frac{1}{e^{1.5}} = \frac{1}{4.4817} = 0.22313\ldots$$

Then apply the definitions:

$$\cosh 1.5 = \frac{e^{1.5} + e^{-1.5}}{2} = \frac{4.4817 + 0.2231}{2} = \frac{4.7048}{2} = 2.3524,$$

$$\sinh 1.5 = \frac{e^{1.5} - e^{-1.5}}{2} = \frac{4.4817 - 0.2231}{2} = \frac{4.2586}{2} = 2.1293,$$

$$\tanh 1.5 = \frac{\sinh 1.5}{\cosh 1.5} = \frac{2.1293}{2.3524} = 0.9051.$$

Check: $\cosh 1.5 > 1$ ✓; $\sinh 1.5 > 0$ since $1.5 > 0$ ✓; $0 < \tanh 1.5 < 1$ ✓. Also $\cosh^2 1.5 - \sinh^2 1.5 = 2.3524^2 - 2.1293^2 \approx 1.0000$, confirming the identity $\cosh^2 x - \sinh^2 x = 1$.

EXERCISE 8 **Numerical evaluation**

Using $e^{0.8} = 2.2255$ (given), evaluate $\cosh 0.8$, $\sinh 0.8$, $\tanh 0.8$, $\operatorname{sech} 0.8$ and $\coth 0.8$ to

4 decimal places.

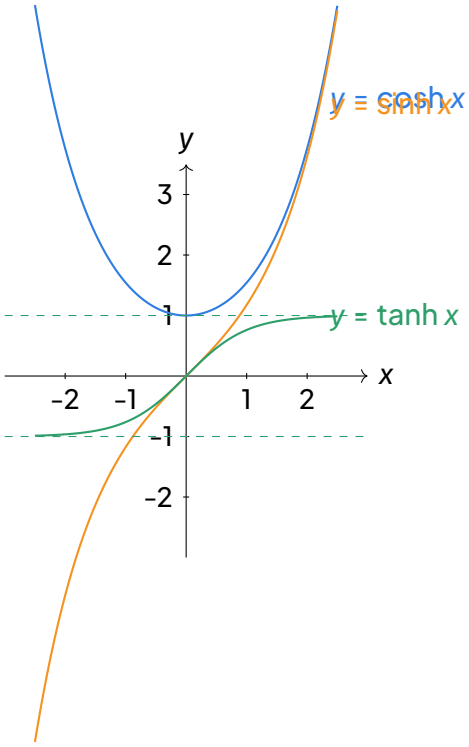
SOLUTION **Exercise 1**

$e^{-0.8} = 1/2.2255 = 0.4493$. $\cosh 0.8 = (2.2255 + 0.4493)/2 = 1.3374$. $\sinh 0.8 = (2.2255 - 0.4493)/2 = 0.8881$. $\tanh 0.8 = 0.8881/1.3374 = 0.6641$. $\operatorname{sech} 0.8 = 1/1.3374 = 0.7477$. $\coth 0.8 = 1/0.6641 = 1.5058$.

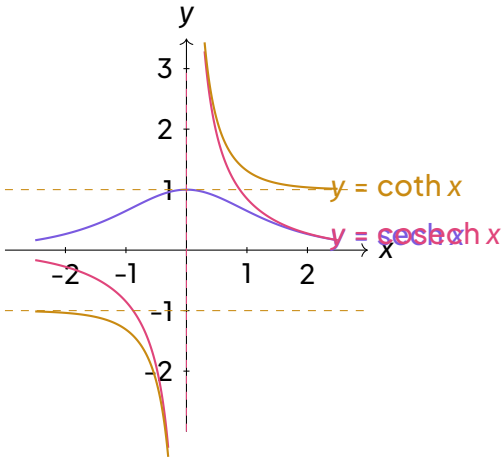
11 GRAPHS OF HYPERBOLIC FUNCTIONS

EXPERIMENT **Sketching the graphs**

We can obtain the graphs by considering the exponential components. For large positive x, e^{-x} is negligible, so $\sinh x \approx \cosh x \approx \frac{1}{2}e^{x}$ (exponential growth). For large negative x, e^{x} is negligible, so $\sinh x \approx -\frac{1}{2}e^{-x}$ (negative exponential) and $\cosh x \approx \frac{1}{2}e^{-x}$ (positive exponential). $\tanh x = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$ tends to ± 1 with horizontal asymptotes.



Graphs of $y = \cosh x$ (blue), $y = \sinh x$ (orange), $y = \tanh x$ (green). Horizontal asymptotes $y = \pm 1$ for $\tanh x$.



Graphs of reciprocal functions: $\operatorname{sech} x$ (purple), $\operatorname{cosech} x$ (pink), $\coth x$ (gold). Vertical asymptote at $x = 0$ for cosech and coth; horizontal asymptotes $y = \pm 1$ for coth.

KEY POINTS **Key features of the graphs**

Function	Domain	Range	Key features
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$	Odd, strictly increasing, inflection at $(0, 0)$
$\cosh x$	$\mathbb{R}$	$[1, +\infty)$	Even, minimum 1 at $x = 0$, strictly decreasing on $(-\infty, 0]$, increasing on $[0, +\infty)$
$\tanh x$	$\mathbb{R}$	$(-1, 1)$	Odd, strictly increasing, horizontal asymptotes $y = \pm 1$
$\operatorname{sech} x$	$\mathbb{R}$	$(0, 1]$	Even, maximum 1 at $x = 0$, horizontal asymptote $y = 0$
$\operatorname{cosech} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	Odd, vertical asymptote $x = 0$, horizontal asymptote $y = 0$
$\coth x$	$\mathbb{R} \setminus \{0\}$	$(-\infty, -1) \cup (1, +\infty)$	Odd, vertical asymptote $x = 0$, horizontal asymptotes $y = \pm 1$

12 HYPERBOLIC IDENTITIES AND OSBORN'S RULE

PROPERTY **Fundamental identity**

$$\cosh^2 x - \sinh^2 x = 1 \qquad (\text{for all } x \in \mathbb{R}).$$

Proof: $\cosh^2 x - \sinh^2 x = \left(\frac{e^x+e^{-x}}{2}\right)^2 - \left(\frac{e^x-e^{-x}}{2}\right)^2 = \frac{e^{2x}+2+e^{-2x}-(e^{2x}-2+e^{-2x})}{4} = 1.$ ■

PROPERTY **Derived identities**

Dividing the fundamental identity by $\cosh^2 x$ and $\sinh^2 x$ (where defined) gives:

$$1 - \tanh^2 x = \operatorname{sech}^2 x, \qquad \coth^2 x - 1 = \operatorname{cosech}^2 x.$$

PROPERTY **Osborn's Rule — statement and justification**

To convert a trigonometric identity into the corresponding hyperbolic identity:

1. Replace cos by cosh and sin by sinh.
2. Change the sign of every term that contains a product of two sines (explicit like $\sin A \sin B$ or implicit like $\sin^2 A$, $\tan^2 A = \frac{\sin^2 A}{\cos^2 A}$).

Justification: The trigonometric functions satisfy $\cos^2\theta + \sin^2\theta = 1$, while hyperbolic functions satisfy $\cosh^2 x - \sinh^2 x = 1$. The sign change compensates for $i^2 = -1$ in the relations $\cosh x = \cos(ix)$, $\sinh x = -i \sin(ix)$.

METHOD **Proving hyperbolic identities**

1. If the identity resembles a trigonometric identity, first try **Osborn's rule** to predict the form.
2. To prove it rigorously, start from the exponential definitions and expand. Work on the more complicated side and reduce it to the other.
3. Key algebraic reflex: $(e^x + e^{-x})^2 = e^{2x} + 2 + e^{-2x}$ and $(e^x - e^{-x})^2 = e^{2x} - 2 + e^{-2x}$.
4. Memorise the fundamental identity $\cosh^2 x - \sinh^2 x = 1$; divide it by $\cosh^2 x$ or $\sinh^2 x$ to obtain the others.

WORKED EXAMPLE 25 **Double angle formulae**

Use Osborn's rule to predict the hyperbolic analogues of $\cos 2x = 2\cos^2 x - 1$, $\cos 2x = 1 - 2\sin^2 x$, $\cos 2x = \cos^2 x - \sin^2 x$, and $\sin 2x = 2 \sin x \cos x$. Then prove $\cosh 2x = 2\cosh^2 x - 1$ from the exponential definitions.

SOLUTION

Solution. Applying Osborn's rule:

- $\cos 2x = 2\cos^2 x - 1 \to \cosh 2x = 2\cosh^2 x - 1$ (no sine product).
- $\cos 2x = 1 - 2\sin^2 x \to \cosh 2x = 1 + 2\sinh^2 x$ (sign change on $\sin^2 x$).
- $\cos 2x = \cos^2 x - \sin^2 x \to \cosh 2x = \cosh^2 x + \sinh^2 x$ (sign change on $\sin^2 x$).
- $\sin 2x = 2 \sin x \cos x \to \sinh 2x = 2 \sinh x \cosh x$ (only one sine factor, no sign change).

Proof of $\cosh 2x = 2\cosh^2 x - 1$:

$$2\cosh^2 x - 1 = 2\left(\frac{e^x + e^{-x}}{2}\right)^2 - 1 = 2 \cdot \frac{e^{2x} + 2 + e^{-2x}}{4} - 1$$
$$= \frac{e^{2x} + 2 + e^{-2x}}{2} - 1 = \frac{e^{2x} + e^{-2x}}{2} = \cosh 2x. \quad \blacksquare$$

WORKED EXAMPLE 26 **Sum and difference formulae**

Prove that $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$ and deduce $\cosh(x + y)$.

SOLUTION

Solution.

$$\sinh x \cosh y + \cosh x \sinh y = \frac{e^x - e^{-x}}{2} \cdot \frac{e^y + e^{-y}}{2} + \frac{e^x + e^{-x}}{2} \cdot \frac{e^y - e^{-y}}{2}$$
$$= \frac{1}{4}\left[(e^{x+y} + e^{x-y} - e^{-x+y} - e^{-x-y}) + (e^{x+y} - e^{x-y} + e^{-x+y} - e^{-x-y})\right]$$
$$= \frac{1}{4}\left[2e^{x+y} - 2e^{-(x+y)}\right] = \frac{e^{x+y} - e^{-(x+y)}}{2} = \sinh(x + y). \quad \blacksquare$$

Similarly, $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$ (note the plus sign, unlike $\cos(x + y)$).

WARNING **Osborn's rule traps**

- The rule applies to identities, not to equations or specific values.
- Implied products of sines: $\tan^2 x = \frac{\sin^2 x}{\cos^2 x}$ contains $\sin^2 x$, so $\tanh^2 x$ gets a sign change relative to $\tan^2 x$. Example: $1 + \tan^2 x = \sec^2 x$ becomes $1 - \tanh^2 x = \operatorname{sech}^2 x$.
- $\sin 2x = 2\sin x \cos x$ has only one sine factor → no sign change: $\sinh 2x = 2\sinh x \cosh x$.

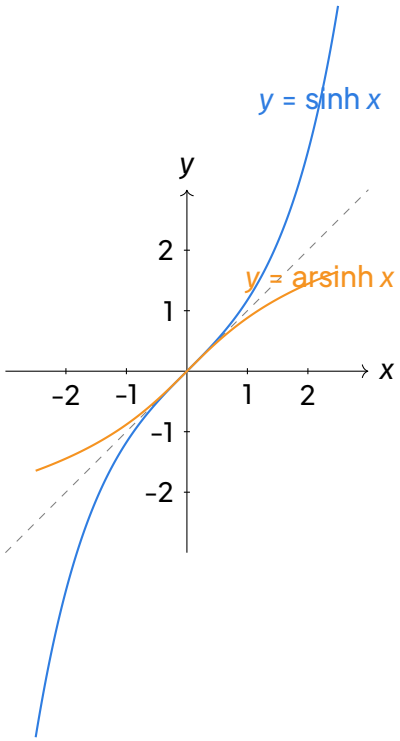
13 INVERSE HYPERBOLIC FUNCTIONS: DEFINITIONS, GRAPHS, AND LOGARITHMIC FORMS

DEFINITION **Inverse hyperbolic functions — principal values**

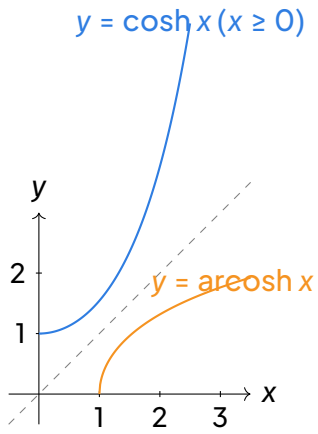
Because sinh and tanh are strictly increasing on $\mathbb{R}$, they are one-to-one. cosh is not one-to-one on $\mathbb{R}$; we restrict it to $[0, +\infty)$ to define its inverse.

- $y = \operatorname{arsinh} x \iff x = \sinh y$, with domain $\mathbb{R}$, range $\mathbb{R}$.
- $y = \operatorname{arcosh} x \iff x = \cosh y$, with domain $[1, +\infty)$, range $[0, +\infty)$.
- $y = \operatorname{artanh} x \iff x = \tanh y$, with domain $(-1, 1)$, range $\mathbb{R}$.

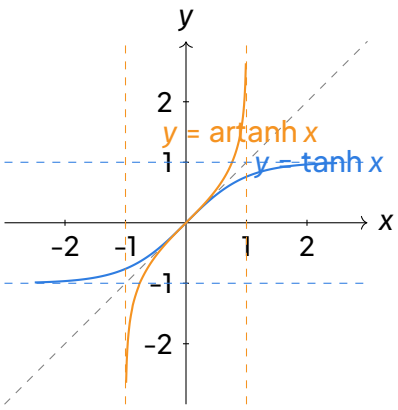
The reciprocal inverses are defined similarly: $\operatorname{arcosech} x$, $\operatorname{arsech} x$ ($0 < x \leq 1$), $\operatorname{arcoth} x$ ($|x| > 1$).



Graphs of $y = \sinh x$ (blue) and its inverse $y = \operatorname{arsinh} x$ (orange), symmetric about $y = x$.



Graphs of $y = \cosh x$ ($x \geq 0$, blue) and its inverse $y = \operatorname{arcosh} x$ (orange). Domain of arcosh is $x \geq 1$.



Graphs of $y = \tanh x$ (blue) and its inverse $y = \operatorname{artanh} x$ (orange). Vertical asymptotes at $x = \pm 1$ for artanh.

METHOD **Deriving logarithmic forms of inverse hyperbolic functions**

To express an inverse hyperbolic function as a logarithm:

1. Set $y = \operatorname{arsinh} x$ (for example), so that $x = \sinh y = \frac{e^y - e^{-y}}{2}$.
2. Multiply through by $2e^y$ to obtain a quadratic in e^y: $e^{2y} - 2xe^y - 1 = 0$.
3. Solve with the quadratic formula: $e^y = x \pm \sqrt{x^2 + 1}$.
4. Reject any negative root, since $e^y > 0$ for all real y.
5. Take natural logarithms: $y = \ln\left(x + \sqrt{x^2 + 1}\right)$.

Standard logarithmic forms (memorise):

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right), \quad x \in \mathbb{R},$$

$$\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right), \quad x \geq 1,$$

$$\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1 + x}{1 - x}\right), \quad |x| < 1.$$

WORKED EXAMPLE 27 **Logarithmic form of arcosh**

Prove that $\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$ for $x \geq 1$, and hence evaluate $\operatorname{arcosh} 2$ in logarithmic form.

SOLUTION

Solution. Let $y = \operatorname{arcosh} x$, so $y \geq 0$ and

$$x = \cosh y = \frac{e^y + e^{-y}}{2}.$$

Multiply by $2e^y$:

$$2xe^y = e^{2y} + 1 \quad \Longrightarrow \quad e^{2y} - 2xe^y + 1 = 0.$$

This is a quadratic in e^y:

$$e^y = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}.$$

Both roots are positive when $x \geq 1$ (their product is 1). We must select the root consistent with $y \geq 0$, i.e. $e^y \geq 1$. Since $x - \sqrt{x^2 - 1} = \dfrac{1}{x + \sqrt{x^2 - 1}} \leq 1$, the minus sign would give $e^y \leq 1$, i.e. $y \leq 0$. As the principal value requires $y \geq 0$, we take the plus sign:

$$e^y = x + \sqrt{x^2 - 1} \quad \Longrightarrow \quad \operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right). \qquad \blacksquare$$

Application: $\operatorname{arcosh} 2 = \ln\left(2 + \sqrt{4 - 1}\right) = \ln(2 + \sqrt{3}) \approx 1.3170$.

WORKED EXAMPLE 28 **Logarithmic form of artanh**

Prove that $\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$ for $|x| < 1$.

SOLUTION

Solution. Let $y = \operatorname{artanh} x$, so $x = \tanh y = \dfrac{e^y - e^{-y}}{e^y + e^{-y}}$.

$$x(e^y + e^{-y}) = e^y - e^{-y} \quad \Longrightarrow \quad xe^y + xe^{-y} = e^y - e^{-y}.$$

Multiply by e^y: $xe^{2y} + x = e^{2y} - 1 \quad \Longrightarrow \quad e^{2y}(x-1) = -x - 1 \quad \Longrightarrow \quad e^{2y} = \frac{1+x}{1-x}$. Since $|x| < 1$, the right-hand side is positive. Taking logarithms:

$$2y = \ln\left(\frac{1+x}{1-x}\right) \quad \Longrightarrow \quad \operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right). \qquad \blacksquare$$

DID YOU KNOW? **Alternative logarithmic forms**

Because $\ln(x - \sqrt{x^2-1}) = -\ln(x + \sqrt{x^2-1})$, we also have $\operatorname{arcosh} x = \pm\ln(x + \sqrt{x^2-1})$, but the principal value takes the + sign. For arsinh, note that $\ln(x - \sqrt{x^2+1})$ is undefined (negative argument) for $x > 0$, so only the + sign works for all x.

14 SOLVING HYPERBOLIC EQUATIONS

METHOD **Solving equations involving hyperbolic functions**

1. Single function: isolate the hyperbolic function, then use the inverse function or its logarithmic form.
2. Mixed cosh and sinh: substitute the exponential definitions and multiply through by e^x to get a quadratic in e^x.
3. Solve the quadratic; reject negative roots because $e^x > 0$.
4. Take logarithms to obtain x.
5. Alternative: if the equation is quadratic in $\cosh x$ or $\sinh x$, factorise or use the quadratic formula directly, remembering $\cosh x \geq 1$.

WORKED EXAMPLE 29 **A GCE-style hyperbolic equation**

Solve the equation $3\cosh x - \sinh x = 3$, giving your answers in exact logarithmic form.

SOLUTION

Solution. Substitute the exponential definitions:

$$3\cdot\frac{e^x+e^{-x}}{2}-\frac{e^x-e^{-x}}{2}=3 \quad\Longrightarrow\quad \frac{3e^x+3e^{-x}-e^x+e^{-x}}{2}=3,$$

$$2e^x+4e^{-x}=6 \quad\Longrightarrow\quad e^x+2e^{-x}=3.$$

Multiply by e^x:

$$e^{2x}-3e^x+2=0 \quad\Longrightarrow\quad (e^x-1)(e^x-2)=0.$$

Hence $e^x=1$ or $e^x=2$ (both positive, so both are valid):

$$\boxed{x=0 \quad \text{or} \quad x=\ln 2.}$$

Check $x=\ln 2$: $e^x=2$, $e^{-x}=\frac{1}{2}$, so $\cosh x=\frac{2+0.5}{2}=1.25$, $\sinh x=\frac{2-0.5}{2}=0.75$, and $3(1.25)-0.75=3.75-0.75=3$ ✓.

WORKED EXAMPLE 30 Quadratic in hyperbolic functions

Solve $2\cosh^2 x-5\sinh x-4=0$.

SOLUTION

Solution. Use $\cosh^2 x=1+\sinh^2 x$:

$$2(1+\sinh^2 x)-5\sinh x-4=0 \quad\Longrightarrow\quad 2\sinh^2 x-5\sinh x-2=0.$$

Factorise: $(2\sinh x+1)(\sinh x-2)=0$. So $\sinh x=-\frac{1}{2}$ or $\sinh x=2$. Using $\operatorname{arsinh} x=\ln(x+\sqrt{x^2+1})$:

$$x=\operatorname{arsinh}\left(-\frac{1}{2}\right)=\ln\left(-\frac{1}{2}+\sqrt{\frac{1}{4}+1}\right)=\ln\left(\frac{\sqrt{5}-1}{2}\right),$$

$$x=\operatorname{arsinh} 2=\ln(2+\sqrt{5}).$$

Both are valid since sinh is onto $\mathbb{R}$.

EXERCISE 9 Solving equations

Solve the following equations, giving exact answers in logarithmic form where appropriate: (a) $\cosh x=\frac{5}{4}$ (b) $\tanh x=\frac{1}{3}$ (c) $\sinh 2x=2\sinh x$ (d) $5\sinh x-3\cosh x=1$.

SOLUTION **Exercise 2**

(a) $x = \pm\operatorname{arcosh}\frac{5}{4} = \pm\ln\left(\frac{5}{4}+\sqrt{\frac{25}{16}-1}\right) = \pm\ln 2$. (b) $x = \operatorname{artanh}\frac{1}{3} = \frac{1}{2}\ln\left(\frac{1+1/3}{1-1/3}\right) = \frac{1}{2}\ln 2$. (c) $2\sinh x\cosh x = 2\sinh x \Rightarrow 2\sinh x(\cosh x - 1) = 0 \Rightarrow \sinh x = 0$ or $\cosh x = 1 \Rightarrow x = 0$. (d) $5\frac{e^x-e^{-x}}{2} - 3\frac{e^x+e^{-x}}{2} = 1 \Rightarrow e^x - 4e^{-x} = 1 \Rightarrow e^{2x} - e^x - 4 = 0 \Rightarrow e^x = \frac{1+\sqrt{17}}{2}$ (positive root) $\Rightarrow x = \ln\left(\frac{1+\sqrt{17}}{2}\right)$.

15 CALCULUS OF HYPERBOLIC FUNCTIONS: DIFFERENTIATION AND INTEGRATION

PROPERTY **Derivatives of basic hyperbolic functions**

$$\frac{d}{dx}(\sinh x) = \cosh x, \qquad \frac{d}{dx}(\cosh x) = \sinh x, \qquad \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x.$$

Proof: $\frac{d}{dx}\sinh x = \frac{d}{dx}\frac{e^x-e^{-x}}{2} = \frac{e^x+e^{-x}}{2} = \cosh x$. Similarly for $\cosh x$. For $\tanh x$, use quotient rule or $\tanh x = \frac{\sinh x}{\cosh x}$.

PROPERTY **Derivatives of inverse hyperbolic functions**

$$\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}, \qquad \frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}\ (x>1), \qquad \frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}\ (|x|<1).$$

Proof (for arsinh): Let $y = \operatorname{arsinh} x$, so $x = \sinh y$. Then $\frac{dx}{dy} = \cosh y = \sqrt{1+\sinh^2 y} = \sqrt{1+x^2}$. Hence $\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}}$.

METHOD **Differentiation reflexes**

1. Basic results: $\frac{d}{dx}(\sinh x) = \cosh x$, $\frac{d}{dx}(\cosh x) = \sinh x$ (no minus sign, unlike cos), $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$.
2. Inverse functions: $\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}$, $\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}$, $\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}$.

3. Combine with the chain rule: for $\sinh u$, $\cosh u$, … where $u = u(x)$, multiply by u'.
4. For artanh of a quotient, the logarithmic form $\frac{1}{2}\ln\frac{1+u}{1-u}$ is often the fastest route.
5. If in doubt, derive: differentiate $y = \operatorname{arsinh} x$ by writing $x = \sinh y$, then $\frac{dx}{dy} = \cosh y = \sqrt{1+\sinh^2 y} = \sqrt{1+x^2}$.

WORKED EXAMPLE 31 **Differentiation practice**

(a) Differentiate $y = x^2\cosh 3x$. (b) Show that $\frac{d}{dx}\operatorname{artanh} x = \frac{1}{1-x^2}$. (c) Find $\frac{dy}{dx}$ if $y = \operatorname{arsinh}(2x)$. (d) Differentiate $y = \ln(\cosh x)$.

SOLUTION

Solution. (a) Product rule with $u = x^2$, $v = \cosh 3x$:

$$\frac{dy}{dx} = 2x\cosh 3x + x^2\cdot 3\sinh 3x = x\big(2\cosh 3x + 3x\sinh 3x\big).$$

(b) Let $y = \operatorname{artanh} x$, so $x = \tanh y$. Differentiate with respect to y:

$$\frac{dx}{dy} = \operatorname{sech}^2 y = 1-\tanh^2 y = 1-x^2.$$

Therefore

$$\frac{dy}{dx} = \frac{1}{dx/dy} = \frac{1}{1-x^2}. \quad \blacksquare$$

(c) Chain rule with $u = 2x$:

$$\frac{dy}{dx} = \frac{1}{\sqrt{(2x)^2+1}}\cdot 2 = \frac{2}{\sqrt{4x^2+1}}.$$

(d) $\frac{dy}{dx} = \frac{1}{\cosh x}\cdot\sinh x = \tanh x$.

PROPERTY **Standard integrals of hyperbolic functions**

$$\int \sinh x\,dx = \cosh x + C, \qquad \int \cosh x\,dx = \sinh x + C, \qquad \int \operatorname{sech}^2 x\,dx = \tanh x + C.$$

$$\int \tanh x\,dx = \ln(\cosh x) + C, \qquad \int \coth x\,dx = \ln|\sinh x| + C.$$

PROPERTY **Integrals leading to inverse hyperbolic functions**

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \operatorname{arsinh}\frac{x}{a} + C, \qquad \int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}\frac{x}{a} + C \ (x > a),$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{a}\operatorname{artanh}\frac{x}{a} + C \ (|x| < a), \qquad \int \frac{dx}{x^2-a^2} = \frac{1}{a}\operatorname{arcoth}\frac{x}{a} + C \ (|x| > a).$$

METHOD **Integration reflexes**

1. Standard integrals: $\int \sinh x\,dx = \cosh x + C$, $\int \cosh x\,dx = \sinh x + C$, $\int \operatorname{sech}^2 x\,dx = \tanh x + C$.
2. Linear argument: $\int \cosh(ax+b)\,dx = \frac{1}{a}\sinh(ax+b) + C$.
3. Recognise the standard forms leading to inverse hyperbolic functions (see above).
4. For $\int \tanh x\,dx$, write $\tanh x = \frac{\sinh x}{\cosh x}$ and use substitution $u = \cosh x$: the answer is $\ln(\cosh x) + C$.
5. Powers: use identities, e.g. $\cosh^2 x = \frac{\cosh 2x + 1}{2}$, $\sinh^2 x = \frac{\cosh 2x - 1}{2}$, exactly as for trigonometric integrals.

WORKED EXAMPLE 32 **Two standard integrals**

Evaluate (a) $\int_0^1 \cosh 2x\,dx$; (b) $\int \frac{dx}{\sqrt{x^2+9}}$, expressing the answer in logarithmic form.

SOLUTION

Solution. (a) Since $\frac{d}{dx}\sinh 2x = 2\cosh 2x$:

$$\int_0^1 \cosh 2x\,dx = \left[\frac{1}{2}\sinh 2x\right]_0^1 = \frac{1}{2}(\sinh 2 - \sinh 0) = \frac{1}{2}\sinh 2 = \frac{e^2 - e^{-2}}{4} \approx 1.8134.$$

(b) This matches the standard form with $a = 3$:

$$\int \frac{dx}{\sqrt{x^2+9}} = \operatorname{arsinh}\frac{x}{3} + C.$$

Using the logarithmic form $\operatorname{arsinh} u = \ln\left(u + \sqrt{u^2+1}\right)$:

$$\operatorname{arsinh}\frac{x}{3} = \ln\left(\frac{x}{3} + \sqrt{\frac{x^2}{9}+1}\right) = \ln\left(\frac{x+\sqrt{x^2+9}}{3}\right).$$

Since $\ln 3$ is a constant, it can be absorbed into C:

$$\int \frac{dx}{\sqrt{x^2+9}} = \ln\left(x+\sqrt{x^2+9}\right) + C'. \quad \blacksquare$$

WORKED EXAMPLE 33 **Integration of powers**

Evaluate $\int \cosh^2 3x\,dx$.

SOLUTION

Solution. Use $\cosh^2\theta = \frac{\cosh 2\theta+1}{2}$ with $\theta = 3x$:

$$\int \cosh^2 3x\,dx = \int \frac{\cosh 6x+1}{2}\,dx = \frac{1}{2}\left(\frac{\sinh 6x}{6}+x\right)+C = \frac{\sinh 6x}{12}+\frac{x}{2}+C.$$

16 ADVANCED APPLICATIONS: HYPERBOLIC SUBSTITUTIONS AND MACLAURIN SERIES

METHOD **Choosing the right hyperbolic substitution**

Match the form of the integrand to the identity that simplifies it:

1. $\sqrt{x^2+a^2}$: put $x = a\sinh t$, because $a^2\sinh^2 t + a^2 = a^2\cosh^2 t$ and $dx = a\cosh t\,dt$.
2. $\sqrt{x^2-a^2}$: put $x = a\cosh t$, because $a^2\cosh^2 t - a^2 = a^2\sinh^2 t$ and $dx = a\sinh t\,dt$ (take $t \geq 0$ so $\sinh t \geq 0$).
3. $a^2 - x^2$ (with $|x| < a$): put $x = a\tanh t$, because $a^2 - a^2\tanh^2 t = a^2\operatorname{sech}^2 t$ and $dx = a\operatorname{sech}^2 t\,dt$.

Then: compute dx in terms of dt, simplify the square root using the identity, integrate in t, and convert back to x using inverse functions or logarithmic forms.

WORKED EXAMPLE 34 **Hyperbolic substitution**

Use the substitution $x = 3\sinh t$ to evaluate $\int \sqrt{x^2+9}\,dx$.

SOLUTION

Solution. With $x = 3\sinh t$: $dx = 3\cosh t\,dt$ and

$$\sqrt{x^2+9} = \sqrt{9\sinh^2 t + 9} = 3\sqrt{\sinh^2 t + 1} = 3\cosh t \quad (\cosh t > 0).$$

Hence

$$\int \sqrt{x^2+9}\,dx = \int 3\cosh t \cdot 3\cosh t\,dt = 9\int \cosh^2 t\,dt.$$

Use the identity $\cosh^2 t = \dfrac{\cosh 2t + 1}{2}$:

$$9\int \cosh^2 t\,dt = \frac{9}{2}\int (\cosh 2t + 1)\,dt = \frac{9}{2}\left(\frac{\sinh 2t}{2} + t\right) + C = \frac{9}{4}\sinh 2t + \frac{9}{2}t + C.$$

Convert back: $t = \operatorname{arsinh}\dfrac{x}{3}$ and

$$\sinh 2t = 2\sinh t\cosh t = 2\cdot\frac{x}{3}\cdot\frac{\sqrt{x^2+9}}{3} = \frac{2x\sqrt{x^2+9}}{9}.$$

Therefore

$$\int \sqrt{x^2+9}\,dx = \frac{x\sqrt{x^2+9}}{2} + \frac{9}{2}\operatorname{arsinh}\frac{x}{3} + C = \frac{x\sqrt{x^2+9}}{2} + \frac{9}{2}\ln\left(x + \sqrt{x^2+9}\right) + C'. \quad \blacksquare$$

Check (differentiate): $\dfrac{d}{dx}\left[\frac{x}{2}\sqrt{x^2+9}\right] = \frac{1}{2}\sqrt{x^2+9} + \frac{x^2}{2\sqrt{x^2+9}} = \frac{2x^2+9}{2\sqrt{x^2+9}}$, and $\dfrac{d}{dx}\left[\frac{9}{2}\operatorname{arsinh}\frac{x}{3}\right] = \frac{9}{2}\cdot\frac{1}{\sqrt{x^2+9}}$.
Sum: $\frac{2x^2+9+9}{2\sqrt{x^2+9}} = \frac{x^2+9}{\sqrt{x^2+9}} = \sqrt{x^2+9}$ ✓.

WORKED EXAMPLE 35 **Another substitution**

Evaluate $\displaystyle\int \frac{dx}{\sqrt{x^2-16}}$ for $x > 4$ using a hyperbolic substitution.

SOLUTION

Solution. Put $x = 4\cosh t$ ($t > 0$). Then $dx = 4\sinh t\,dt$ and $\sqrt{x^2-16} = \sqrt{16\cosh^2 t - 16} = 4\sinh t$.

$$\int \frac{dx}{\sqrt{x^2-16}} = \int \frac{4\sinh t}{4\sinh t}\,dt = \int dt = t + C = \operatorname{arcosh}\frac{x}{4} + C = \ln\left(\frac{x}{4} + \sqrt{\frac{x^2}{16} - 1}\right) + C = \ln\left(x + \sqrt{x^2-16}\right) + C'.$$

METHOD **Building a Maclaurin series for hyperbolic functions**

1. Recall the Maclaurin formula: $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$
2. Exploit the cyclic derivatives: for sinh and cosh, derivatives alternate between the two functions, and at $x = 0$: $\sinh 0 = 0$, $\cosh 0 = 1$.
3. Hence even powers survive for cosh, odd powers for sinh:

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \qquad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}.$$

4. Shortcut: take the series of $e^x = \sum \frac{x^n}{n!}$ and use the definitions; the odd/even terms separate automatically:

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{1}{2}\sum_{n=0}^{\infty} \frac{x^n - (-x)^n}{n!} = \sum_{n\ \text{odd}} \frac{x^n}{n!}.$$

5. For approximations, state clearly the order of the neglected terms, e.g. $\cosh x \approx 1 + \frac{x^2}{2}$ for small x.

WORKED EXAMPLE 36 **Series expansion and approximation**

(a) Obtain the Maclaurin expansion of cosh x up to the term in x^4. (b) Hence show that for small x, $\cosh 2x - 1 - 2x^2 \approx \frac{2x^4}{3}$, and use this to estimate $\lim_{x\to 0} \frac{\cosh 2x - 1}{x^2}$.

SOLUTION

Solution. (a) Let $f(x) = \cosh x$. The derivatives cycle:

$$f'(x) = \sinh x, \quad f''(x) = \cosh x, \quad f'''(x) = \sinh x, \quad f^{(4)}(x) = \cosh x.$$

At $x = 0$: $f(0) = 1$, $f'(0) = 0$, $f''(0) = 1$, $f'''(0) = 0$, $f^{(4)}(0) = 1$. Therefore

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \cdots$$

(b) Replace x by $2x$:

$$\cosh 2x = 1 + \frac{(2x)^2}{2} + \frac{(2x)^4}{24} + \cdots = 1 + 2x^2 + \frac{16x^4}{24} + \cdots = 1 + 2x^2 + \frac{2x^4}{3} + \cdots$$

Hence $\cosh 2x - 1 - 2x^2 \approx \frac{2x^4}{3}$ for small x, as required. For the limit:

$$\frac{\cosh 2x - 1}{x^2} = \frac{2x^2 + \frac{2}{3}x^4 + \cdots}{x^2} = 2 + \frac{2}{3}x^2 + \cdots \xrightarrow[x\to 0]{} 2.$$

$$\lim_{x\to 0}\frac{\cosh 2x-1}{x^2}=2.$$

Consistency check: the same result follows from the identity $\cosh 2x-1=2\sinh^2 x$ and $\sinh x\approx x$, giving $\frac{2\sinh^2 x}{x^2}\to 2$ ✓.

WORKED EXAMPLE 37 **Maclaurin series of** $\tanh x$

Find the Maclaurin expansion of $\tanh x$ up to the term in x^5.

SOLUTION

Solution. Use $\tanh x=\frac{\sinh x}{\cosh x}=\frac{x+\frac{x^3}{6}+\frac{x^5}{120}+\cdots}{1+\frac{x^2}{2}+\frac{x^4}{24}+\cdots}$. Perform polynomial long division or use $(1+u)^{-1}\approx 1-u+u^2$ with $u=\frac{x^2}{2}+\frac{x^4}{24}$:

$$\tanh x=\left(x+\frac{x^3}{6}+\frac{x^5}{120}\right)\left(1-\frac{x^2}{2}-\frac{x^4}{24}+\frac{x^4}{4}+\cdots\right)=x-\frac{x^3}{3}+\frac{2x^5}{15}+\cdots$$

WARNING **Frequent errors to avoid in GCE examinations**

- Writing $\frac{d}{dx}(\cosh x)=-\sinh x$: the minus sign belongs to cos, not to cosh.
- Forgetting to reject negative roots of e^x when solving hyperbolic equations.
- Using arcosh formulas for $x<1$ or artanh for $|x|\geq 1$: always check domains.
- Applying Osborn's rule but forgetting to change the sign of implied products of sines such as $\tan^2 x=\frac{\sin^2 x}{\cos^2 x}$.
- Confusing arsinh with $\frac{1}{\sinh}$: the inverse function is not the reciprocal.
- In integration, forgetting the factor $\frac{1}{a}$ when the argument is $ax+b$.
- In hyperbolic substitutions, forgetting to change the differential dx and the limits (if definite integral).
- In Maclaurin series, mixing up the factorial denominators: sinh has odd factorials, cosh has even factorials.

DID YOU KNOW? **Hyperbolic functions in Cameroon and beyond**

The catenary curve $y = a\cosh(x/a)$ appears in the design of suspension bridges (like the Wouri Bridge in Douala) and power lines. The shape of a hanging chain minimizes potential energy. In relativity, rapidity ϕ is defined by $\tanh\phi = v/c$, and Lorentz transformations become hyperbolic rotations. The Gateway Arch in St. Louis is a weighted catenary, not a parabola.

EXERCISE 10 **Mixed practice — GCE style**

1. Prove that $\sinh 3x = 3\sinh x + 4\sinh^3 x$. 2. Solve $\cosh 2x - 3\sinh x = 0$, giving answers in exact logarithmic form. 3. Find $\displaystyle\int_0^{\ln 2} \frac{e^x}{\sqrt{e^{2x}+4}}\,dx$. 4. Use the substitution $x = 2\tanh t$ to evaluate $\displaystyle\int \frac{dx}{(4-x^2)^{3/2}}$. 5. Obtain the Maclaurin series of $\sinh x\cosh x$ up to x^5 and hence deduce the series for $\sinh 2x$.

SOLUTION **Exercise 3**

1. $\sinh 3x = \sinh(2x+x) = \sinh 2x\cosh x + \cosh 2x\sinh x = (2\sinh x\cosh x)\cosh x + (1+2\sinh^2 x)\sinh x = 2\sinh x(1+\sinh^2 x) + \sinh x + 2\sinh^3 x = 3\sinh x + 4\sinh^3 x$. 2. $\cosh 2x = 1 + 2\sinh^2 x$, so $1 + 2\sinh^2 x - 3\sinh x = 0 \Rightarrow (2\sinh x - 1)(\sinh x - 1) = 0$. $\sinh x = \frac{1}{2}$ or 1. $x = \operatorname{arsinh}\frac{1}{2} = \ln\left(\frac{1+\sqrt{5}}{2}\right)$ or $x = \operatorname{arsinh}1 = \ln(1+\sqrt{2})$. 3. Substitute $u = e^x$, $du = e^x dx$. Limits: $x = 0 \to u = 1$, $x = \ln 2 \to u = 2$. $\int_1^2 \frac{du}{\sqrt{u^2+4}} = \left[\operatorname{arsinh}\frac{u}{2}\right]_1^2 = \operatorname{arsinh}1 - \operatorname{arsinh}\frac{1}{2} = \ln(1+\sqrt{2}) - \ln\left(\frac{1+\sqrt{5}}{2}\right) = \ln\left(\frac{2(1+\sqrt{2})}{1+\sqrt{5}}\right)$. 4. $x = 2\tanh t$, $dx = 2\operatorname{sech}^2 t\,dt$, $4 - x^2 = 4\operatorname{sech}^2 t$, $(4-x^2)^{3/2} = 8\operatorname{sech}^3 t$. Integral $= \int \frac{2\operatorname{sech}^2 t}{8\operatorname{sech}^3 t}\,dt = \frac{1}{4}\int \cosh t\,dt = \frac{1}{4}\sinh t + C$. Since $\tanh t = x/2$, $\sinh t = \frac{x/2}{\sqrt{1-x^2/4}} = \frac{x}{\sqrt{4-x^2}}$. Answer: $\frac{x}{4\sqrt{4-x^2}} + C$. 5. $\sinh x\cosh x = \left(x + \frac{x^3}{6} + \frac{x^5}{120}\right)\left(1 + \frac{x^2}{2} + \frac{x^4}{24}\right) = x + \frac{2x^3}{3} + \frac{17x^5}{120} + \cdots$. But $\sinh x\cosh x = \frac{1}{2}\sinh 2x$, so $\sinh 2x = 2x + \frac{4x^3}{3} + \frac{17x^5}{60} + \cdots$. Check: direct substitution $2x$ into sinh series gives $2x + \frac{8x^3}{6} + \frac{32x^5}{120} = 2x + \frac{4x^3}{3} + \frac{4x^5}{15} = 2x + \frac{4x^3}{3} + \frac{16x^5}{60}$. Discrepancy? Re-check multiplication: $\frac{1}{6}\cdot\frac{1}{2} = \frac{1}{12}$, $\frac{1}{120}\cdot 1 = \frac{1}{120}$, $\frac{1}{6}\cdot\frac{1}{24} = \frac{1}{144}$... Better: $\sinh 2x = 2x + \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} = 2x + \frac{8x^3}{6} + \frac{32x^5}{120} = 2x + \frac{4x^3}{3} + \frac{4x^5}{15}$. So $\sinh x\cosh x = x + \frac{2x^3}{3} + \frac{2x^5}{15} + \cdots$. The product expansion must yield this. Let's do carefully: $(x + x^3/6 + x^5/120)(1 + x^2/2 + x^4/24) = x + x^3/2 + x^5/24 + x^3/6 + x^5/12 + x^5/120 = x + (1/2 + 1/6)x^3 + (1/24 + 1/12 + 1/120)x^5 = x + (2/3)x^3 + (5/120 + 10/120 + 1/120)x^5 = x + 2x^3/3 + 16x^5/120 = x + 2x^3/3 + 2x^5/15$. Correct.

17 EXERCISES

17.1 I check my knowledge

EXERCISE 11 **Multiple choice**

For each question, choose the single correct answer, justifying briefly.

1. $\cosh x$ is equal to:
   (a) $\frac{e^x - e^{-x}}{2}$
   (b) $\frac{e^x + e^{-x}}{2}$
   (c) $\frac{e^x + e^{-x}}{e^x - e^{-x}}$
   (d) $\frac{2}{e^x + e^{-x}}$
2. The domain of the function $x \mapsto \operatorname{arcosh} x$ is:
   (a) $\mathbb{R}$
   (b) $]0, +\infty[$
   (c) $[1, +\infty[$
   (d) $]-1, 1[$
3. $\cosh^2 x - \sinh^2 x$ equals:
   (a) -1
   (b) 0
   (c) 1
   (d) $\cosh 2x$
4. $\int \sinh x \, dx$ equals:
   (a) $\cosh x + C$
   (b) $-\cosh x + C$
   (c) $\operatorname{sech}^2 x + C$
   (d) $\ln(\cosh x) + C$

EXERCISE 12 **True or false**

State whether each assertion is true or false, and justify your answer carefully.

1. The function sinh is an even function.
2. For all real x, $\cosh x \geq 1$.
3. $\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$ for all real x.
4. $\frac{d}{dx}(\tanh x) = 1 + \tanh^2 x$.

EXERCISE 13 **Definitions**

1. Give the definitions of $\sinh x$, $\cosh x$ and $\tanh x$ in terms of the exponential function.
2. State Osborn's rule and illustrate it by writing the hyperbolic identity corresponding to $\cos 2x = 1 - 2\sin^2 x$.
3. Define $\operatorname{artanh} x$ and state its domain.

EXERCISE 14 **Direct calculations**

1. Given that $\sinh x = \frac{3}{4}$, find the exact values of $\cosh x$ and $\tanh x$.
2. Evaluate $\cosh(\ln 2)$ and $\sinh(\ln 3)$ exactly.
3. Differentiate $f(x) = \sinh(2x) + \cosh^2 x$ with respect to x.
4. Evaluate $\int_0^{\ln 2} \cosh x \, dx$, giving your answer as a rational number.

17.2 I practise

EXERCISE 15 **Proving identities**

1. Prove that $\cosh 2x = 2\cosh^2 x - 1$.
2. Prove that $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$.
3. Hence express $\sinh 2x$ in terms of $\sinh x$ and $\cosh x$.

EXERCISE 16 **Solving a hyperbolic equation**

Solve for x, giving your answers in exact logarithmic form:

$$3\sinh x - 4\cosh x = 5.$$

EXERCISE 17 **Logarithmic form of an inverse function**

1. Starting from $y = \operatorname{arcosh} x$, show that $e^{2y} - 2xe^{y} + 1 = 0$.
2. Hence prove that $\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$ for $x \geq 1$, explaining why the other root of the quadratic is rejected.
3. Use this result to evaluate $\operatorname{arcosh} 2$ in logarithmic form.

EXERCISE 18 **Integration by hyperbolic substitution**

1. Using the substitution $x = 3\cosh t$, evaluate

$$\int \frac{dx}{\sqrt{x^2 - 9}}, \qquad x > 3,$$

and express the result in terms of an inverse hyperbolic function.

2. Using a suitable hyperbolic substitution, evaluate $\displaystyle\int \frac{dx}{\sqrt{x^2 + 4}}$.

17.3 I prepare for the GCE

EXERCISE 19 **Maclaurin series and approximation** **(14 marks)**

(a) Write down the Maclaurin expansions of $\sinh x$ and $\cosh x$ up to the term in x^5. (4 marks)

(b) Hence, or otherwise, find the Maclaurin expansion of $\tanh x$ up to the term in x^5. (5 marks)

(c) Use your expansion to estimate $\tanh 0.2$, giving your answer to 5 decimal places. (3 marks)

(d) Given that the true value of $\tanh 0.2$ is 0.197375, comment on the accuracy of your es-

timate. (2 marks)

EXERCISE 20 **Calculus with hyperbolic functions** **(16 marks)**

(a) Using the substitution $u = e^x$, prove that

$$\int \operatorname{sech} x \, dx = 2 \arctan(e^x) + C.$$

(5 marks)

(b) Show that $\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1 - x^2}$ for $|x| < 1$. (4 marks)

(c) Hence evaluate $\int_0^{1/2} \frac{dx}{1 - x^2}$, giving your answer in exact logarithmic form. (3 marks)

(d) Solve the equation $\operatorname{artanh} x = \ln 2$, giving x as an exact fraction. (4 marks)

EXERCISE 21 **The catenary: a hanging chain** **(18 marks)**

A uniform cable hanging between two pylons in Douala is modelled by the curve

$$y = 10 \cosh\left(\frac{x}{10}\right), \qquad -10 \le x \le 10,$$

where x and y are measured in metres.

(a) Show that $1 + \left(\frac{dy}{dx}\right)^2 = \cosh^2\left(\frac{x}{10}\right)$. (4 marks)

(b) The arc length of a curve $y = f(x)$ between $x = a$ and $x = b$ is given by

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx.$$

Show that the length of the cable between $x = -10$ and $x = 10$ is $20 \sinh(1)$ metres, and find this length correct to 2 decimal places. (6 marks)

(c) Find the height of the cable above its lowest point at the pylons $x = \pm 10$, giving your answer in terms of $\cosh 1$. (3 marks)

(d) Sketch the curve, indicating its axis of symmetry, its minimum point and its behaviour as $|x|$ increases. (5 marks)

18 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple choice**

1. Answer (b). By definition, $\cosh x = \dfrac{e^x + e^{-x}}{2}$. Option (a) is $\sinh x$, option (c) is $\coth x$, option (d) is $\operatorname{sech} x$.
2. Answer (c). Since $\cosh : [0, +\infty[\to [1, +\infty[$ is a bijection, its inverse arcosh has domain $[1, +\infty[$.
3. Answer (c). The fundamental identity gives $\cosh^2 x - \sinh^2 x = 1$ for all real x. (Note: $\cosh 2x = \cosh^2 x + \sinh^2 x$, so (d) is wrong.)
4. Answer (a). Since $\dfrac{d}{dx}(\cosh x) = \sinh x$, an antiderivative of $\sinh x$ is $\cosh x$, so $\displaystyle\int \sinh x \, dx = \cosh x + C$.

SOLUTION **Exercise 2 — True or false**

1. False. $\sinh(-x) = \dfrac{e^{-x} - e^x}{2} = -\sinh x$, so sinh is odd, not even. (It is cosh that is even.)
2. True. $\cosh x - 1 = \dfrac{e^x + e^{-x} - 2}{2} = \dfrac{(e^{x/2} - e^{-x/2})^2}{2} \geq 0$, with equality only at $x = 0$. Hence $\cosh x \geq 1$ for all $x \in \mathbb{R}$.
3. True. Setting $y = \operatorname{arsinh} x$, we have $x = \sinh y$, so $e^{2y} - 2xe^y - 1 = 0$, giving $e^y = x + \sqrt{x^2 + 1}$ (the other root $x - \sqrt{x^2 + 1} < 0$ is rejected since $e^y > 0$). Hence $y = \ln\left(x + \sqrt{x^2 + 1}\right)$, valid for all $x \in \mathbb{R}$.
4. False. $\dfrac{d}{dx}(\tanh x) = \operatorname{sech}^2 x = 1 - \tanh^2 x$, not $1 + \tanh^2 x$ (that is the trigonometric formula for tan).

SOLUTION **Exercise 3 — Definitions**

1. $\sinh x = \dfrac{e^x - e^{-x}}{2}$, $\cosh x = \dfrac{e^x + e^{-x}}{2}$, $\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$.
2. Osborn's rule: to obtain a hyperbolic identity from a trigonometric one, replace each trigonometric function by the corresponding hyperbolic function (sin → sinh, cos → cosh, tan → tanh), and change the sign of every product (or implied product, such as

a square) of two sines. Applying this to $\cos 2x = 1 - 2\sin^2 x$: the term $\sin^2 x$ is a product of two sines, so its sign changes:

$$\cosh 2x = 1 + 2\sinh^2 x.$$

3. $\operatorname{artanh} x$ is the inverse function of $\tanh : \mathbb{R} \to]-1,1[$: for $x \in]-1,1[$, $y = \operatorname{artanh} x \iff x = \tanh y$. Its domain is $]-1,1[$.

SOLUTION **Exercise 4 — Direct calculations**

1. From $\cosh^2 x - \sinh^2 x = 1$: $\cosh^2 x = 1 + \frac{9}{16} = \frac{25}{16}$, so $\cosh x = \frac{5}{4}$ (positive root since $\cosh x \geq 1$). Then $\tanh x = \frac{\sinh x}{\cosh x} = \frac{3/4}{5/4} = \frac{3}{5}$.

2. $\cosh(\ln 2) = \frac{2 + \frac{1}{2}}{2} = \frac{5}{4}$; $\quad \sinh(\ln 3) = \frac{3 - \frac{1}{3}}{2} = \frac{4}{3}$.

3. Let $f(x) = \sinh 2x + \cosh^2 x$. Then

   $f'(x) = 2\cosh(2x) + 2\cosh x \sinh x = 2\cosh 2x + \sinh 2x$ (using $2\sinh x \cosh x = \sinh 2x$).

4. $\displaystyle\int_0^{\ln 2} \cosh x \, dx = \left[\sinh x\right]_0^{\ln 2} = \sinh(\ln 2) - 0 = \frac{2 - \frac{1}{2}}{2} = \frac{3}{4}$.

SOLUTION **Exercise 5 — Proving identities**

1. Starting from the right-hand side:

$$2\cosh^2 x - 1 = 2\left(\frac{e^x + e^{-x}}{2}\right)^2 - 1 = \frac{e^{2x} + 2 + e^{-2x}}{2} - 1 = \frac{e^{2x} + e^{-2x}}{2} = \cosh 2x. \quad \blacksquare$$

2. From the definitions:

$$\sinh x \cosh y + \cosh x \sinh y = \frac{(e^x - e^{-x})(e^y + e^{-y}) + (e^x + e^{-x})(e^y - e^{-y})}{4}$$
$$= \frac{2e^{x+y} - 2e^{-(x+y)}}{4} = \frac{e^{x+y} - e^{-(x+y)}}{2} = \sinh(x + y). \quad \blacksquare$$

3. Setting $y = x$ in part (b): $\sinh 2x = \sinh x \cosh x + \cosh x \sinh x = 2\sinh x \cosh x$.

SOLUTION **Exercise 6 — Solving a hyperbolic equation**

Solve $3\sinh x - 4\cosh x = 5$ for $x \in \mathbb{R}$.
Write in exponential form:

$$3 \cdot \frac{e^x - e^{-x}}{2} - 4 \cdot \frac{e^x + e^{-x}}{2} = 5 \implies -e^x - 7e^{-x} = 10.$$

Multiplying by e^x (with $u = e^x > 0$):

$$-u^2 - 7 = 10u \implies u^2 + 10u + 7 = 0.$$

The discriminant is $\Delta = 100 - 28 = 72$, so

$$u = \frac{-10 \pm \sqrt{72}}{2} = -5 \pm 3\sqrt{2}.$$

Since $3\sqrt{2} \approx 4.243 < 5$, both roots $-5 + 3\sqrt{2}$ and $-5 - 3\sqrt{2}$ are strictly negative. But $u = e^x > 0$ for all real x. Hence the quadratic has no admissible root, and the original equation has no real solution.
Verification (optional): The function $f(x) = 3\sinh x - 4\cosh x$ has derivative $f'(x) = 3\cosh x - 4\sinh x$. Setting $f'(x) = 0$ gives $\tanh x = \frac{3}{4}$, so $\sinh x = \frac{3}{\sqrt{7}}$, $\cosh x = \frac{4}{\sqrt{7}}$ (using $\cosh^2 x - \sinh^2 x = 1$). The maximum value is $f_{\max} = \frac{9}{\sqrt{7}} - \frac{16}{\sqrt{7}} = -\sqrt{7} \approx -2.65$. Since $f_{\max} < 5$, the equation $f(x) = 5$ is indeed impossible in $\mathbb{R}$.
Conclusion: The solution set is $\varnothing$.

SOLUTION **Exercise 7 — Logarithmic form of an inverse function**

1. $y = \operatorname{arcosh} x \iff x = \cosh y = \frac{e^y + e^{-y}}{2}$. Multiplying by $2e^y$: $2xe^y = e^{2y} + 1$, i.e. $e^{2y} - 2xe^y + 1 = 0$. ■

2. This is a quadratic in e^y: $e^y = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}$. The two roots are positive and their product is $(x + \sqrt{x^2 - 1})(x - \sqrt{x^2 - 1}) = 1$, so $x - \sqrt{x^2 - 1} = \frac{1}{x + \sqrt{x^2 - 1}} \le 1$, giving $y = \ln(x - \sqrt{x^2 - 1}) \le 0$. Since arcosh takes values in $[0, +\infty[$ (we chose the principal branch $y \ge 0$), we reject this root and keep

$$\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right), \qquad x \ge 1. \qquad ■$$

3. $\operatorname{arcosh} 2 = \ln\left(2 + \sqrt{4 - 1}\right) = \ln(2 + \sqrt{3})$.

SOLUTION **Exercise 8 — Integration by hyperbolic substitution**

1. Let $x = 3\cosh t$, $t > 0$ (valid since $x > 3$). Then $dx = 3\sinh t\,dt$ and $\sqrt{x^2-9} = \sqrt{9\cosh^2 t - 9} = 3\sinh t$ (positive as $t > 0$). Hence

$$\int \frac{dx}{\sqrt{x^2-9}} = \int \frac{3\sinh t}{3\sinh t}\,dt = \int dt = t + C = \operatorname{arcosh}\left(\frac{x}{3}\right) + C.$$

2. The form $\sqrt{x^2+a^2}$ calls for a sinh substitution: let $x = 2\sinh t$, $dx = 2\cosh t\,dt$, $\sqrt{x^2+4} = 2\cosh t$. Then

$$\int \frac{dx}{\sqrt{x^2+4}} = \int \frac{2\cosh t}{2\cosh t}\,dt = t + C = \operatorname{arsinh}\left(\frac{x}{2}\right) + C = \ln\left(\frac{x+\sqrt{x^2+4}}{2}\right) + C.$$

SOLUTION **Exercise 9 — Maclaurin series and approximation** **(14 marks)**

(a) (4 marks) Since $\frac{d}{dx}\sinh x = \cosh x$ and $\frac{d}{dx}\cosh x = \sinh x$, with $\sinh 0 = 0$, $\cosh 0 = 1$, the Maclaurin series are:

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = x + \frac{x^3}{6} + \frac{x^5}{120} + \cdots \quad \text{(M1 A1)}$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \cdots \quad \text{(M1 A1)}$$

(b) (5 marks) Write $\tanh x = \frac{\sinh x}{\cosh x}$ and use long division (or set $\tanh x = a_1 x + a_3 x^3 + a_5 x^5$, since tanh is odd, and solve $\sinh x = \cosh x \cdot \tanh x$):

$$x + \frac{x^3}{6} + \frac{x^5}{120} = \left(1 + \frac{x^2}{2} + \frac{x^4}{24}\right)\left(a_1 x + a_3 x^3 + a_5 x^5\right).$$

Comparing coefficients: $a_1 = 1$; $\frac{a_1}{2} + a_3 = \frac{1}{6} \Rightarrow a_3 = -\frac{1}{3}$; $\frac{a_1}{24} + \frac{a_3}{2} + a_5 = \frac{1}{120} \Rightarrow a_5 = \frac{1}{120} - \frac{1}{24} + \frac{1}{6} = \frac{2}{15}$. (M1 M1 A1 A1 A1)

$$\tanh x = x - \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$$

(c) (3 marks) With $x = 0.2$: (M1 for substitution)

$$\tanh 0.2 \approx 0.2 - \frac{0.008}{3} + \frac{2(0.00032)}{15} = 0.2 - 0.0026667 + 0.0000427 = 0.19738 \text{ (5 d.p.).} \quad \text{(A1 A1)}$$

(d) (2 marks) The estimate 0.19738 agrees with the true value 0.197375 to 5 decimal places (error $\approx 10^{-6}$). (B1) The approximation is excellent because $|x| = 0.2$ is small, so the neglected term in x^7 is of order 10^{-7}. (B1)

SOLUTION **Exercise 10 — Calculus with hyperbolic functions** **(16 marks)**

(a) (5 marks) $\operatorname{sech} x = \dfrac{2}{e^x + e^{-x}} = \dfrac{2e^x}{e^{2x}+1}$. Let $u = e^x$, $du = e^x dx$: (M1 M1)

$$\int \operatorname{sech} x\,dx = \int \frac{2e^x}{e^{2x}+1}\,dx = \int \frac{2\,du}{u^2+1} = 2\arctan u + C = 2\arctan(e^x) + C. \quad \text{(M1 A1 A1)} \blacksquare$$

(b) (4 marks) $y = \operatorname{artanh} x \iff x = \tanh y$. Differentiating with respect to y: $\dfrac{dx}{dy} = \operatorname{sech}^2 y = 1 - \tanh^2 y = 1 - x^2$. (M1 A1) Hence $\dfrac{dy}{dx} = \dfrac{1}{1-x^2}$, valid for $|x| < 1$ (where $1 - x^2 \neq 0$). (M1 A1) $\blacksquare$

(c) (3 marks) From (b), an antiderivative of $\dfrac{1}{1-x^2}$ is $\operatorname{artanh} x$: (M1)

$$\int_0^{1/2} \frac{dx}{1-x^2} = \left[\operatorname{artanh} x\right]_0^{1/2} = \operatorname{artanh}\frac{1}{2} = \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) = \frac{1}{2}\ln 3. \quad \text{(M1 A1)}$$

(d) (4 marks) $\operatorname{artanh} x = \dfrac{1}{2}\ln\dfrac{1+x}{1-x} = \ln 2 \Rightarrow \ln\dfrac{1+x}{1-x} = \ln 4 \Rightarrow \dfrac{1+x}{1-x} = 4$. (M1 A1) Then $1+x = 4-4x$, so $5x = 3$ and $x = \dfrac{3}{5}$. (M1 A1) Check: $\frac{3}{5} \in]-1, 1[$, valid.

SOLUTION **Exercise 11 — The catenary: a hanging chain** **(18 marks)**

(a) (4 marks) $y = 10\cosh\left(\dfrac{x}{10}\right) \Rightarrow \dfrac{dy}{dx} = 10 \cdot \dfrac{1}{10}\sinh\left(\dfrac{x}{10}\right) = \sinh\left(\dfrac{x}{10}\right)$. (M1 A1) Then

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \sinh^2\left(\frac{x}{10}\right) = \cosh^2\left(\frac{x}{10}\right),$$

by the fundamental identity $\cosh^2 u - \sinh^2 u = 1$. (M1 A1) $\blacksquare$

(b) (6 marks) Since $\cosh\left(\frac{x}{10}\right) > 0$, $\sqrt{1 + (y')^2} = \cosh\left(\frac{x}{10}\right)$. (M1) Hence

$$L = \int_{-10}^{10} \cosh\left(\frac{x}{10}\right) dx = \left[10\sinh\left(\tfrac{x}{10}\right)\right]_{-10}^{10} = 10\sinh 1 - 10\sinh(-1) = 20\sinh 1, \quad \text{(M1 M1 A1)}$$

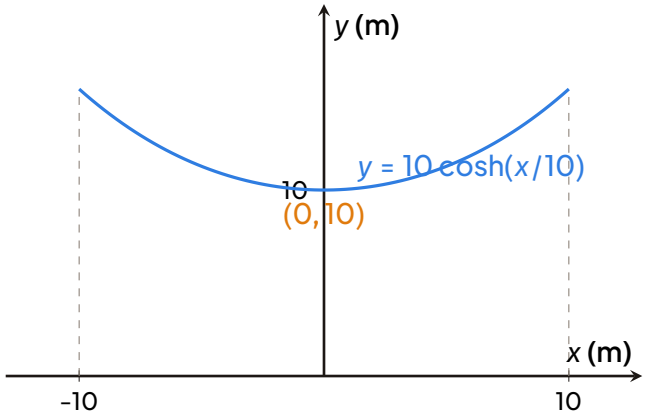
using $\sinh(-1) = -\sinh 1$. Numerically, $\sinh 1 = \dfrac{e - e^{-1}}{2} \approx 1.17520$, so $L \approx 23.50$ m (2 d.p.). (M1 A1)

(c) (3 marks) The lowest point is at $x = 0$: $y(0) = 10\cosh 0 = 10$ m. (B1) At the pylons, $y(\pm 10) = 10\cosh 1$. (B1) The height above the lowest point is $10\cosh 1 - 10 = 10(\cosh 1 - 1) \approx 5.43$ m. (B1)

(d) (5 marks) Expected features: (B1 each)

- curve symmetric about the y-axis (since cosh is even);
- minimum point $(0, 10)$ clearly marked;

- curve passing through $(\pm 10, 10\cosh 1) \approx (\pm 10, 15.43)$;
- $y \to +\infty$ rapidly (exponentially) as $|x|$ increases;
- smooth U-shape, always above $y = 10$.



The catenary $y = 10\cosh(x/10)$: symmetric about the y-axis, minimum at $(0, 10)$.

19 SUMMARY SHEET

Hyperbolic & Inverse Hyperbolic Functions

Definitions $\cosh x$, $\sinh x$, $\tanh x$
- Exponential forms
- Numerical values

Graphs
- $y = \cosh x$: catenary
- $y = \sinh x$: odd

Identities & Osborn's rule
- $\cosh^2 x - \sinh^2 x = 1$

Inverse functions
- Log forms e.g. $\operatorname{arsinh} x$
- Graphs & domains

Hyperbolic equations
- Solve via e^x

Calculus
- Derivatives
- Integrals
- Substitutions

Maclaurin series
- $\sinh x$, $\cosh x$

Mind map of the chapter: hyperbolic and inverse hyperbolic functions.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions.

- $\cosh x = \dfrac{e^x + e^{-x}}{2}$ **(even)**, $\sinh x = \dfrac{e^x - e^{-x}}{2}$ **(odd)**, $\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^{2x} - 1}{e^{2x} + 1}$.
- $\operatorname{sech} x = \dfrac{1}{\cosh x}$, $\operatorname{cosech} x = \dfrac{1}{\sinh x}$, $\coth x = \dfrac{1}{\tanh x}$.
- **Special values:** $\cosh 0 = 1$, $\sinh 0 = 0$, $\tanh 0 = 0$; $\cosh x \geq 1$ **for all** x; $-1 < \tanh x < 1$.
- **Inverse functions:** $y = \operatorname{arsinh} x \iff x = \sinh y$ **(domain** $\mathbb{R}$**)**; $y = \operatorname{arcosh} x \iff x = \cosh y$, $y \geq 0$ **(domain** $x \geq 1$**)**; $y = \operatorname{artanh} x \iff x = \tanh y$ **(domain** $|x| < 1$**)**.

Formulas to know by heart.

- **Fundamental identity:** $\cosh^2 x - \sinh^2 x = 1$; **hence** $1 - \tanh^2 x = \operatorname{sech}^2 x$ **and** $\coth^2 x - 1 = \operatorname{cosech}^2 x$.
- **Addition:** $\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$; $\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$.
- **Double angle:** $\cosh 2x = \cosh^2 x + \sinh^2 x = 2\cosh^2 x - 1 = 1 + 2\sinh^2 x$; $\sinh 2x = 2 \sinh x \cosh x$.
- **Osborn's rule**: take any trigonometric identity, replace cos → cosh, sin → sinh, and change the sign of every product of two sines (or implied product, e.g. $\tan^2$, $\sin^2$).

- **Logarithmic forms:** $\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$, **all** x; $\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$, $x \geq 1$; $\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$, $|x| < 1$.

Calculus results.

- **Derivatives:** $(\sinh x)' = \cosh x$; $(\cosh x)' = \sinh x$ **(no minus sign!)**; $(\tanh x)' = \operatorname{sech}^2 x$.
- **Inverse derivatives:** $(\operatorname{arsinh} x)' = \frac{1}{\sqrt{x^2+1}}$; $(\operatorname{arcosh} x)' = \frac{1}{\sqrt{x^2-1}}$ $(x > 1)$; $(\operatorname{artanh} x)' = \frac{1}{1-x^2}$ $(|x| < 1)$.
- **Integrals:** $\int \sinh x\,dx = \cosh x + C$; $\int \cosh x\,dx = \sinh x + C$; $\int \operatorname{sech}^2 x\,dx = \tanh x + C$.
- **Standard integral forms:** $\int \frac{dx}{\sqrt{x^2+a^2}} = \operatorname{arsinh}\frac{x}{a} + C$; $\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}\frac{x}{a} + C$ $(x > a)$; $\int \frac{dx}{a^2-x^2} = \frac{1}{a}\operatorname{artanh}\frac{x}{a} + C$ $(|x| < a)$.
- **Maclaurin series (valid for all x):** $\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$; $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$.

Methods.

- To evaluate numerically: rewrite in terms of e^x and use the calculator's e^x key.
- To solve a hyperbolic equation: either use identities to reduce to one function, or set $u = e^x$ to obtain a quadratic in u; keep only $u > 0$, then $x = \ln u$.
- To prove an identity: start from the exponential definitions, or quote Osborn's rule applied to the corresponding trig identity.
- For $\int dx/\sqrt{x^2+a^2}$ substitute $x = a\sinh u$; for $\int dx/\sqrt{x^2-a^2}$ substitute $x = a\cosh u$; for $\int dx/(a^2-x^2)$ substitute $x = a\tanh u$.
- For Maclaurin expansions: differentiate repeatedly at 0, or add/subtract the series of e^x and e^{-x}.

Common mistakes.

- Writing $\cosh^2 x + \sinh^2 x = 1$: the correct identity has a minus sign.
- Putting a minus sign in $(\cosh x)'$: it is $+\sinh x$.
- Forgetting the domains: $\operatorname{arcosh} x$ needs $x \geq 1$; $\operatorname{artanh} x$ needs $|x| < 1$.
- Taking the negative root in arcosh: by definition $\operatorname{arcosh} x \geq 0$.
- Rejecting valid roots: when solving with $u = e^x$, only $u > 0$ is acceptable, but every positive u gives a solution.
- Misapplying Osborn's rule: the sign change applies to products of two sines, including hidden ones like $\tan^2 x = \sin^2 x/\cos^2 x$.

GCE tips.

- Quote the exponential definitions first in any proof question; examiners award marks for them.
- Show the substitution $u = e^x$ explicitly when solving equations, and state why negative roots are rejected.
- In integration, identify the standard form before substituting, and do not forget the constant C and the domain condition.
- Sketch graphs quickly: cosh is a U-shape through $(0, 1)$, sinh passes through the origin, tanh has asymptotes $y = \pm 1$; inverse graphs are reflections in $y = x$.

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